Linear Algebra

Lecture notes, Chapters 1–6 and Appendices

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Chapter 1. Vector Spaces

1.1 Introduction

Motivation. Forces and velocities add by the parallelogram law and can be stretched by real factors. Eight algebraic rules satisfied by these two operations are all that linear algebra uses, so any system obeying them (tuples, matrices, functions, polynomials) can be treated by the same theory.

Two vectors x, y acting at a point P add by the **parallelogram law**: $x + y$ is the diagonal of the parallelogram with sides x and y. With P at the origin, if x ends at (a_1, a_2) and y at (b_1, b_2), then $x + y$ ends at $(a_1 + b_1, a_2 + b_2)$, and tx ends at (ta_1, ta_2).

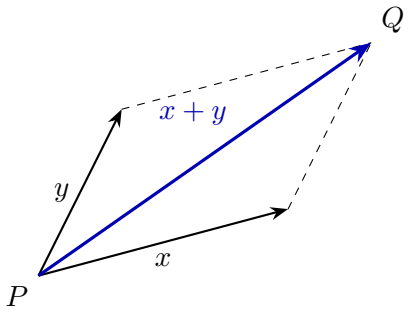


Figure. *Parallelogram law: $x + y$ is the diagonal through P.*

Geometric vectors satisfy, for all vectors x, y, z and reals a, b:

1. $x + y = y + x$; 2. $(x + y) + z = x + (y + z)$; 3. there is 0 with $x + 0 = x$; 4. each x has y with $x + y = 0$;

5. $1x = x$; 6. $(ab)x = a(bx)$; 7. $a(x + y) = ax + ay$; 8. $(a + b)x = ax + bx$.

Lines and planes. If u, v end at A, B, the line through A and B is $x = u + t(v - u)$, $t \in \mathbb{R}$. If u, v start at A and end at B, C (three noncollinear points), the plane through A, B, C is $x = A + su + tv$, $s, t \in \mathbb{R}$.

Example 1

For $A = (-2, 0, 1)$, $B = (4, 5, 3)$: $B - A = (6, 5, 2)$, and the line is $x = (-2, 0, 1) + t(6, 5, 2)$.

Example 2

For $A = (1, 0, 2)$, $B = (-3, -2, 4)$, $C = (1, 8, -5)$: $B - A = (-4, -2, 2)$, $C - A = (0, 8, -7)$, and the plane is $x = (1, 0, 2) + s(-4, -2, 2) + t(0, 8, -7)$.

1.2 Vector spaces

Motivation. The definition abstracts the eight rules. Many familiar collections (tuples, matrices, functions, polynomials, sequences) satisfy them, and a handful of consequences, such as cancellation and uniqueness of 0 and $-x$, follow from the axioms alone.

Definition (Vector space)

A **vector space** V over a field $\mathbb{F}$ (see Appendix C) is a set with an addition $V \times V \to V$ and a scalar multiplication $\mathbb{F} \times V \to V$ such that for all $x, y, z \in V$, $a, b \in \mathbb{F}$:
(VS 1) $x + y = y + x$; (VS 2) $(x + y) + z = x + (y + z)$;
(VS 3) there is $0 \in V$ with $x + 0 = x$ for all x; (VS 4) for each x there is y with $x + y = 0$;
(VS 5) $1x = x$; (VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$; (VS 8) $(a + b)x = ax + bx$.
Elements of $\mathbb{F}$ are **scalars**, elements of V **vectors**.

An n**-tuple** $(a_1, \ldots, a_n)$ has entries in $\mathbb{F}$; tuples are equal iff all entries agree. An $m \times n$ **matrix** A has entries A_{ij} (i-th row, j-th column); A_{ii} are the **diagonal entries**, O is the **zero matrix**, and A is **square** if $m = n$.

Example 1

$\mathbb{F}^n$, the n-tuples with coordinatewise operations, is a vector space; e.g. in $\mathbb{C}^2$, $(1+i, 2) + (2 - 3i, 4i) = (3 - 2i, 2 + 4i)$ and $i(1 + i, 2) = (-1 + i, 2i)$. Vectors of $\mathbb{F}^n$ are often written as columns; $\mathbb{F}^1 = \mathbb{F}$.

Example 2

$\mathrm{M}_{m\times n}(\mathbb{F})$, with $(A + B)_{ij} = A_{ij} + B_{ij}$ and $(cA)_{ij} = cA_{ij}$, is a vector space.

Example 3

For a nonempty set S, the functions $\mathcal{F}(S, \mathbb{F})$ with $(f+g)(s) = f(s)+g(s)$ and $(cf)(s) = cf(s)$ form a vector space.

A **polynomial** $f(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_k \in \mathbb{F}$ has **degree** the largest k with $a_k \neq 0$; the **zero polynomial** has degree -1. Over an infinite field we may regard polynomials as functions $\mathbb{F} \to \mathbb{F}$.

Example 4

$\mathsf{P}(\mathbb{F})$, the polynomials with coefficientwise addition and scalar multiplication (pad the shorter one with zero coefficients), is a vector space.

Example 5

The sequences $\{a_n\}$ in $\mathbb{F}$ with only finitely many nonzero terms, with termwise operations, form a vector space.

Example 6

On $S = \mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 - b_2)$ and $c(a_1, a_2) = (ca_1, ca_2)$, (VS 1), (VS 2) and (VS 8) fail: S is not a vector space.

Example 7

On $S = \mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, 0)$ and $c(a_1, a_2) = (ca_1, 0)$, (VS 3) (hence (VS 4)) and (VS 5) fail.

Theorem 1.1 (Cancellation law for vector addition)

If $x + z = y + z$ in V, then $x = y$.

Proof. Take v with $z + v = 0$ (VS 4). Then $x = x + 0 = x + (z + v) = (x + z) + v = (y + z) + v = y + (z + v) = y + 0 = y$ by (VS 2), (VS 3). ■

Corollary 1

The vector 0 of (VS 3) is unique.

Proof [Supplement]. If $0'$ also satisfies (VS 3), then $0 + 0' = 0 = 0 + 0$ (by (VS 1), (VS 3)), so $0' = 0$ by Theorem 1.1. ■

Corollary 2

The vector y of (VS 4) is unique.

Proof [Supplement]. If $x+y = 0 = x+y'$, then $y+x = y'+x$ by (VS 1), so $y = y'$ by Theorem 1.1. ■

0 is the **zero vector**, and the y of (VS 4) is the **additive inverse** $-x$.

Theorem 1.2

In any vector space V:
(a) $0x = 0$ for each $x \in V$.
(b) $(-a)x = -(ax) = a(-x)$ for each $a \in \mathbb{F}$, $x \in V$.
(c) $a0 = 0$ for each $a \in \mathbb{F}$.

Proof.

(a) $0x + 0x = (0 + 0)x = 0x = 0x + 0 = 0 + 0x$, so $0x = 0$ by Theorem 1.1.

(b) $ax+(-a)x = (a+(-a))x = 0x = 0$, so $(-a)x = -(ax)$ by Corollary 2. In particular $(-1)x = -x$, and $a(-x) = a((-1)x) = (a(-1))x = (-a)x$ by (VS 6).

(c) [Supplement] $a0 + a0 = a(0 + 0) = a0 = a0 + 0$, so $a0 = 0$ by Theorem 1.1. ■

1.3 Subspaces

Motivation. A subset that is itself a vector space under the inherited operations is a subspace. Three closure checks suffice. Intersections of subspaces are subspaces; unions usually are not, which is why sums are introduced.

Definition (Subspace)

A subset W of V is a **subspace** if it is a vector space over $\mathbb{F}$ with the operations of V. V and the **zero subspace** $\{0\}$ are subspaces.

(VS 1, 2, 5–8) are inherited by any subset, so W is a subspace iff it is closed under addition and scalar multiplication, has a zero vector, and contains additive inverses.

Theorem 1.3

$W \subseteq V$ is a subspace iff (a) $0 \in W$,
(b) $x + y \in W$ for $x, y \in W$, and (c) $cx \in W$ for $c \in \mathbb{F}$, $x \in W$.

Proof.

Step 1 (only if). (b), (c) hold since W is a vector space under these operations. If $0'$ is the zero of W, then $x + 0' = x = x + 0$ for $x \in W$, so $0' = 0$ by Theorem 1.1, and $0 \in W$.

Step 2 (if). It remains to see that inverses lie in W: for $x \in W$, $-x = (-1)x \in W$ by (c) and Theorem 1.2. ■

The **transpose** A^t of $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has $(A^t)_{ij} = A_{ji}$; $(aA + bB)^t = aA^t + bB^t$. A is **symmetric** if $A^t = A$ (so A is square). The symmetric matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$: $O^t = O$, $(A + B)^t = A + B$, $(aA)^t = aA$.

Example 1

$\mathsf{P}_n(\mathbb{F})$, the polynomials of degree $\leq n$, is a subspace of $\mathsf{P}(\mathbb{F})$.

Example 2

$C(\mathbb{R})$, the continuous real functions, is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Example 3

M is **diagonal** if $M_{ij} = 0$ for $i \neq j$. The diagonal $n \times n$ matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$.

Example 4

The **trace** is $\mathrm{tr}(M) = M_{11} + \cdots + M_{nn}$. It is linear, so the trace-zero matrices form a subspace.

Example 5

The matrices with nonnegative entries do not form a subspace of $\mathrm{M}_{m\times n}(\mathbb{R})$: multiplying by -1 leaves the set.

Theorem 1.4

Any intersection of subspaces of V is a subspace of V.

Proof. Let W be the intersection of a collection $\mathcal{C}$ of subspaces. Each contains 0, so $0 \in W$. If $x, y \in W$ and $a \in \mathbb{F}$, then $x + y$ and ax lie in every member of $\mathcal{C}$, hence in W. Apply Theorem 1.3. ■

A union of two subspaces is a subspace iff one contains the other. The **sum** $W_1 + W_2 = \{x + y : x \in W_1, y \in W_2\}$ is the smallest subspace containing both. V is the **direct sum** $W_1 \oplus W_2$ if $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{0\}$.

1.4 Linear combinations and systems of linear equations

Motivation. Deciding whether a vector is a linear combination of others is solving a system of linear equations. The set of all linear combinations, the span, is the smallest subspace containing the given vectors.

Definition (Linear combination)

For nonempty $S \subseteq V$, v is a **linear combination** of vectors of S if $v = a_1u_1 + \cdots + a_nu_n$ for some finitely many $u_i \in S$ and scalars a_i (the **coefficients**). $0 = 0u$ is a linear combination of any nonempty S.

Example 1

Vitamin contents of foods are vectors in $\mathbb{R}^5$. A combination of foods has the corresponding linear combination of their vitamin vectors; e.g. the vitamin vector of some food is a sum of those of several others.

Solving systems by elimination. Three operations do not change the solution set of a system of linear equations: interchanging two equations, multiplying an equation by a nonzero constant, and adding a multiple of one equation to another. Using them, bring the system to a form where (i) the first nonzero coefficient of each equation is 1, (ii) that variable occurs in no other equation, and (iii) these leading variables move to the right from equation to equation. The solutions can then be read off; an equation $0 = c$ with $c \neq 0$ signals inconsistency.

Example 2

Is $2x^3 - 2x^2 + 12x - 6$ a linear combination of $x^3 - 2x^2 - 5x - 3$ and $3x^3 - 5x^2 - 4x - 9$? Comparing coefficients in $a(x^3 - 2x^2 - 5x - 3) + b(3x^3 - 5x^2 - 4x - 9)$ gives

$$a + 3b = 2, \quad -2a - 5b = -2, \quad -5a - 4b = 12, \quad -3a - 9b = -6,$$

with solution $a = -4$, $b = 2$. So yes. For $3x^3 - 2x^2 + 7x + 8$ the corresponding system reduces to an equation $0 = 1$, so it is not a linear combination.

Definition (Span)

The **span** $\operatorname{span}(S)$ of nonempty $S \subseteq V$ is the set of all linear combinations of vectors in S; $\operatorname{span}(\emptyset) = \{0\}$.

Theorem 1.5

$\operatorname{span}(S)$ is a subspace of V, and every subspace containing S contains $\operatorname{span}(S)$.

Proof. For $S = \emptyset$ this is clear. Otherwise $0 = 0z \in \operatorname{span}(S)$ for $z \in S$. If $x = \sum a_iu_i$ and $y = \sum b_jv_j$, then $x + y = \sum a_iu_i + \sum b_jv_j$ and $cx = \sum (ca_i)u_i$ are again linear combinations, so $\operatorname{span}(S)$ is a subspace by Theorem 1.3. A subspace $W \supseteq S$ is closed under addition and scalar multiplication, so it contains every $\sum a_iu_i$ with $u_i \in S$. ∎

Definition (Generating set)

S **generates** (**spans**) V if $\operatorname{span}(S) = V$.

Example 3

$(1,1,0),(1,0,1),(0,1,1)$ generate $\mathbb{R}^3$:

$$(a_1,a_2,a_3)=\tfrac{1}{2}(a_1+a_2-a_3)(1,1,0)+\tfrac{1}{2}(a_1-a_2+a_3)(1,0,1)+\tfrac{1}{2}(-a_1+a_2+a_3)(0,1,1).$$

Example 4

x^2+3x-2, $2x^2+5x-3$, $-x^2-4x+4$ generate $\mathsf{P}_2(\mathbb{R})$: every ax^2+bx+c is a combination of them (solve the 3×3 system).

Example 5

The matrices $\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$ generate $\mathrm{M}_{2\times 2}(\mathbb{R})$. On the other hand, matrices all having equal diagonal entries cannot generate $\mathrm{M}_{2\times 2}(\mathbb{R})$, since every linear combination of them again has equal diagonal entries.

1.5 Linear dependence and linear independence

Motivation. A generating set may be wasteful: some vector may already be a combination of the others. Linear independence means no redundancy, i.e. 0 has only the trivial representation.

Definition (Linear dependence)

$S\subseteq V$ is **linearly dependent** if there are finitely many distinct $u_1,\dots,u_n\in S$ and scalars $a_1,\dots,a_n$, not all zero, with $a_1u_1+\cdots+a_nu_n=0$ (a **nontrivial representation** of 0). Any set containing 0 is dependent $(1\cdot 0=0)$.

Example 1

$S=\{(1,3,-4,2),(2,2,-4,0),(1,-3,2,-4),(-1,0,1,0)\}\subseteq\mathbb{R}^4$ is dependent: $4(1,3,-4,2)-3(2,2,-4,0)+2(1,-3,2,-4)+0(-1,0,1,0)=0$.

Example 2

In $\mathrm{M}_{2\times 3}(\mathbb{R})$, $\left(\begin{smallmatrix}1&-3&2\\-4&0&5\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}-3&7&4\\6&-2&-7\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}-2&3&11\\-1&-3&2\end{smallmatrix}\right)$ are dependent: $5M_1+3M_2-2M_3=O$.

Definition (Linear independence)

S is **linearly independent** if it is not linearly dependent: $\sum a_iu_i=0$ with distinct $u_i\in S$ forces all $a_i=0$.

$\emptyset$ is independent; $\{u\}$ is independent iff $u\neq 0$ (if $au=0$, $a\neq 0$, then $u=a^{-1}(au)=0$).

Example 3

$\{(1,0,0,-1),(0,1,0,-1),(0,0,1,-1),(0,0,0,1)\}$ is independent: $a_1(1,0,0,-1)+\cdots+a_4(0,0,0,1)=(a_1,a_2,a_3,a_4-a_1-a_2-a_3)=0$ forces $a_1=a_2=a_3=a_4=0$.

Example 4

For $p_k(x) = x^k + x^{k+1} + \cdots + x^n$ $(0 \le k \le n)$, $\{p_0, \ldots, p_n\}$ is independent: in $\sum a_k p_k = 0$ the coefficient of x^m is $a_0 + \cdots + a_m$, so $a_0 = 0$, then $a_1 = 0$, etc.

Theorem 1.6

If $S_1 \subseteq S_2 \subseteq V$ and S_1 is linearly dependent, so is S_2.

Proof. A nontrivial representation of 0 by vectors of S_1 is one by vectors of S_2. ■

Corollary

If $S_1 \subseteq S_2 \subseteq V$ and S_2 is linearly independent, so is S_1.

Proof. Contrapositive of Theorem 1.6. ■

Theorem 1.7

Let S be linearly independent and $v \notin S$. Then $S \cup \{v\}$ is linearly dependent iff $v \in \text{span}(S)$.

Proof.

Step 1 (only if). Take $a_1 u_1 + \cdots + a_n u_n = 0$ nontrivial with distinct $u_i \in S \cup \{v\}$. Since S is independent, some u_i, say u_1, equals v and $a_1 \neq 0$. Then $v = -a_1^{-1}(a_2 u_2 + \cdots + a_n u_n) \in \text{span}(S)$.

Step 2 (if). If $v = b_1 v_1 + \cdots + b_m v_m$ with distinct $v_i \in S$, then $b_1 v_1 + \cdots + b_m v_m + (-1)v = 0$ is nontrivial, with $v \neq v_i$. ■

1.6 Bases and dimension

Motivation. A basis is a generating set with no redundancy, and gives unique coordinates. The replacement theorem shows that independent sets never outnumber generating sets. So all bases have the same size, the dimension, and finite-dimensional spaces behave like $\mathbb{F}^n$.

Definition (Basis)

A **basis** β for V is a linearly independent subset that generates V.

Example 1

$\emptyset$ is a basis for the zero space.

Example 2

$e_1 = (1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$ form the **standard basis** of $\mathbb{F}^n$.

Example 3

The matrices E^{ij} (single 1 in position (i, j)) form a basis of $\text{M}_{m \times n}(\mathbb{F})$.

Example 4

$\{1, x, \ldots, x^n\}$ is the **standard basis** of $\mathsf{P}_n(\mathbb{F})$.

Example 5

$\{1, x, x^2, \ldots\}$ is a basis of $\mathsf{P}(\mathbb{F})$.

Theorem 1.8

$\beta = \{u_1, \ldots, u_n\}$ is a basis for V iff each $v \in V$ can be written uniquely as $v = a_1u_1 + \cdots + a_nu_n$.

Proof.

Step 1 (only if). Existence holds since β generates V. If $v = \sum a_iu_i = \sum b_iu_i$, then $\sum(a_i - b_i)u_i = 0$, and independence gives $a_i = b_i$.

Step 2 (if) [Supplement]. β generates V. Uniqueness for $v = 0$ (with $0 = \sum 0u_i$) means $\sum a_iu_i = 0$ forces all $a_i = 0$, so β is independent. ■

So a basis of n vectors gives each v a unique n-tuple of coordinates; V "is" $\mathbb{F}^n$ (made precise in Section 2.4).

Theorem 1.9

If V is generated by a finite set S, then some subset of S is a basis for V. Hence V has a finite basis.

Proof. If $S = \emptyset$ or $S = \{0\}$, then $V = \{0\}$ and $\emptyset$ is a basis. Otherwise S contains some $u_1 \neq 0$, and $\{u_1\}$ is independent. Extend $\beta = \{u_1, \ldots, u_k\} \subseteq S$ by vectors of S as long as it stays independent; since S is finite this stops at an independent $\beta \subseteq S$ such that $\beta \cup \{v\}$ is dependent for every $v \in S - \beta$. By Theorem 1.7, $S \subseteq \text{span}(\beta)$, so $V = \text{span}(S) \subseteq \text{span}(\beta)$ by Theorem 1.5. Hence β is a basis. ■

Example 6

From $S = \{(2,-3,5), (8,-12,20), (1,0,-2), (0,2,-1), (7,2,0)\}$ keep $(2,-3,5)$, discard $(8,-12,20) = 4(2,-3,5)$, keep $(1,0,-2)$ and $(0,2,-1)$ (each independent of the previous ones), and discard $(7,2,0) = 2(2,-3,5) + 3(1,0,-2) + 4(0,2,-1)$. So $\{(2,-3,5), (1,0,-2), (0,2,-1)\}$ is a basis of $\mathbb{R}^3$.

Theorem 1.10 (Replacement theorem)

Let V be generated by a set G of exactly n vectors and let L be linearly independent with exactly m vectors. Then $m \leq n$, and there is $H \subseteq G$ with exactly $n - m$ vectors such that $L \cup H$ generates V.

Proof. Induction on m; $m = 0$ is trivial with $H = G$. Let $L = \{v_1, \ldots, v_{m+1}\}$ be independent. By Corollary , $\{v_1, \ldots, v_m\}$ is independent, so by induction $m \leq n$ and some $\{u_1, \ldots, u_{n-m}\} \subseteq G$ makes $\{v_1, \ldots, v_m, u_1, \ldots, u_{n-m}\}$ generate V. Write

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_{n-m}u_{n-m} = v_{m+1}. \tag{1.1}$$

If $n - m = 0$, or all $b_i = 0$, then $v_{m+1} \in \text{span}\{v_1, \ldots, v_m\}$, contradicting Theorem 1.7. So $n \geq m + 1$ and some b_i, say b_1, is nonzero. Solving (1.1) for u_1 shows $u_1 \in \text{span}(L \cup H)$ with $H = \{u_2, \ldots, u_{n-m}\}$

$(n-(m+1)$ vectors). Then $\operatorname{span}(L \cup H)$ contains $v_1, \ldots, v_m, u_1, \ldots, u_{n-m}$, hence equals V by Theorem 1.5. ■

Corollary 1

If V has a finite basis, every basis of V has the same number of vectors.

Proof. Let β be a basis with n vectors. Any independent γ has at most n vectors (apply Theorem 1.10 to finite subsets of γ), so γ is finite with $m \leq n$ vectors. Reversing roles, $n \leq m$. ■

Definition (Dimension)

V is **finite-dimensional** if it has a finite basis; the common number of vectors in a basis is its **dimension** $\dim(V)$. Otherwise V is **infinite-dimensional**.

Example 7

$\dim\{0\} = 0$.

Example 8

$\dim \mathbb{F}^n = n$.

Example 9

$\dim \mathrm{M}_{m\times n}(\mathbb{F}) = mn$.

Example 10

$\dim \mathsf{P}_n(\mathbb{F}) = n+1$.

Example 11

$\mathbb{C}$ over $\mathbb{C}$ has dimension 1 (basis $\{1\}$).

Example 12

$\mathbb{C}$ over $\mathbb{R}$ has dimension 2 (basis $\{1, i\}$). So the dimension depends on the field.

Corollary 2

Let $\dim(V) = n$.
(a) Every finite generating set has at least n vectors, and a generating set of exactly n vectors is a basis.
(b) Every linearly independent set of exactly n vectors is a basis.
(c) Every linearly independent subset of V extends to a basis.

Proof. Let β be a basis (n vectors).

(a) A finite generating set G contains a basis H by Theorem 1.9, and $|H| = n$ by Corollary 1. So $|G| \geq n$, and if $|G| = n$, then $G = H$.

(b) Apply Theorem 1.10 with $G = \beta$ and L independent, $|L| = n$: some $H \subseteq \beta$ with 0 vectors makes $L = L \cup H$ generate V.

(c) If L is independent with m vectors ($m \leq n$), Theorem 1.10 gives $H \subseteq \beta$ of $n - m$ vectors with $L \cup H$ generating V. Then $|L \cup H| \leq n$, and by (a) $L \cup H$ has exactly n vectors and is a basis. ■

Example 13

By Example 4 of Section 1.4 and (a), $\{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4x + 4\}$ is a basis of $\mathsf{P}_2(\mathbb{R})$.

Example 14

By Example 5 of Section 1.4 and (a), the four matrices there form a basis of $\mathrm{M}_{2\times 2}(\mathbb{R})$.

Example 15

By Example 3 of Section 1.5 and (b), $\{(1,0,0,-1), (0,1,0,-1), (0,0,1,-1), (0,0,0,1)\}$ is a basis of $\mathbb{R}^4$.

Example 16

By Example 4 of Section 1.5 and (b), $\{p_0, \ldots, p_n\}$ is a basis of $\mathsf{P}_n(\mathbb{F})$.

Summary. If $\dim V = n$: every basis has n vectors; independent sets have at most n vectors and extend to bases; generating sets have at least n vectors and shrink to bases; a set of exactly n vectors is a basis as soon as it is independent *or* generating.

Theorem 1.11

If W is a subspace of a finite-dimensional V, then W is finite-dimensional and $\dim W \leq \dim V$; if $\dim W = \dim V$, then $W = V$.

Proof. Let $\dim V = n$. If $W = \{0\}$, we are done. Otherwise choose $x_1 \neq 0$ in W and keep adding vectors $x_{k+1} \in W$ so that $\{x_1, \ldots, x_{k+1}\}$ stays independent. By Theorem 1.10 this stops after at most n steps, at an independent $\{x_1, \ldots, x_k\}$ to which no vector of W can be added. By Theorem 1.7 it generates W, so it is a basis, and $\dim W = k \leq n$. If $k = n$, it is a basis of V by Corollary 2(b), so $W = V$. ■

Example 17

$W = \{(a_1, \ldots, a_5) \in \mathbb{F}^5 : a_1 + a_3 + a_5 = 0,\ a_2 = a_4\}$ has basis $\{(-1,0,1,0,0), (-1,0,0,0,1), (0,1,0,1,0)\}$, so $\dim W = 3$.

Example 18

The diagonal $n \times n$ matrices have basis $\{E^{11}, \ldots, E^{nn}\}$; dimension n.

Example 19

The symmetric $n \times n$ matrices have basis $\{A^{ij} : 1 \le i \le j \le n\}$, where A^{ij} has 1 in positions (i,j) and (j,i) and 0 elsewhere; dimension $\frac{1}{2}n(n+1)$.

Corollary

If W is a subspace of a finite-dimensional V, any basis for W extends to a basis for V.

Proof. A basis of W is an independent subset of V; apply Corollary 2(c) of Theorem 1.10. ■

Example 20

The polynomials $a_{18}x^{18} + a_{16}x^{16} + \cdots + a_2x^2 + a_0$ form a subspace of $\mathsf{P}_{18}(\mathbb{F})$ with basis $\{1, x^2, \ldots, x^{18}\}$, dimension 10.

Subspaces of $\mathbb{R}^3$. Their dimensions are $0, 1, 2, 3$: the origin, lines through 0, planes through 0, and $\mathbb{R}^3$.

1.6.1 The Lagrange interpolation formula

Let $c_0, \ldots, c_n$ be distinct scalars of an infinite field $\mathbb{F}$, and

$$f_i(x) = \prod_{k \ne i} \frac{x - c_k}{c_i - c_k} \qquad (0 \le i \le n),$$

the **Lagrange polynomials**, of degree n, with $f_i(c_j) = \delta_{ij}$ (1 if $i = j$, else 0). They are independent: if $\sum a_i f_i = 0$, evaluating at c_j gives $a_j = 0$. Since $\dim \mathsf{P}_n(\mathbb{F}) = n+1$, $\beta = \{f_0, \ldots, f_n\}$ is a basis by Corollary 2(b). Evaluating $g = \sum b_i f_i$ at c_j gives $b_j = g(c_j)$, so

$$g = \sum_{i=0}^{n} g(c_i) f_i \qquad (g \in \mathsf{P}_n(\mathbb{F})),$$

the **Lagrange interpolation formula**. Hence for any $b_0, \ldots, b_n$ the unique polynomial of degree $\le n$ with $g(c_j) = b_j$ is $\sum b_i f_i$. In particular a nonzero polynomial of degree $\le n$ has at most n zeros.

1.7 Maximal linearly independent subsets *

Motivation. Infinite-dimensional spaces such as $\mathsf{P}(\mathbb{F})$ or $C(\mathbb{R})$ also have bases, but finding one in general needs an axiom of set theory. The Maximal Principle (equivalent to the Axiom of Choice) produces a maximal independent set, and maximality forces it to generate.

Definition (Maximal element; chain)

A member M of a family $\mathcal{F}$ of sets is **maximal** if no member of $\mathcal{F}$ other than M contains M. A collection $\mathcal{C}$ of sets is a **chain** if for all $A, B \in \mathcal{C}$, $A \subseteq B$ or $B \subseteq A$.

Example 1

In the power set of S, S is the (unique) maximal element.

Example 2

If S, T are disjoint and nonempty, the union of their power sets has two maximal elements S and T.

Example 3

The finite subsets of an infinite set S have no maximal element: add an element not yet present.

Example 4

$A_n = \{1, \ldots, n\}$, $n \geq 1$, form a chain.

Maximal Principle. Let $\mathcal{F}$ be a family of sets. If for every chain $\mathcal{C} \subseteq \mathcal{F}$ some member of $\mathcal{F}$ contains every member of $\mathcal{C}$, then $\mathcal{F}$ has a maximal member. (It is equivalent to the Axiom of Choice and is assumed here.)

Definition (Maximal linearly independent subset)

$B \subseteq S$ is a **maximal linearly independent subset of** S if (a) B is independent and (b) the only independent subset of S containing B is B.

Example 5

By Example 2 of Section 1.4, $\{x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$ is a maximal linearly independent subset of $S = \{2x^3 - 2x^2 + 12x - 6, x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$; so is any two-element subset of S.

A basis β of V is a maximal independent subset of V: for $v \notin \beta$, $\beta \cup \{v\}$ is dependent by Theorem 1.7.

Theorem 1.12

If S generates V and β is a maximal linearly independent subset of S, then β is a basis for V.

Proof. It suffices to show $S \subseteq \text{span}(\beta)$ (Theorem 1.5). If $v \in S - \text{span}(\beta)$, then $\beta \cup \{v\}$ is independent by Theorem 1.7, contradicting maximality. ■

Theorem 1.13

If S is a linearly independent subset of V, some maximal linearly independent subset of V contains S.

Proof. Let $\mathcal{F}$ be the family of independent subsets of V containing S. For a chain $\mathcal{C} \subseteq \mathcal{F}$, let U be the union of its members (or $U = S$ if $\mathcal{C} = \emptyset$). Then $U \supseteq S$ and U contains every member of $\mathcal{C}$. U is independent: if $u_1, \ldots, u_n \in U$ are distinct with $\sum a_i u_i = 0$, say $u_i \in A_i \in \mathcal{C}$, then since $\mathcal{C}$ is a chain, one A_k contains all A_i. Since A_k is independent, all $a_i = 0$. So $U \in \mathcal{F}$, and the Maximal Principle gives a maximal member of $\mathcal{F}$. It is a maximal independent subset of V, since any independent set containing it also contains S. ■

Corollary

Every vector space has a basis.

Proof. Apply Theorem 1.13 to $S = \emptyset$, and Theorem 1.12 with $S = V$. ■

All bases of an infinite-dimensional space also have the same cardinality (cited).

Chapter 2. Linear Transformations and Matrices

2.1 Linear transformations, null spaces, and ranges

Motivation. Maps preserving addition and scalar multiplication are the morphisms of vector spaces. Their null space and range are subspaces, their dimensions add up to $\dim V$, and a linear map is pinned down by its values on a basis.

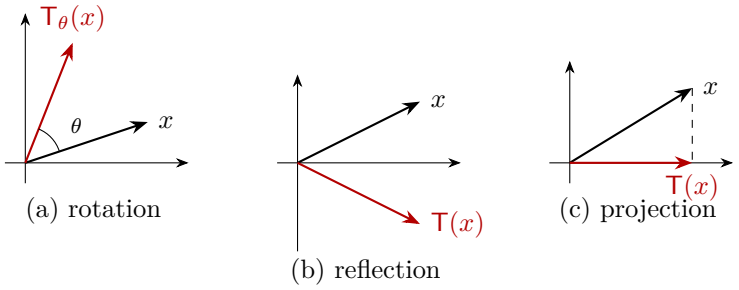
Definition (Linear transformation)

$\mathsf{T}: V \to W$ is **linear** if $\mathsf{T}(x+y) = \mathsf{T}(x) + \mathsf{T}(y)$ and $\mathsf{T}(cx) = c\mathsf{T}(x)$ for all $x, y \in V$, $c \in \mathbb{F}$.

For linear T: $\mathsf{T}(0) = 0$; $\mathsf{T}(cx+y) = c\mathsf{T}(x) + \mathsf{T}(y)$ (and this characterises linearity); $\mathsf{T}(x-y) = \mathsf{T}(x) - \mathsf{T}(y)$; $\mathsf{T}(\sum a_i x_i) = \sum a_i \mathsf{T}(x_i)$.

Example 1

$\mathsf{T}(a_1, a_2) = (2a_1 + a_2, a_1)$ is linear: with $c \in \mathbb{R}$, $x = (b_1, b_2)$, $y = (d_1, d_2)$, $\mathsf{T}(cx+y) = (2(cb_1 + d_1) + cb_2 + d_2,\ cb_1 + d_1) = c\mathsf{T}(x) + \mathsf{T}(y)$.



(a) rotation (b) reflection (c) projection

Figure. *Examples 2–4: rotation by θ, reflection about the x-axis, projection on the x-axis.*

Example 2

The rotation T_θ by angle θ is $\mathsf{T}_\theta(a_1, a_2) = (a_1 \cos\theta - a_2 \sin\theta,\ a_1 \sin\theta + a_2 \cos\theta)$ (write $(a_1, a_2) = r(\cos\alpha, \sin\alpha)$, then the image is $r(\cos(\alpha+\theta), \sin(\alpha+\theta))$). It is linear.

Example 3

The **reflection about the x-axis** $\mathsf{T}(a_1, a_2) = (a_1, -a_2)$ is linear.

Example 4

The **projection on the x-axis** $\mathsf{T}(a_1, a_2) = (a_1, 0)$ is linear.

Example 5

$\mathsf{T}: \mathrm{M}_{m\times n}(\mathbb{F}) \to \mathrm{M}_{n\times m}(\mathbb{F})$, $\mathsf{T}(A) = A^t$, is linear.

Example 6

Differentiation $\mathsf{T}(f) = f'$ from $\mathsf{P}_n(\mathbb{R})$ to $\mathsf{P}_{n-1}(\mathbb{R})$ is linear: $(ag+h)' = ag' + h'$.

Example 7

$\mathsf{T}(f) = \int_a^b f(t)\,dt$ from $C(\mathbb{R})$ to $\mathbb{R}$ is linear.

The **identity** $\mathsf{I}_V(x) = x$ and the **zero transformation** $\mathsf{T}_0(x) = 0$ are linear.

Definition (Null space, range)

For linear $\mathsf{T} : V \to W$, the **null space** (kernel) is $\mathrm{N}(\mathsf{T}) = \{x \in V : \mathsf{T}(x) = 0\}$ and the **range** (image) is $\mathrm{R}(\mathsf{T}) = \{\mathsf{T}(x) : x \in V\}$.

Example 8

$\mathrm{N}(\mathsf{I}) = \{0\}$, $\mathrm{R}(\mathsf{I}) = V$; $\mathrm{N}(\mathsf{T}_0) = V$, $\mathrm{R}(\mathsf{T}_0) = \{0\}$.

Example 9

For $\mathsf{T}(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ on $\mathbb{R}^3$: $\mathrm{N}(\mathsf{T}) = \{(a, a, 0) : a \in \mathbb{R}\}$ and $\mathrm{R}(\mathsf{T}) = \mathbb{R}^2$.

Theorem 2.1

For linear $\mathsf{T} : V \to W$, $\mathrm{N}(\mathsf{T})$ and $\mathrm{R}(\mathsf{T})$ are subspaces of V and W.

Proof. $\mathsf{T}(0) = 0$, so 0 lies in both. If $x, y \in \mathrm{N}(\mathsf{T})$, $c \in \mathbb{F}$, then $\mathsf{T}(x+y) = 0$ and $\mathsf{T}(cx) = 0$. If $\mathsf{T}(x), \mathsf{T}(y) \in \mathrm{R}(\mathsf{T})$, then $\mathsf{T}(x)+\mathsf{T}(y) = \mathsf{T}(x+y)$ and $c\mathsf{T}(x) = \mathsf{T}(cx)$ lie in $\mathrm{R}(\mathsf{T})$. Apply Theorem 1.3. ■

Theorem 2.2

If $\beta = \{v_1, \ldots, v_n\}$ is a basis for V and $\mathsf{T} : V \to W$ is linear, then $\mathrm{R}(\mathsf{T}) = \mathrm{span}(\mathsf{T}(\beta)) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\}$.

Proof. $\mathsf{T}(v_i) \in \mathrm{R}(\mathsf{T})$, a subspace, so $\mathrm{span}(\mathsf{T}(\beta)) \subseteq \mathrm{R}(\mathsf{T})$ by Theorem 1.5. Conversely $w = \mathsf{T}(v)$ with $v = \sum a_i v_i$ gives $w = \sum a_i \mathsf{T}(v_i)$. ■

(The same holds for an infinite basis.)

Example 10

For $\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathrm{M}_{2\times 2}(\mathbb{R})$, $\mathsf{T}(f) = \begin{pmatrix} f(1)-f(2) & 0 \\ 0 & f(0) \end{pmatrix}$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(1), \mathsf{T}(x), \mathsf{T}(x^2)\} = \mathrm{span}\left\{\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} -1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} -3 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right\}$, of dimension 2.

Definition (Nullity, rank)

If $\mathrm{N}(\mathsf{T})$, $\mathrm{R}(\mathsf{T})$ are finite-dimensional, $\mathrm{nullity}(\mathsf{T}) = \dim \mathrm{N}(\mathsf{T})$ and $\mathrm{rank}(\mathsf{T}) = \dim \mathrm{R}(\mathsf{T})$.

Theorem 2.3 (Dimension theorem)

If $\mathsf{T}: V \to W$ is linear and V is finite-dimensional, then $\text{nullity}(\mathsf{T}) + \text{rank}(\mathsf{T}) = \dim(V)$.

Proof. Let $\dim V = n$ and let $\{v_1, \dots, v_k\}$ be a basis of $\mathrm{N}(\mathsf{T})$; extend it to a basis $\beta = \{v_1, \dots, v_n\}$ of V (Corollary). We claim $S = \{\mathsf{T}(v_{k+1}), \dots, \mathsf{T}(v_n)\}$ is a basis of $\mathrm{R}(\mathsf{T})$ with $n-k$ distinct elements.

Step 1 (spans). By Theorem 2.2, $\mathrm{R}(\mathsf{T}) = \text{span}\{\mathsf{T}(v_1), \dots, \mathsf{T}(v_n)\} = \text{span}(S)$, as $\mathsf{T}(v_i) = 0$ for $i \le k$.

Step 2 (independent). If $\sum_{i>k} b_i \mathsf{T}(v_i) = 0$, then $\sum_{i>k} b_i v_i \in \mathrm{N}(\mathsf{T})$, so $\sum_{i>k} b_i v_i = \sum_{i \le k} c_i v_i$. Independence of β gives all $b_i = 0$ (and shows the $\mathsf{T}(v_i)$, $i > k$, are distinct). ∎

Theorem 2.4

Linear T is one-to-one iff $\mathrm{N}(\mathsf{T}) = \{0\}$.

Proof. If T is one-to-one and $\mathsf{T}(x) = 0 = \mathsf{T}(0)$, then $x = 0$. Conversely, if $\mathrm{N}(\mathsf{T}) = \{0\}$ and $\mathsf{T}(x) = \mathsf{T}(y)$, then $\mathsf{T}(x - y) = 0$, so $x = y$. ∎

Theorem 2.5

If $\dim V = \dim W < \infty$ and $\mathsf{T}: V \to W$ is linear, the following are equivalent: (a) T is one-to-one;
(b) T is onto;
(c) $\text{rank}(\mathsf{T}) = \dim(V)$.

Proof. By Theorems 2.3 and 2.4: T one-to-one $\iff$ $\mathrm{N}(\mathsf{T}) = \{0\}$ $\iff$ $\text{nullity}(\mathsf{T}) = 0$ $\iff$ $\text{rank}(\mathsf{T}) = \dim V = \dim W$ $\iff$ $\mathrm{R}(\mathsf{T}) = W$, the last step by Theorem 1.11. ∎

Both finite-dimensionality and linearity are essential here.

Example 11

$\mathsf{T}: \mathsf{P}_2(\mathbb{R}) \to \mathsf{P}_3(\mathbb{R})$, $\mathsf{T}(f) = 2f' + \int_0^x 3f(t)\,dt$: $\mathrm{R}(\mathsf{T}) = \text{span}\{3x,\ 2 + \frac{3}{2}x^2,\ 4x + x^3\}$ has rank $3 < 4$, so T is not onto; nullity $= 3 - 3 = 0$, so T is one-to-one.

Example 12

$\mathsf{T}(a_1, a_2) = (a_1 + a_2, a_1)$ on $\mathbb{F}^2$ has $\mathrm{N}(\mathsf{T}) = \{0\}$; by Theorem 2.5 it is also onto.

A one-to-one linear T maps S to an independent set iff S is independent.

Example 13

$\mathsf{T}(a_0 + a_1 x + a_2 x^2) = (a_0, a_1, a_2)$ is linear and one-to-one. $S = \{2 - x + 3x^2,\ x + x^2,\ 1 - 2x^2\}$ is independent since $\mathsf{T}(S) = \{(2, -1, 3), (0, 1, 1), (1, 0, -2)\}$ is.

Theorem 2.6

Let $\{v_1, \dots, v_n\}$ be a basis for V. For any $w_1, \dots, w_n \in W$ there is exactly one linear $\mathsf{T}: V \to W$ with $\mathsf{T}(v_i) = w_i$.

Proof. For $x = \sum a_i v_i$ (unique coefficients, Theorem 1.8) set $\mathsf{T}(x) = \sum a_i w_i$.

Step 1 (linear). If $y = \sum b_i v_i$, then $cx + y = \sum (ca_i + b_i) v_i$, so $\mathsf{T}(cx + y) = \sum (ca_i + b_i) w_i = c\mathsf{T}(x) + \mathsf{T}(y)$.

Step 2 (values). $\mathsf{T}(v_i) = w_i$.

Step 3 (unique). A linear U with $\mathsf{U}(v_i) = w_i$ satisfies $\mathsf{U}(x) = \sum a_i \mathsf{U}(v_i) = \sum a_i w_i = \mathsf{T}(x)$. ■

Corollary

If $\{v_1, \ldots, v_n\}$ is a basis for V and linear $\mathsf{U}, \mathsf{T} : V \to W$ agree on each v_i, then $\mathsf{U} = \mathsf{T}$.

Proof. Uniqueness in Theorem 2.6. ■

Example 14

If $\mathsf{U} : \mathbb{R}^2 \to \mathbb{R}^2$ is linear with $\mathsf{U}(1,2) = (3,3)$ and $\mathsf{U}(1,1) = (1,3)$, then $\mathsf{U} = \mathsf{T}$ for $\mathsf{T}(a_1, a_2) = (2a_2 - a_1, 3a_1)$, since T takes the same values on the basis $\{(1,2),(1,1)\}$.

2.2 The matrix representation of a linear transformation

Motivation. Once ordered bases are fixed, coordinates turn vectors into columns and linear maps into matrices: the j-th column of $[\mathsf{T}]_\beta^\gamma$ lists the coordinates of $\mathsf{T}(v_j)$. Sums and scalar multiples of maps correspond to those of matrices.

Definition (Ordered basis; coordinate vector)

An **ordered basis** is a basis with a fixed order. For $\beta = \{u_1, \ldots, u_n\}$ and $x = \sum a_i u_i$, the **coordinate vector** is $[x]_\beta = (a_1, \ldots, a_n)^t \in \mathbb{F}^n$. The **standard ordered bases** are $\{e_1, \ldots, e_n\}$ of $\mathbb{F}^n$ and $\{1, x, \ldots, x^n\}$ of $\mathsf{P}_n(\mathbb{F})$.

Example 1

$\beta = \{e_1, e_2, e_3\}$ and $\gamma = \{e_2, e_1, e_3\}$ are different ordered bases of $\mathbb{F}^3$.

Example 2

For $f = 4 + 6x - 7x^2$ and the standard β of $\mathsf{P}_2(\mathbb{R})$, $[f]_\beta = (4, 6, -7)^t$.

Definition (Matrix representation)

Let $\beta = \{v_1, \ldots, v_n\}$, $\gamma = \{w_1, \ldots, w_m\}$ be ordered bases of V, W and $\mathsf{T} : V \to W$ linear, with $\mathsf{T}(v_j) = \sum_{i=1}^m a_{ij} w_i$. The $m \times n$ matrix $A = (a_{ij}) = [\mathsf{T}]_\beta^\gamma$ is the **matrix representation** of T; its j-th column is $[\mathsf{T}(v_j)]_\gamma$. If $V = W$, $\beta = \gamma$, write $[\mathsf{T}]_\beta$.

By the Corollary to Theorem 2.6, $[\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma$ implies $\mathsf{U} = \mathsf{T}$.

Example 3

$\mathsf{T}(a_1, a_2) = (a_1 + 3a_2, 0, 2a_1 - 4a_2)$ in the standard bases: $\mathsf{T}(e_1) = (1,0,2)$, $\mathsf{T}(e_2) = (3,0,-4)$, so $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}$. With $\gamma' = \{e_3, e_2, e_1\}$ the rows are reversed.

Example 4

$\mathsf{T}(f) = f'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$, standard bases: $\mathsf{T}(1) = 0$, $\mathsf{T}(x) = 1$, $\mathsf{T}(x^2) = 2x$, $\mathsf{T}(x^3) = 3x^2$, so $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

Definition (Sum and scalar multiple of maps)

$(\mathsf{T} + \mathsf{U})(x) = \mathsf{T}(x) + \mathsf{U}(x)$ and $(a\mathsf{T})(x) = a\mathsf{T}(x)$.

Theorem 2.7

Let $\mathsf{T}, \mathsf{U} : V \to W$ be linear.
(a) $a\mathsf{T} + \mathsf{U}$ is linear for all $a \in \mathbb{F}$.
(b) With these operations, the linear maps $V \to W$ form a vector space over $\mathbb{F}$.

Proof.

(a) $(a\mathsf{T} + \mathsf{U})(cx + y) = a\mathsf{T}(cx + y) + \mathsf{U}(cx + y) = c(a\mathsf{T} + \mathsf{U})(x) + (a\mathsf{T} + \mathsf{U})(y)$.

(b) By (a) the linear maps form a subset of $\mathcal{F}(V, W)$ containing T_0 and closed under the operations; the axioms hold pointwise since they hold in W. ■

$\mathcal{L}(V, W)$ denotes this space, and $\mathcal{L}(V) = \mathcal{L}(V, V)$.

Theorem 2.8

For finite-dimensional V, W with ordered bases β, γ and linear $\mathsf{T}, \mathsf{U} : V \to W$:
(a) $[\mathsf{T} + \mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma + [\mathsf{U}]_\beta^\gamma$;
(b) $[a\mathsf{T}]_\beta^\gamma = a[\mathsf{T}]_\beta^\gamma$.

Proof. If $\mathsf{T}(v_j) = \sum_i a_{ij} w_i$ and $\mathsf{U}(v_j) = \sum_i b_{ij} w_i$, then $(\mathsf{T}+\mathsf{U})(v_j) = \sum_i (a_{ij}+b_{ij}) w_i$ and $(a\mathsf{T})(v_j) = \sum_i a a_{ij} w_i$. ■

Example 5

For $\mathsf{T}(a_1, a_2) = (a_1 + 3a_2, 0, 2a_1 - 4a_2)$ and $\mathsf{U}(a_1, a_2) = (a_1 - a_2, 2a_1, 3a_1 + 2a_2)$, $(\mathsf{T}+\mathsf{U})(a_1, a_2) = (2a_1 + 2a_2, 2a_1, 5a_1 - 2a_2)$, and in the standard bases its matrix $\begin{pmatrix} 2 & 2 \\ 2 & 0 \\ 5 & -2 \end{pmatrix}$ is the sum of $\begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 3 & 2 \end{pmatrix}$.

2.3 Composition of linear transformations and matrix multiplication

Motivation. Matrix multiplication is defined exactly so that composing maps corresponds to multiplying their matrices. Associativity of matrix products then comes for free from associativity of composition.

Theorem 2.9

If $\mathsf{T} : V \to W$ and $\mathsf{U} : W \to Z$ are linear, so is $\mathsf{UT} : V \to Z$.

Proof. $\mathsf{UT}(ax + y) = \mathsf{U}(a\mathsf{T}(x) + \mathsf{T}(y)) = a\mathsf{UT}(x) + \mathsf{UT}(y)$. ■

Theorem 2.10

For $\mathsf{T}, \mathsf{U}_1, \mathsf{U}_2 \in \mathcal{L}(V)$:
(a) $\mathsf{T}(\mathsf{U}_1 + \mathsf{U}_2) = \mathsf{T}\mathsf{U}_1 + \mathsf{T}\mathsf{U}_2$ and $(\mathsf{U}_1 + \mathsf{U}_2)\mathsf{T} = \mathsf{U}_1\mathsf{T} + \mathsf{U}_2\mathsf{T}$;
(b) $\mathsf{T}(\mathsf{U}_1\mathsf{U}_2) = (\mathsf{T}\mathsf{U}_1)\mathsf{U}_2$;
(c) $\mathsf{T}\mathsf{I} = \mathsf{I}\mathsf{T} = \mathsf{T}$;
(d) $a(\mathsf{U}_1\mathsf{U}_2) = (a\mathsf{U}_1)\mathsf{U}_2 = \mathsf{U}_1(a\mathsf{U}_2)$.

Proof [Supplement]. Evaluate at x. (a) $\mathsf{T}(\mathsf{U}_1 x + \mathsf{U}_2 x) = \mathsf{T}\mathsf{U}_1 x + \mathsf{T}\mathsf{U}_2 x$ by linearity of T; $(\mathsf{U}_1 + \mathsf{U}_2)(\mathsf{T}x) = \mathsf{U}_1\mathsf{T}x + \mathsf{U}_2\mathsf{T}x$ by definition.

(b) Composition of functions is associative.

(c) Clear.

(d) $a\mathsf{U}_1(\mathsf{U}_2 x) = (a\mathsf{U}_1)(\mathsf{U}_2 x) = \mathsf{U}_1(a\mathsf{U}_2 x)$ by linearity of U_1. ■

Let $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ with ordered bases $\alpha = \{v_j\}$, $\beta = \{w_k\}$, $\gamma = \{z_i\}$, $A = [\mathsf{U}]^\gamma_\beta$, $B = [\mathsf{T}]^\beta_\alpha$. Then

$$\mathsf{U}\mathsf{T}(v_j) = \mathsf{U}\Big(\sum_k B_{kj} w_k\Big) = \sum_k B_{kj} \sum_i A_{ik} z_i = \sum_i \Big(\sum_k A_{ik} B_{kj}\Big) z_i.$$

Definition (Matrix product)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $B \in \mathrm{M}_{n\times p}(\mathbb{F})$, $AB \in \mathrm{M}_{m\times p}(\mathbb{F})$ has $(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}$.

Example 1

$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 4 & -1 \end{pmatrix}\begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \end{pmatrix}$. Matrix multiplication is not commutative: $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ but $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$. Also $(AB)^t = B^t A^t$, since both have (i,j) entry $\sum_k A_{jk}B_{ki}$.

Theorem 2.11

With α, β, γ ordered bases of finite-dimensional V, W, Z and linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$: $[\mathsf{U}\mathsf{T}]^\gamma_\alpha = [\mathsf{U}]^\gamma_\beta [\mathsf{T}]^\beta_\alpha$.

Proof. The computation preceding the definition: the (i,j) entry of $[\mathsf{U}\mathsf{T}]^\gamma_\alpha$ is $\sum_k A_{ik}B_{kj}$. ■

Corollary

For $\mathsf{T}, \mathsf{U} \in \mathcal{L}(V)$ and an ordered basis β, $[\mathsf{U}\mathsf{T}]_\beta = [\mathsf{U}]_\beta [\mathsf{T}]_\beta$.

Proof. Theorem 2.11 with $\alpha = \beta = \gamma$. ■

Example 2

$\mathsf{U}(f) = f'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$ and $\mathsf{T}(f) = \int_0^x f(t)\,dt$ from $\mathsf{P}_2(\mathbb{R})$ to $\mathsf{P}_3(\mathbb{R})$ give $\mathsf{U}\mathsf{T} = \mathsf{I}$. Indeed, in standard bases

$$[\mathsf{U}][\mathsf{T}] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} = I_3.$$

Definition (Kronecker delta; identity matrix)

$\delta_{ij} = 1$ if $i = j$ and 0 otherwise; the $n \times n$ **identity matrix** is $(I_n)_{ij} = \delta_{ij}$.

Theorem 2.12

Let A be $m \times n$, B, C be $n \times p$, D, E be $q \times m$. Then
(a) $A(B + C) = AB + AC$ and $(D + E)A = DA + EA$;
(b) $a(AB) = (aA)B = A(aB)$;
(c) $I_m A = A = AI_n$;
(d) $[\mathsf{I}_V]_\beta = I_n$ for an n-dimensional V with ordered basis β.

Proof.

(a) $(A(B+C))_{ij} = \sum_k A_{ik}(B_{kj} + C_{kj}) = (AB)_{ij} + (AC)_{ij}$; the other identity is similar.

(b) [Supplement] All three have entries $a\sum_k A_{ik}B_{kj}$.

(c) $(I_m A)_{ij} = \sum_k \delta_{ik}A_{kj} = A_{ij}$, and similarly for AI_n.

(d) [Supplement] $\mathsf{I}_V(v_j) = v_j$, whose coordinate vector is e_j. ■

Corollary

$A(\sum_{i=1}^k a_i B_i) = \sum_i a_i AB_i$ and $(\sum_i a_i C_i)A = \sum_i a_i C_i A$.

Proof. Induction on k using Theorem 2.12(a), (b). ■

Powers: $A^1 = A$, $A^k = A^{k-1}A$, $A^0 = I$. Note $A^2 = O$ is possible with $A \neq O$, e.g. $A = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, so there is no cancellation law.

Theorem 2.13

Let A be $m \times n$, B be $n \times p$, and u_j, v_j the j-th columns of AB and B. Then (a) $u_j = Av_j$; (b) $v_j = Be_j$.

Proof.

(a) $(u_j)_i = (AB)_{ij} = \sum_k A_{ik}B_{kj} = (Av_j)_i$.

(b) Apply (a) with $A = B$, $B = I_p$: Be_j is the j-th column of $BI_p = B$. ■

Similarly the i-th row of AB is (row i of A)$\cdot B$.

Theorem 2.14

For linear $\mathsf{T} : V \to W$ with ordered bases β, γ and $u \in V$: $[\mathsf{T}(u)]_\gamma = [\mathsf{T}]_\beta^\gamma [u]_\beta$.

Proof. Let $f : \mathbb{F} \to V$, $f(a) = au$, and $g : \mathbb{F} \to W$, $g(a) = a\mathsf{T}(u)$, with the basis $\alpha = \{1\}$ of $\mathbb{F}$. Then $g = \mathsf{T}f$, and $[f]_\alpha^\beta = [u]_\beta$, $[g]_\alpha^\gamma = [\mathsf{T}(u)]_\gamma$. By Theorem 2.11, $[\mathsf{T}(u)]_\gamma = [\mathsf{T}f]_\alpha^\gamma = [\mathsf{T}]_\beta^\gamma [f]_\alpha^\beta = [\mathsf{T}]_\beta^\gamma [u]_\beta$. ■

Example 3

For $\mathsf{T}(f) = f'$ on $\mathsf{P}_3(\mathbb{R})$ and $p = 2 - 4x + x^2 + 3x^3$: $[\mathsf{T}(p)]_\gamma = [\mathsf{T}]_\beta^\gamma (2, -4, 1, 3)^t = (-4, 2, 9)^t$, matching $p' = -4 + 2x + 9x^2$.

Definition (Left-multiplication transformation)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $\mathsf{L}_A : \mathbb{F}^n \to \mathbb{F}^m$ is $\mathsf{L}_A(x) = Ax$.

Example 4

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, $\mathsf{L}_A(1,3,-1)^t = (6,1)^t$.

Theorem 2.15

L_A is linear. For $B \in \mathrm{M}_{m\times n}(\mathbb{F})$ and standard ordered bases β, γ of $\mathbb{F}^n, \mathbb{F}^m$:
(a) $[\mathsf{L}_A]_\beta^\gamma = A$;
(b) $\mathsf{L}_A = \mathsf{L}_B$ iff $A = B$;
(c) $\mathsf{L}_{A+B} = \mathsf{L}_A + \mathsf{L}_B$ and $\mathsf{L}_{aA} = a\mathsf{L}_A$;
(d) if $\mathsf{T} : \mathbb{F}^n \to \mathbb{F}^m$ is linear, there is a unique C with $\mathsf{T} = \mathsf{L}_C$, namely $C = [\mathsf{T}]_\beta^\gamma$;
(e) if E is $n \times p$, then $\mathsf{L}_{AE} = \mathsf{L}_A\mathsf{L}_E$;
(f) if $m = n$, then $\mathsf{L}_{I_n} = \mathsf{I}_{\mathbb{F}^n}$.

Proof. Linearity follows from Theorem 2.12.

(a) The j-th column of $[\mathsf{L}_A]_\beta^\gamma$ is $\mathsf{L}_A(e_j) = Ae_j$, the j-th column of A (Theorem 2.13).

(b) If $\mathsf{L}_A = \mathsf{L}_B$, then $A = [\mathsf{L}_A]_\beta^\gamma = [\mathsf{L}_B]_\beta^\gamma = B$ by (a).

(c) [Supplement] $(A+B)x = Ax + Bx$ and $(aA)x = a(Ax)$.

(d) Put $C = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.14, $\mathsf{T}(x) = [\mathsf{T}(x)]_\gamma = C[x]_\beta = Cx = \mathsf{L}_C(x)$. Uniqueness is (b).

(e) For each j, $(AE)e_j = A(Ee_j)$ by Theorem 2.13; so L_{AE} and $\mathsf{L}_A\mathsf{L}_E$ agree on a basis, and are equal by the Corollary to Theorem 2.6.

(f) [Supplement] $I_n x = x$. ■

Theorem 2.16

If $A(BC)$ is defined, so is $(AB)C$, and $A(BC) = (AB)C$.

Proof. By Theorem 2.15(e) and associativity of composition, $\mathsf{L}_{A(BC)} = \mathsf{L}_A\mathsf{L}_{BC} = \mathsf{L}_A(\mathsf{L}_B\mathsf{L}_C) = (\mathsf{L}_A\mathsf{L}_B)\mathsf{L}_C = \mathsf{L}_{(AB)C}$; apply (b). ■

Incidence matrices. For a relation on $\{1,\dots,n\}$ (e.g. "i can send a message to j"), the **incidence matrix** has $A_{ij} = 1$ if i relates to j and 0 otherwise. Then $(A^2)_{ij}$ counts the two-stage paths $i \to k \to j$, and $(A^k)_{ij}$ counts paths of k steps. In a **dominance relation** (for $i \neq j$ exactly one of $i \to j$, $j \to i$; $A_{ii} = 0$), the row sums of $A + A^2$ measure the power of each member.

2.4 Invertibility and isomorphisms

Motivation. An invertible linear map is an isomorphism: it identifies two spaces completely. Finite-dimensional spaces over $\mathbb{F}$ are classified by a single number, the dimension, and $\mathcal{L}(V,W) \cong \mathrm{M}_{m\times n}(\mathbb{F})$.

Definition (Inverse)

$\mathsf{U} : W \to V$ is an **inverse** of linear $\mathsf{T} : V \to W$ if $\mathsf{TU} = \mathsf{I}_W$ and $\mathsf{UT} = \mathsf{I}_V$; T is then **invertible**, the inverse is unique and denoted T^{-1}.

For invertible functions: $(\mathsf{TU})^{-1} = \mathsf{U}^{-1}\mathsf{T}^{-1}$; $(\mathsf{T}^{-1})^{-1} = \mathsf{T}$; and a function is invertible iff it is one-to-one and onto. So if $\dim V = \dim W < \infty$, linear T is invertible iff $\operatorname{rank}(\mathsf{T}) = \dim V$ (Theorem 2.5).

Example 1

$\mathsf{T}(a+bx) = (a, a+b)$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$ has $\mathsf{T}^{-1}(c,d) = c + (d-c)x$.

Theorem 2.17

If $\mathsf{T} : V \to W$ is linear and invertible, then T^{-1} is linear.

Proof. For $y_1, y_2 \in W$ let $x_i = \mathsf{T}^{-1}(y_i)$. Then $\mathsf{T}^{-1}(cy_1 + y_2) = \mathsf{T}^{-1}(c\mathsf{T}(x_1) + \mathsf{T}(x_2)) = \mathsf{T}^{-1}(\mathsf{T}(cx_1 + x_2)) = cx_1 + x_2 = c\mathsf{T}^{-1}(y_1) + \mathsf{T}^{-1}(y_2)$. ■

Definition (Invertible matrix)

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is **invertible** if $AB = BA = I$ for some B. Then B is unique ($C = CI = C(AB) = (CA)B = B$), the **inverse** A^{-1}.

Example 2

$\left(\begin{smallmatrix} 5 & 7 \\ 2 & 3 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 3 & -7 \\ -2 & 5 \end{smallmatrix}\right)$.

Lemma

Let $\mathsf{T} : V \to W$ be an invertible linear map. Then V is finite-dimensional iff W is, and then $\dim V = \dim W$.

Proof. If V has a basis $\beta = \{x_1, \ldots, x_n\}$, then $\operatorname{span}\mathsf{T}(\beta) = \mathrm{R}(\mathsf{T}) = W$ (Theorem 2.2), so W is finite-dimensional; the converse uses T^{-1}. Then $\operatorname{nullity}\mathsf{T} = 0$ and $\operatorname{rank}\mathsf{T} = \dim W$, so $\dim V = \dim W$ by Theorem 2.3. ■

Theorem 2.18

Let V, W be finite-dimensional with ordered bases β, γ and $\mathsf{T} : V \to W$ linear. Then T is invertible iff $[\mathsf{T}]_\beta^\gamma$ is invertible, and then $[\mathsf{T}^{-1}]_\gamma^\beta = ([\mathsf{T}]_\beta^\gamma)^{-1}$.

Proof.

Step 1 (only if). By the Lemma $\dim V = \dim W = n$, so $[\mathsf{T}]_\beta^\gamma$ is $n \times n$. By Theorems 2.11 and 2.12(d), $I_n = [\mathsf{I}_V]_\beta = [\mathsf{T}^{-1}\mathsf{T}]_\beta = [\mathsf{T}^{-1}]_\gamma^\beta [\mathsf{T}]_\beta^\gamma$, and likewise $[\mathsf{T}]_\beta^\gamma [\mathsf{T}^{-1}]_\gamma^\beta = I_n$.

Step 2 (if). Let $A = [\mathsf{T}]_\beta^\gamma$ have inverse B; then $n = \dim V = \dim W$ (A square). By Theorem 2.6 there is a linear $\mathsf{U} : W \to V$ with $\mathsf{U}(w_j) = \sum_i B_{ij} v_i$, i.e. $[\mathsf{U}]_\gamma^\beta = B$. Then $[\mathsf{UT}]_\beta = BA = I_n = [\mathsf{I}_V]_\beta$, so $\mathsf{UT} = \mathsf{I}_V$, and likewise $\mathsf{TU} = \mathsf{I}_W$. ■

Example 3

For T of Example 1 in standard bases, $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $[\mathsf{T}^{-1}] = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$, which are inverse to each other.

Corollary 1

For $\mathsf{T} \in \mathcal{L}(V)$, V finite-dimensional with ordered basis β: T is invertible iff $[\mathsf{T}]_\beta$ is, and $[\mathsf{T}^{-1}]_\beta = ([\mathsf{T}]_\beta)^{-1}$.

Proof. Theorem 2.18 with $W = V$, $\gamma = \beta$. ■

Corollary 2

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is invertible iff L_A is, and $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$.

Proof. Theorem 2.18 with standard bases and Theorem 2.15(a), (d): $[(\mathsf{L}_A)^{-1}] = A^{-1}$, so $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$. ■

Definition (Isomorphism)

V is **isomorphic** to W ($V \cong W$) if there is an invertible linear $\mathsf{T} : V \to W$, an **isomorphism**. "Is isomorphic to" is an equivalence relation.

Example 4

$\mathsf{T}(a_1, a_2) = a_1 + a_2 x$ is an isomorphism $\mathbb{F}^2 \to \mathsf{P}_1(\mathbb{F})$.

Example 5

$\mathsf{T}(f) = \begin{pmatrix} f(1) & f(2) \\ f(3) & f(4) \end{pmatrix}$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathrm{M}_{2\times 2}(\mathbb{R})$ is linear, and $\mathsf{T}(f) = O$ only for $f = 0$ (a cubic with four zeros is zero; Lagrange). Since the dimensions agree (4), T is an isomorphism by Theorem 2.5.

Theorem 2.19

Finite-dimensional V, W over $\mathbb{F}$ are isomorphic iff $\dim V = \dim W$.

Proof. "Only if" is the Lemma. If $\dim V = \dim W = n$ with bases $\{v_i\}$, $\{w_i\}$, Theorem 2.6 gives linear T with $\mathsf{T}(v_i) = w_i$. Then $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{w_i\} = W$, so T is onto, hence one-to-one by Theorem 2.5: an isomorphism. ■

Corollary

$V \cong \mathbb{F}^n$ iff $\dim V = n$.

Proof. $\dim \mathbb{F}^n = n$; apply Theorem 2.19 (if $V \cong \mathbb{F}^n$, V is finite-dimensional by the Lemma). ■

Theorem 2.20

Let $\dim V = n$, $\dim W = m$ with ordered bases β, γ. Then $\Phi : \mathcal{L}(V, W) \to \mathrm{M}_{m\times n}(\mathbb{F})$, $\Phi(\mathsf{T}) = [\mathsf{T}]_\beta^\gamma$, is an isomorphism.

Proof. Φ is linear by Theorem 2.8. Given A, Theorem 2.6 gives a unique linear T with $\mathsf{T}(v_j) = \sum_i A_{ij} w_i$, i.e. a unique T with $\Phi(\mathsf{T}) = A$. So Φ is bijective. ■

Corollary

$\dim \mathcal{L}(V,W) = mn$.

Proof. Theorems 2.20 and 2.19. ■

Definition (Standard representation)

For an ordered basis β of n-dimensional V, $\phi_\beta : V \to \mathbb{F}^n$, $\phi_\beta(x) = [x]_\beta$, is the **standard representation** of V with respect to β.

Example 6

For $\beta = \{(1,0),(0,1)\}$, $\gamma = \{(1,2),(3,4)\}$ and $x = (1,-2)$: $\phi_\beta(x) = (1,-2)^t$ and $\phi_\gamma(x) = (-5,2)^t$, since $(1,-2) = -5(1,2) + 2(3,4)$.

Theorem 2.21

ϕ_β is an isomorphism.

Proof [Supplement]. ϕ_β is linear (coordinates of $cx+y$ are $c[x]_\beta + [y]_\beta$ by uniqueness in Theorem 1.8) and $\phi_\beta(x) = 0$ only for $x = 0$. As $\dim V = \dim \mathbb{F}^n$, Theorem 2.5 applies. ■

$$\begin{array}{ccc} V & \xrightarrow{\mathsf{T}} & W \\ {\scriptstyle\phi_\beta}\downarrow & & \downarrow{\scriptstyle\phi_\gamma} \\ \mathbb{F}^n & \xrightarrow[\mathsf{L}_A]{} & \mathbb{F}^m \end{array}$$

Figure. *With $A = [\mathsf{T}]_\beta^\gamma$, the square commutes: $\mathsf{L}_A\phi_\beta = \phi_\gamma\mathsf{T}$ (Theorem 2.14). Computations in V, W become matrix computations.*

Example 7

For $\mathsf{T}(p) = p'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$ and $p = 2 + x - 3x^2 + 5x^3$: $\mathsf{L}_A\phi_\beta(p) = A(2,1,-3,5)^t = (1,-6,15)^t = \phi_\gamma(p')$.

2.5 The change of coordinate matrix

Motivation. Changing the basis changes coordinates by an invertible matrix Q and the matrix of an operator by conjugation $Q^{-1}[\mathsf{T}]Q$. Choosing a clever basis can make $[\mathsf{T}]$ simple; this is the theme of Chapter 5.

Theorem 2.22

Let β, β' be ordered bases of finite-dimensional V and $Q = [\mathsf{I}_V]_{\beta'}^{\beta}$. Then (a) Q is invertible; (b) $[v]_\beta = Q[v]_{\beta'}$ for all $v \in V$.

Proof.

(a) I_V is invertible, so Q is by Theorem 2.18.

(b) $[v]_\beta = [\mathsf{I}_V(v)]_\beta = [\mathsf{I}_V]^\beta_{\beta'}[v]_{\beta'}$ by Theorem 2.14. ■

Q is the **change of coordinate matrix** (from β'- to β-coordinates); its j-th column is $[x'_j]_\beta$, and Q^{-1} changes β- into β'-coordinates.

Example 1

In $\mathbb{R}^2$ with $\beta = \{(1,1),(1,-1)\}$, $\beta' = \{(2,4),(3,1)\}$: $(2,4) = 3(1,1) - (1,-1)$ and $(3,1) = 2(1,1) + (1,-1)$, so $Q = \left(\begin{smallmatrix} 3 & 2 \\ -1 & 1 \end{smallmatrix}\right)$.

A linear map $\mathsf{T} : V \to V$ is a **linear operator**.

Theorem 2.23

Let T be a linear operator on finite-dimensional V with ordered bases β, β', and $Q = [\mathsf{I}_V]^\beta_{\beta'}$. Then $[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q$.

Proof. $\mathsf{T} = \mathsf{IT} = \mathsf{TI}$, so by Theorem 2.11, $Q[\mathsf{T}]_{\beta'} = [\mathsf{I}]^\beta_{\beta'}[\mathsf{T}]^{\beta'}_{\beta'} = [\mathsf{T}]^\beta_{\beta'} = [\mathsf{T}]^\beta_\beta[\mathsf{I}]^\beta_{\beta'} = [\mathsf{T}]_\beta Q$. Multiply by Q^{-1}. ■

Example 2

For $\mathsf{T}(a,b) = (3a - b, a + 3b)$ and β, β' of Example 1: $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 3 \end{smallmatrix}\right)$, $Q^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 1 & -2 \\ 1 & 3 \end{smallmatrix}\right)$, and

$$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q = \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix}.$$

Check: $\mathsf{T}(2,4) = (2,14) = 4(2,4) - 2(3,1)$.

Example 3

The reflection T about the line $y = 2x$: with $\beta' = \{(1,2),(-2,1)\}$, $\mathsf{T}(1,2) = (1,2)$ and $\mathsf{T}(-2,1) = -(-2,1)$, so $[\mathsf{T}]_{\beta'} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. With β standard, $Q = \left(\begin{smallmatrix} 1 & -2 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$[\mathsf{T}]_\beta = Q[\mathsf{T}]_{\beta'}Q^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \mathsf{T}(a,b) = \tfrac{1}{5}(-3a + 4b,\ 4a + 3b).$$

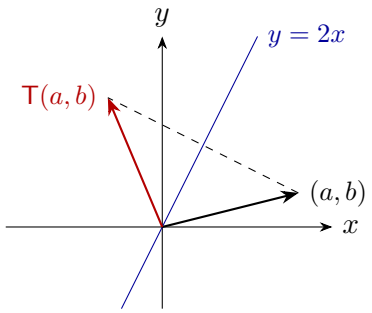


Figure. *Example 3: reflection about $y = 2x$.*

Corollary

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ and an ordered basis γ of $\mathbb{F}^n$, $[\mathsf{L}_A]_\gamma = Q^{-1}AQ$, where the j-th column of Q is the j-th vector of γ.

Proof. Theorem 2.23 with β standard: $[\mathsf{L}_A]_\beta = A$ and $Q = [\mathsf{I}]_\gamma^\beta$ has columns the vectors of γ. ■

Example 4

For $\gamma = \{(-1,0,0),(2,1,0),(1,1,1)\}$,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad [\mathsf{L}_A]_\gamma = Q^{-1}AQ = \begin{pmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{pmatrix}.$$

Definition (Similar matrices)

B is **similar** to A if $B = Q^{-1}AQ$ for some invertible Q; similarity is an equivalence relation. Theorem 2.23: matrices of one operator in two bases are similar.

2.6 Dual spaces *

Motivation. Linear functionals, the linear maps $V \to \mathbb{F}$, form the dual space V^*, of the same dimension as V. Coordinate functions give the dual basis; transposes of matrices are matrices of dual maps; and V is naturally isomorphic to V^{**}.

A **linear functional** on V is a linear map $V \to \mathbb{F}$.

Example 1

For continuous g on $[0,2\pi]$, $h(x) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$ is a linear functional; for $g(t) = \sin nt$ or $\cos nt$ it gives Fourier coefficients.

Example 2

$\mathrm{tr} : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is a linear functional.

Example 3

For an ordered basis $\beta = \{x_1,\ldots,x_n\}$, the **$i$-th coordinate function** $f_i(x) = a_i$, where $[x]_\beta = (a_1,\ldots,a_n)^t$, is a linear functional, and $f_i(x_j) = \delta_{ij}$.

Definition (Dual space)

$V^* = \mathcal{L}(V,\mathbb{F})$. If $\dim V < \infty$, $\dim V^* = \dim V \cdot \dim \mathbb{F} = \dim V$ (Corollary to Theorem 2.20), so $V \cong V^*$. The **double dual** is $V^{**} = (V^*)^*$.

Theorem 2.24

For an ordered basis $\beta = \{x_1, \dots, x_n\}$ with coordinate functions f_i, $\beta^* = \{f_1, \dots, f_n\}$ is an ordered basis of V^*, and $f = \sum_{i=1}^n f(x_i) f_i$ for every $f \in V^*$.

Proof. $\dim V^* = n$, so it suffices that β^* generates V^* (Corollary 2 of Theorem 1.10). For $g = \sum f(x_i) f_i$, $g(x_j) = \sum_i f(x_i)\delta_{ij} = f(x_j)$, so $g = f$ by the Corollary to Theorem 2.6. ■

β^*, characterised by $f_i(x_j) = \delta_{ij}$, is the **dual basis** of β.

Example 4

For $\beta = \{(2,1),(3,1)\}$: solving $f_1(2,1) = 1$, $f_1(3,1) = 0$, $f_2(2,1) = 0$, $f_2(3,1) = 1$ gives $f_1(x,y) = -x + 3y$ and $f_2(x,y) = x - 2y$.

Theorem 2.25

Let V, W be finite-dimensional with ordered bases β, γ. For linear $\mathsf{T} : V \to W$, the map $\mathsf{T}^t : W^* \to V^*$, $\mathsf{T}^t(g) = g\mathsf{T}$, is linear and $[\mathsf{T}^t]_{\gamma^*}^{\beta^*} = ([\mathsf{T}]_\beta^\gamma)^t$.

Proof. $g\mathsf{T}$ is linear (Theorem 2.9), and $\mathsf{T}^t(ag + h) = (ag + h)\mathsf{T} = a\, g\mathsf{T} + h\mathsf{T}$. Let $\beta = \{x_s\}$, $\gamma = \{y_k\}$, $\beta^* = \{f_s\}$, $\gamma^* = \{g_j\}$, $A = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.24, $\mathsf{T}^t(g_j) = g_j\mathsf{T} = \sum_s (g_j\mathsf{T})(x_s) f_s$, so the (i,j) entry of $[\mathsf{T}^t]_{\gamma^*}^{\beta^*}$ is

$$(g_j\mathsf{T})(x_i) = g_j\Big(\sum_k A_{ki} y_k\Big) = A_{ji}.$$

■

Example 5

$\mathsf{T}(p) = (p(0), p(2))$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$, standard bases: $[\mathsf{T}]_\beta^\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 2 \end{smallmatrix}\right)$. Directly, $\mathsf{T}^t(g_1) = g_1\mathsf{T}$ sends $p \mapsto p(0)$, i.e. f_1, and $\mathsf{T}^t(g_2)$ sends $a + bx \mapsto a + 2b$, i.e. $f_1 + 2f_2$. So $[\mathsf{T}^t] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right) = ([\mathsf{T}]_\beta^\gamma)^t$.

For $x \in V$ define $\hat{x} \in V^{**}$ by $\hat{x}(f) = f(x)$; $\hat{x}$ is linear in f.

Lemma

If $\dim V < \infty$, $x \in V$ and $\hat{x}(f) = 0$ for all $f \in V^*$, then $x = 0$.

Proof. If $x \neq 0$, extend $\{x\}$ to an ordered basis $\beta = \{x_1 = x, \dots, x_n\}$; the dual basis gives f_1 with $f_1(x) = 1 \neq 0$. ■

Theorem 2.26

For finite-dimensional V, $\psi : V \to V^{**}$, $\psi(x) = \hat{x}$, is an isomorphism.

Proof. ψ is linear: $\widehat{cx + y}(f) = f(cx + y) = c\hat{x}(f) + \hat{y}(f)$. By the Lemma it is one-to-one, and $\dim V^{**} = \dim V$, so it is an isomorphism by Theorem 2.5. ■

(ψ uses no basis: it is a *natural* isomorphism.)

Corollary

If V is finite-dimensional, every ordered basis of V^* is the dual basis of some basis of V.

Proof. Let $\{f_1, \dots, f_n\}$ be a basis of V^*, with dual basis $\{\hat{x}_1, \dots, \hat{x}_n\}$ in V^{**} (Theorems 2.24, 2.26). Then $\delta_{ij} = \hat{x}_i(f_j) = f_j(x_i)$, so $\{f_j\}$ is the dual basis of $\{x_1, \dots, x_n\}$, which is a basis because ψ is an isomorphism. ■

2.7 Homogeneous linear differential equations with constant coefficients *

Motivation. The solutions of $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y = 0$ form the null space of a differential operator $p(D)$. Linear algebra shows this space has dimension exactly n, and factoring the auxiliary polynomial $p(t)$ produces an explicit basis of exponentials.

A spring with constant k and mass m satisfies $y'' + \frac{k}{m}y = 0$.

Example 1

$y(t) = \sin\sqrt{k/m}\,t$ solves $y'' + \frac{k}{m}y = 0$, while $y(t) = t$ does not.

A **homogeneous linear differential equation with constant coefficients** is

$$a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \qquad a_i \in \mathbb{C},\ a_n \neq 0. \tag{2.1}$$

Dividing by a_n we assume $a_n = 1$. Solutions are sought among complex-valued functions of a real variable.

Definition (Derivative of complex functions)

$x = x_1 + ix_2 \in \mathcal{F}(\mathbb{R}, \mathbb{C})$ is **differentiable** if x_1, x_2 are, with $x' = x_1' + ix_2'$.

Example 2

$x(t) = \cos 2t + i \sin 2t$ has $x'(t) = -2\sin 2t + 2i\cos 2t$; and $x^2 = \cos 4t + i \sin 4t$.

Theorem 2.27

Every solution of (2.1) has derivatives of all orders.

Proof. If x is a solution, $x^{(n)} = -\sum_{i<n} a_i x^{(i)}$ is differentiable (each $x^{(i)}$, $i < n$, is). Inductively, if $x^{(k)}$ exists for all $k \leq m$ with $m \geq n$, then $x^{(m)} = -\sum_{i<n} a_i x^{(i+m-n)}$ is differentiable. ■

Example 3

For $y'' + 4y = 0$, $y'' = -4y$, so $y''' = -4y'$, and so on.

Definition (C^∞; differential operators)

C^∞ is the subspace of $\mathcal{F}(\mathbb{R}, \mathbb{C})$ of functions with derivatives of all orders. $D : C^\infty \to C^\infty$, $D(x) = x'$, is linear. For a complex polynomial $p(t) = \sum a_i t^i$ of positive degree, $p(D) = \sum a_i D^i$ is a **differential operator** of order $\deg p$. For (2.1) (with $a_n = 1$), $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ is the **auxiliary polynomial**, and (2.1) reads $p(D)(y) = 0$.

Theorem 2.28

The solutions of (2.1) are exactly $\mathrm{N}(p(D))$.

Proof. By Theorem 2.27 all solutions lie in C^∞, where the equation is $p(D)(y) = 0$. ■

Corollary

The solution set of (2.1) is a subspace of C^∞, the **solution space**.

Proof. Theorems 2.28 and 2.1. ■

Definition (Complex exponential)

For $c = a+ib$, $e^c = e^a(\cos b + i \sin b)$ (**Euler's formula**: $e^{ib} = \cos b + i \sin b$). Then $e^{c+d} = e^c e^d$ and $e^{-c} = 1/e^c$. $f(t) = e^{ct}$ is an **exponential function**.

Theorem 2.29

$\frac{d}{dt} e^{ct} = ce^{ct}$.

Proof [Supplement]. With $c = a + ib$, $e^{ct} = e^{at}\cos bt + ie^{at}\sin bt$, whose derivative is $e^{at}(a\cos bt - b\sin bt) + ie^{at}(a\sin bt + b\cos bt) = (a+ib)e^{at}(\cos bt + i\sin bt)$. ■

Theorem 2.30

The solution space of $y' + a_0 y = 0$ has dimension 1 and basis $\{e^{-a_0 t}\}$.

Proof. $e^{-a_0 t}$ is a solution by Theorem 2.29. If x is a solution, then $(e^{a_0 t}x)' = e^{a_0 t}(a_0 x + x') = 0$. A complex function with zero derivative is constant (apply the real fact to its real and imaginary parts). So $x = ke^{-a_0 t}$. ■

Corollary

For $c \in \mathbb{C}$, $\mathrm{N}(D - c\mathsf{I})$ has basis $\{e^{ct}\}$.

Proof. Theorem 2.30 with $a_0 = -c$. ■

Theorem 2.31

If c is a zero of the auxiliary polynomial $p(t)$, then e^{ct} is a solution.

Proof. Write $p(t) = q(t)(t-c)$. Polynomials in D commute, so $p(D) = q(D)(D - c\mathsf{I})$ and $p(D)(e^{ct}) = q(D)((D - c\mathsf{I})e^{ct}) = q(D)(0) = 0$ by the Corollary. ■

Example 4

$y'' - 3y' + 2y = 0$: $p(t) = (t-1)(t-2)$, so e^t and e^{2t} are solutions.

Theorem 2.32

For any differential operator $p(D)$ of order n, $\mathrm{N}(p(D))$ is an n-dimensional subspace of C^∞.

Lemma 1

$D - c\mathsf{I} : C^\infty \to C^\infty$ is onto for every $c \in \mathbb{C}$.

Proof. Let $v \in C^\infty$ and $w = ve^{-ct}$. Its real and imaginary parts have antiderivatives, giving $W \in C^\infty$ with $W' = w$. Put $u = We^{ct}$. Then $(D - c\mathsf{I})u = W'e^{ct} + cWe^{ct} - cWe^{ct} = v$. ■

Lemma 2

Let T, U be linear operators on V with U onto and $\mathrm{N}(\mathsf{T})$, $\mathrm{N}(\mathsf{U})$ finite-dimensional. Then $\mathrm{N}(\mathsf{TU})$ is finite-dimensional and $\dim \mathrm{N}(\mathsf{TU}) = \dim \mathrm{N}(\mathsf{T}) + \dim \mathrm{N}(\mathsf{U})$.

Proof. Let $\{u_1, \dots, u_p\}$ be a basis of $\mathrm{N}(\mathsf{T})$ and $\{w_1, \dots, w_q\}$ one of $\mathrm{N}(\mathsf{U})$. As U is onto, choose v_i with $\mathsf{U}(v_i) = u_i$. We show $\beta = \{v_1, \dots, v_p, w_1, \dots, w_q\}$ is a basis of $\mathrm{N}(\mathsf{TU})$ with $p + q$ distinct vectors.

Step 1 (inside). $\mathsf{TU}(v_i) = \mathsf{T}(u_i) = 0$ and $\mathsf{TU}(w_j) = \mathsf{T}(0) = 0$.

Step 2 (spans). If $\mathsf{TU}(v) = 0$, then $\mathsf{U}(v) = \sum a_i u_i = \mathsf{U}(\sum a_i v_i)$, so $v - \sum a_i v_i \in \mathrm{N}(\mathsf{U})$ equals $\sum b_j w_j$.

Step 3 (independent). If $\sum a_i v_i + \sum b_j w_j = 0$, applying U gives $\sum a_i u_i = 0$, so all $a_i = 0$, then all $b_j = 0$. (Distinctness follows similarly.) ■

Proof of Theorem 2.32. Induction on n. For $n = 1$, $p(D)$ is a nonzero multiple of $D - c\mathsf{I}$, and Theorem 2.30 applies. For $n > 1$, factor $p(t) = q(t)(t - c)$ over $\mathbb{C}$ with $\deg q = n - 1$, so $p(D) = q(D)(D - c\mathsf{I})$. By Lemma 1, $D - c\mathsf{I}$ is onto; by induction and Theorem 2.30, $\dim \mathrm{N}(q(D)) = n - 1$ and $\dim \mathrm{N}(D - c\mathsf{I}) = 1$. Lemma 2 gives $\dim \mathrm{N}(p(D)) = n$. ■

Corollary

The solution space of any nth-order equation (2.1) is an n-dimensional subspace of C^∞.

Proof. Theorems 2.28 and 2.32. ■

Theorem 2.33

For distinct $c_1, \dots, c_n \in \mathbb{C}$, $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is linearly independent.

Proof. Induction on n; $n = 1$ is clear. If $\sum_i b_i e^{c_i t} = 0$, apply $D - c_n\mathsf{I}$: $\sum_{i<n} b_i(c_i - c_n)e^{c_i t} = 0$. By induction $b_i(c_i - c_n) = 0$, so $b_i = 0$ for $i < n$, and then $b_n e^{c_n t} = 0$ gives $b_n = 0$. ■

Corollary

If the auxiliary polynomial of an nth-order equation (2.1) has n distinct zeros $c_1, \dots, c_n$, then $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is a basis of the solution space.

Proof. They are n independent solutions (Theorems 2.31, 2.33) in an n-dimensional space. ■

Example 5

$y'' + 5y' + 4y = 0$: $p(t) = (t + 1)(t + 4)$, so the solutions are $b_1 e^{-t} + b_2 e^{-4t}$.

Example 6

$y'' + 9y = 0$: $p(t) = (t - 3i)(t + 3i)$, basis $\{e^{3it}, e^{-3it}\}$. Since $\cos 3t = \frac{1}{2}(e^{3it} + e^{-3it})$ and $\sin 3t = \frac{1}{2i}(e^{3it} - e^{-3it})$, $\{\cos 3t, \sin 3t\}$ is also a basis.

Lemma

If the auxiliary polynomial is $(t-c)^n$, then $\beta = \{e^{ct}, te^{ct}, \ldots, t^{n-1}e^{ct}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). $(D - c\mathsf{I})(t^k e^{ct}) = kt^{k-1}e^{ct}$, so $(D - c\mathsf{I})^n(t^k e^{ct}) = 0$ for $k < n$.

Step 2 (independent). If $\sum_k b_k t^k e^{ct} = 0$, multiply by e^{-ct}: the polynomial $\sum b_k t^k$ vanishes for all real t, so all $b_k = 0$. Since the space has dimension n (Theorem 2.32), β is a basis. ■

Example 7

$y^{(4)} - 4y^{(3)} + 6y^{(2)} - 4y' + y = 0$: $p(t) = (t-1)^4$, so the solutions are $(b_1 + b_2 t + b_3 t^2 + b_4 t^3)e^t$.

Theorem 2.34

If the auxiliary polynomial is $(t - c_1)^{n_1} \cdots (t - c_k)^{n_k}$ with distinct c_i, then $\{e^{c_1 t}, te^{c_1 t}, \ldots, t^{n_1-1}e^{c_1 t}, \ldots, e^{c_k t}, \ldots, t^{n_k-1}e^{c_k t}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). Factors of $p(D)$ commute, so each $t^j e^{c_i t}$ $(j < n_i)$, killed by $(D - c_i\mathsf{I})^{n_i}$ (Lemma), is killed by $p(D)$.

Step 2 (independent) [Supplement]. There are $n = \sum n_i$ functions, so it suffices to show independence. Suppose $\sum_i q_i(t)e^{c_i t} = 0$ with polynomials $\deg q_i < n_i$; we show all $q_i = 0$ by induction on k. For $a \neq 0$, $(D - c\mathsf{I})(q(t)e^{(c+a)t}) = (q' + aq)e^{(c+a)t}$, and $q \mapsto q' + aq$ preserves degree, hence is injective on polynomials. Apply $(D - c_k\mathsf{I})^{n_k}$: the k-th term vanishes (Step 1), and the others become $\tilde{q}_i e^{c_i t}$ with $\deg \tilde{q}_i = \deg q_i$. By induction all $\tilde{q}_i = 0$, so $q_i = 0$ for $i < k$. Then $q_k e^{c_k t} = 0$, so $q_k = 0$. ■

Example 8

$y''' - 4y'' + 5y' - 2y = 0$: $p(t) = (t-1)^2(t-2)$, so the solutions are $b_1 e^t + b_2 te^t + b_3 e^{2t}$.

Chapter 3. Elementary Matrix Operations and Systems of Linear Equations

3.1 Elementary matrix operations and elementary matrices

Motivation. Three simple row (or column) operations underlie Gaussian elimination. Each is multiplication by an invertible elementary matrix, so they can be analysed with the matrix algebra of Chapter 2.

Definition (Elementary operations)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, an **elementary row [column] operation** is (1) interchanging two rows [columns]; (2) multiplying a row [column] by a nonzero scalar; (3) adding a scalar multiple of a row [column] to another row [column]. These are of **type** 1, 2, 3.

Example 1

For $A = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$: interchanging rows $1, 2$ gives $B = \left(\begin{smallmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$; multiplying column 2 by 3 gives C; adding $4\times$row 3 to row 1 gives $M = \left(\begin{smallmatrix} 17 & 2 & 7 & 12 \\ 2 & 1 & -1 & 3 \\ 4 & 0 & 1 & 2 \end{smallmatrix}\right)$.

Definition (Elementary matrix)

An $n \times n$ **elementary matrix** is obtained from I_n by one elementary operation; its type is that of the operation.

Theorem 3.1

If B is obtained from $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ by an elementary row [column] operation, then $B = EA$ $[B = AE]$, where E is the $m \times m$ $[n \times n]$ elementary matrix obtained from I_m $[I_n]$ by the same operation. Conversely, if E is elementary, EA $[AE]$ is obtained from A by the corresponding row [column] operation.

Proof [Supplement]. By Theorem 2.13 (in its row form), the i-th row of EA is (row i of E)$A = \sum_k E_{ik}$(row k of A). Row i of E is e_i^t unless the operation changed it. For type 1 (swap rows p, q), row p of E is e_q^t, so row p of EA is row q of A, and vice versa. For type 2 (row p times c), row p of EA is c times row p of A. For type 3 (add c·row q to row p), row p of E is $e_p^t + ce_q^t$, so row p of EA is row p plus c times row q of A. All other rows are unchanged. Column operations follow by transposing: $(AE)^t = E^tA^t$, and E^t is the elementary matrix of the transposed operation. ■

Example 2

In Example 1, $B = EA$ with $E = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, and $C = AE'$ with $E' = \mathrm{diag}(1, 3, 1, 1)$.

Theorem 3.2

Elementary matrices are invertible, and the inverse of an elementary matrix is elementary of the same type.

Proof. An elementary E comes from I_n by an operation, which can be undone by an operation $\bar{E}$ of the same type: swap the same rows again; multiply by c^{-1}; add $-c$ times the row. By Theorem 3.1, $\bar{E}E = I_n$, so E is invertible with $E^{-1} = \bar{E}$ (a square matrix with a one-sided inverse is invertible, since $\mathsf{L}_{\bar{E}}\mathsf{L}_E = \mathsf{I}$ forces L_E to be one-to-one, hence invertible by Theorem 2.5). ■

3.2 The rank of a matrix and matrix inverses

Motivation. The rank of a matrix is the dimension of its column space. Elementary operations preserve it, reduce every matrix to $\begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$, and show that row rank equals column rank. The same operations, applied to $(A \mid I)$, compute inverses.

Definition (Rank of a matrix)

$\operatorname{rank}(A) = \operatorname{rank}(\mathsf{L}_A)$. Thus an $n \times n$ matrix is invertible iff its rank is n (Theorem 2.5 and Corollary 2 of Theorem 2.18).

Theorem 3.3

For linear $\mathsf{T} : V \to W$ between finite-dimensional spaces with ordered bases β, γ: $\operatorname{rank}(\mathsf{T}) = \operatorname{rank}([\mathsf{T}]_\beta^\gamma)$.

Proof [Supplement]. With $A = [\mathsf{T}]_\beta^\gamma$, $\mathsf{L}_A\phi_\beta = \phi_\gamma\mathsf{T}$ (Theorem 2.14). As ϕ_β is onto, $\mathrm{R}(\mathsf{L}_A) = \mathrm{R}(\mathsf{L}_A\phi_\beta) = \phi_\gamma(\mathrm{R}(\mathsf{T}))$, and the isomorphism ϕ_γ preserves dimension. ■

Theorem 3.4

If P, Q are invertible $m \times m$ and $n \times n$, then (a) $\operatorname{rank}(AQ) = \operatorname{rank}(A)$,
(b) $\operatorname{rank}(PA) = \operatorname{rank}(A)$,
(c) $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$.

Proof.

(a) L_Q is onto, so $\mathrm{R}(\mathsf{L}_{AQ}) = \mathsf{L}_A\mathsf{L}_Q(\mathbb{F}^n) = \mathsf{L}_A(\mathbb{F}^n) = \mathrm{R}(\mathsf{L}_A)$.

(b) [Supplement] $\mathrm{R}(\mathsf{L}_{PA}) = \mathsf{L}_P(\mathrm{R}(\mathsf{L}_A))$, and the isomorphism L_P preserves dimension.

(c) Combine (a) and (b). ■

Corollary

Elementary row and column operations preserve rank.

Proof. Theorems 3.1, 3.2, 3.4. ■

Theorem 3.5

The rank of a matrix is the maximum number of linearly independent columns, i.e. the dimension of the span of its columns.

Proof. $\mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{\mathsf{L}_A(e_j)\} = \operatorname{span}\{a_1, \ldots, a_n\}$ by Theorems 2.2 and 2.13(b), where $a_j = Ae_j$ is the j-th column. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{smallmatrix}\right)$: the first two columns are independent and the third is their sum, so $\operatorname{rank} A = 2$.

Example 2

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{smallmatrix}\right)$, subtract row 1 from rows $2, 3$ to get $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & -2 & 2 \\ 0 & -1 & 1 \end{smallmatrix}\right)$, then subtract multiples of column 1 to clear row 1: the result has two independent columns, so $\operatorname{rank} A = 2$.

Theorem 3.6

If $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has rank r, then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into

$$D = \begin{pmatrix} I_r & O_1 \\ O_2 & O_3 \end{pmatrix} \qquad (O_i \text{ zero matrices}).$$

Proof. If $A = O$, then $r = 0$ and $D = A$. Otherwise we show by induction on m that A can be transformed into D for some r; by Corollary , $r = \operatorname{rank} D = \operatorname{rank} A$, and clearly $r \le m, n$.

Step 1 (a pivot). Some entry $A_{ij} \neq 0$; by row and column interchanges move it to position $(1, 1)$, and scale row 1 to make it 1.

Step 2 (clear). Subtract multiples of row 1 from the other rows and multiples of column 1 from the other columns, getting $\left(\begin{smallmatrix} 1 & 0 \\ 0 & B' \end{smallmatrix}\right)$ with B' of size $(m-1) \times (n-1)$.

Step 3 (recurse). If $m = 1$, we are done. Otherwise, by induction (or trivially if $B' = O$), operations on the last $m-1$ rows and $n-1$ columns bring B' to $\left(\begin{smallmatrix} I_{r-1} & O \\ O & O \end{smallmatrix}\right)$ without disturbing the first row and column. ∎

Example 3

$A = \left(\begin{smallmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 4 & 4 & 8 & 0 \\ 8 & 2 & 0 & 10 & 2 \\ 6 & 3 & 2 & 9 & 1 \end{smallmatrix}\right)$: interchanging rows $1, 2$, scaling, and clearing as in the proof leads to $D = \left(\begin{smallmatrix} I_3 & O \\ O & O \end{smallmatrix}\right)$, so $\operatorname{rank} A = 3$.

Corollary 1

If $\operatorname{rank} A = r$, there are invertible B $(m \times m)$ and C $(n \times n)$ with $D = BAC$.

Proof. By Theorem 3.6, $D = E_p \cdots E_1 A G_1 \cdots G_q$ with elementary E_i, G_j (Theorem 3.1); take $B = E_p \cdots E_1$ and $C = G_1 \cdots G_q$, invertible by Theorem 3.2. ∎

Corollary 2

For any $A \in \mathrm{M}_{m\times n}(\mathbb{F})$:
(a) $\operatorname{rank}(A^t) = \operatorname{rank}(A)$;
(b) the rank of A is the maximum number of linearly independent rows, i.e. the dimension of the row space;
(c) the rows and the columns of A span subspaces of the same dimension, equal to $\operatorname{rank} A$.

Proof.

(a) With $D = BAC$, $D^t = C^t A^t B^t$ with B^t, C^t invertible, so by Theorem 3.4, $\operatorname{rank} A^t = \operatorname{rank} D^t = r = \operatorname{rank} A$ (D^t visibly has rank r).

(b) The rows of A are the columns of A^t; apply Theorem 3.5 and (a).

(c) Theorem 3.5 and (b). ■

Corollary 3

Every invertible matrix is a product of elementary matrices.

Proof. If A is $n \times n$ invertible, $\operatorname{rank} A = n$ and $D = I_n$, so $I_n = E_p \cdots E_1 A G_1 \cdots G_q$ and $A = E_1^{-1} \cdots E_p^{-1} G_q^{-1} \cdots G_1^{-1}$, a product of elementary matrices by Theorem 3.2. ■

Theorem 3.7

For linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ (finite-dimensional) and matrices with AB defined:
(a) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{U})$;
(b) $\operatorname{rank}(\mathsf{UT}) \le \operatorname{rank}(\mathsf{T})$;
(c) $\operatorname{rank}(AB) \le \operatorname{rank}(A)$;
(d) $\operatorname{rank}(AB) \le \operatorname{rank}(B)$.

Proof.

(a) $\mathrm{R}(\mathsf{UT}) = \mathsf{U}(\mathsf{T}(V)) \subseteq \mathsf{U}(W) = \mathrm{R}(\mathsf{U})$.

(c) Apply (a) to $\mathsf{L}_A \mathsf{L}_B = \mathsf{L}_{AB}$.

(d) $\operatorname{rank}(AB) = \operatorname{rank}(B^t A^t) \le \operatorname{rank}(B^t) = \operatorname{rank}(B)$ by (c) and Corollary 2.

(b) With bases α, β, γ, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$: $\operatorname{rank}(\mathsf{UT}) = \operatorname{rank}(AB) \le \operatorname{rank}(B) = \operatorname{rank}(\mathsf{T})$ by Theorems 3.3, 2.11 and (d). ■

Example 4

To find a rank, simplify by operations until independence is visible.
(a) $\left(\begin{smallmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & -1 & 1 \end{smallmatrix}\right)$ has two independent rows: rank 2.
(b) $\left(\begin{smallmatrix} 1 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 3 & 0 & 0 \end{smallmatrix}\right)$: row 3 = row 1 − row 2, rank 2.
(c) $\left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{smallmatrix}\right)$: subtracting gives rows $(1, 2, 1), (0, -3, -3), (0, 3, 3)$, rank 2.

The inverse of a matrix. The **augmented matrix** $(A \mid B)$ of an $m \times n$ A and $m \times p$ B is the $m \times (n + p)$ matrix with the columns of A followed by those of B; $M(A \mid B) = (MA \mid MB)$. If A is invertible, $A^{-1} = E_p \cdots E_1$ (Corollary 3), so the row operations $E_1, \ldots, E_p$ turn $(A \mid I_n)$ into $(I_n \mid A^{-1})$. Conversely, if row operations turn $(A \mid I_n)$ into $(I_n \mid B)$, then $E_p \cdots E_1 A = I$ and $B = E_p \cdots E_1 = A^{-1}$. If instead a row of zeros appears in the left half, $\operatorname{rank} A < n$ and A is not invertible.

Example 5

For $A = \left(\begin{smallmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{smallmatrix}\right)$, row-reducing $(A \mid I_3)$ gives

$$A^{-1} = \begin{pmatrix} \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{pmatrix}.$$

Example 6

For $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 5 & 4 \end{pmatrix}$, reducing $(A \mid I_3)$ produces a zero row on the left ($\operatorname{rank} A = 2$), so A is not invertible.

Example 7

$\mathsf{T}(f) = f + f' + f''$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}]_\beta = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ with inverse $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$. So T is invertible (Corollary 1 of Theorem 2.18) and $\mathsf{T}^{-1}(a_0 + a_1x + a_2x^2) = (a_0 - a_1) + (a_1 - 2a_2)x + a_2x^2$.

3.3 Systems of linear equations: theoretical aspects

Motivation. A system $Ax = b$ is a question about L_A: solutions of $Ax = 0$ form the null space, solutions of $Ax = b$ are a translate of it, and solvability means b lies in the column space, i.e. adjoining b does not raise the rank.

A system of m linear equations in n unknowns is $Ax = b$, with **coefficient matrix** $A \in \mathrm{M}_{m\times n}(\mathbb{F})$; a **solution** is $s \in \mathbb{F}^n$ with $As = b$. The system is **consistent** if its solution set is nonempty.

Example 1

(a) $x_1 + x_2 = 3$, $x_1 - x_2 = 1$ has the unique solution $(2, 1)$.
(b) $x_1 + x_2 + x_3 = 1$ (one equation, three unknowns) has infinitely many solutions, e.g. $(1, 0, 0)$ and $(0, 0, 1)$.
(c) $x_1 + x_2 = 0$, $x_1 + x_2 = 1$ has no solution.

Definition (Homogeneous system)

$Ax = b$ is **homogeneous** if $b = 0$; it always has the solution 0.

Theorem 3.8

The solution set K of a homogeneous system $Ax = 0$ (m equations, n unknowns) is $\mathrm{N}(\mathsf{L}_A)$, a subspace of $\mathbb{F}^n$ of dimension $n - \operatorname{rank}(A)$.

Proof. $K = \{s : As = 0\} = \mathrm{N}(\mathsf{L}_A)$, and $\dim \mathrm{N}(\mathsf{L}_A) = n - \operatorname{rank} \mathsf{L}_A$ by Theorem 2.3. ∎

Corollary

If $m < n$, then $Ax = 0$ has a nonzero solution.

Proof. $\operatorname{rank} A \le m < n$, so $\dim K > 0$. ∎

Example 2

(a) $x_1 + 2x_2 + x_3 = 0$, $x_1 - x_2 - x_3 = 0$: $\operatorname{rank} A = 2$, so $\dim K = 1$, and $K = \operatorname{span}\{(1, -2, 3)\}$.
(b) $x_1 - 2x_2 + x_3 = 0$: $\dim K = 2$, with basis $\{(2, 1, 0), (-1, 0, 1)\}$.

Theorem 3.9

Let K be the solution set of $Ax = b$ and K_H that of $Ax = 0$. For any solution s of $Ax = b$, $K = \{s\} + K_H = \{s + k : k \in K_H\}$.

Proof. If $w \in K$, then $A(w - s) = b - b = 0$, so $w = s + (w - s) \in \{s\} + K_H$. Conversely $A(s + k) = b + 0 = b$. ■

Example 3

(a) $x_1+2x_2+x_3 = 7$, $x_1-x_2-x_3 = -4$ has solution $s = (1, 1, 4)$, so $K = \{(1, 1, 4)+t(1, -2, 3) : t \in \mathbb{F}\}$.
(b) $x_1 - 2x_2 + x_3 = 4$: $K = \{(4, 0, 0) + t_1(2, 1, 0) + t_2(-1, 0, 1)\}$.

Theorem 3.10

Let $Ax = b$ have n equations and n unknowns. If A is invertible, the unique solution is $A^{-1}b$. Conversely, if the system has exactly one solution, A is invertible.

Proof. If A is invertible, $A(A^{-1}b) = b$, and $As = b$ implies $s = A^{-1}b$. Conversely, if s is the only solution, then by Theorem 3.9 $K_H = \{0\}$, so $\mathrm{N}(\mathsf{L}_A) = \{0\}$ and A is invertible by Theorems 2.5 and 2.18. ■

Example 4

$2x_2 + 4x_3 = 2$, $2x_1 + 4x_2 + 2x_3 = 3$, $3x_1 + 3x_2 + x_3 = 1$ has the invertible coefficient matrix of Example 5 of Section 3.2, so the unique solution is

$$A^{-1}\begin{pmatrix}2\\3\\1\end{pmatrix} = \begin{pmatrix}-\frac{7}{8}\\ \frac{5}{4}\\ -\frac{1}{8}\end{pmatrix}.$$

Theorem 3.11

$Ax = b$ is consistent iff $\operatorname{rank}(A) = \operatorname{rank}(A \mid b)$.

Proof. Consistency means $b \in \mathrm{R}(\mathsf{L}_A) = \operatorname{span}\{a_1, \ldots, a_n\}$ (columns), i.e. $\operatorname{span}\{a_1, \ldots, a_n\} = \operatorname{span}\{a_1, \ldots, a_n, b\}$, i.e. (Theorem 1.11, the first span lying in the second) the two spans have equal dimension: $\operatorname{rank} A = \operatorname{rank}(A \mid b)$ by Theorem 3.5. ■

Example 5

For $x_1 + x_2 = 0$, $x_1 + x_2 = 1$: $\operatorname{rank} A = 1$ but $\operatorname{rank}(A \mid b) = 2$, so the system is inconsistent.

Example 6

Is $(3, 3, 2) \in \mathrm{R}(\mathsf{T})$ for $\mathsf{T}(a_1, a_2, a_3) = (a_1+a_2+a_3,\ a_1-a_2+a_3,\ a_1+a_3)$? With $A = \left(\begin{smallmatrix}1 & 1 & 1\\ 1 & -1 & 1\\ 1 & 0 & 1\end{smallmatrix}\right)$, $\operatorname{rank} A = 2$ and $\operatorname{rank}(A \mid b) = 3$. So no.

3.3.1 An application: Leontief's input–output models

Closed model. Three producers (food, clothing, housing) consume fractions $A_{ij} \geq 0$ of commodity j; each column of A sums to 1. With incomes p, survival requires the **equilibrium condition** $Ap = p$, i.e. $(I - A)p = 0$ with $p \geq 0$. (For vectors, $b \geq c$ [$b > c$] means entrywise; b is **nonnegative** [**positive**] if $b \geq 0$ [$b > 0$].) The weaker $Ap \leq p$ already forces $Ap = p$: otherwise $\sum_i p_i > \sum_i \sum_j A_{ij} p_j = \sum_j p_j$. For

$$A = \begin{pmatrix} 0.40 & 0.20 & 0.20 \\ 0.10 & 0.70 & 0.20 \\ 0.50 & 0.10 & 0.60 \end{pmatrix}, \qquad p = \begin{pmatrix} 0.25 \\ 0.35 \\ 0.40 \end{pmatrix}$$

solves it: incomes in the ratio $5 : 7 : 8$.

Theorem 3.12

Let A be an $n \times n$ input–output matrix (nonnegative entries, columns summing to 1) of the form $A = \left(\begin{smallmatrix} B & C \\ D & E \end{smallmatrix}\right)$, with D a positive $1 \times (n-1)$ vector and C a positive $(n-1) \times 1$ vector. Then $(I - A)x = 0$ has a one-dimensional solution space generated by a nonnegative vector.

Proof [Supplement]. The book cites the proof. Write $x = (y, x_n)$ with $y \in \mathbb{R}^{n-1}$.

Step 1 (reduction). The columns of $I - A$ sum to 0, so the last equation is minus the sum of the others. Hence $(I - A)x = 0$ iff $(I - B)y = x_n C$.

Step 2 ($I - B$ is invertible). The column sums of B are $1 - D_j < 1$. If $(I - B)y = 0$ with $y \neq 0$, then $|y_i| \leq \sum_j B_{ij}|y_j|$, and summing over i gives $\sum_i |y_i| \leq \sum_j (1 - D_j)|y_j| < \sum_j |y_j|$, a contradiction.

Step 3 (the solution space). By Steps 1–2 the solutions are $x_n((I - B)^{-1}C, 1)$: a one-dimensional space generated by $v = (y_0, 1)$, where $(I - B)y_0 = C$.

Step 4 ($y_0 \geq 0$). Let $N = \{i : (y_0)_i < 0\}$ and suppose $N \neq \emptyset$. For $i \in N$, $(y_0)_i = \sum_j B_{ij}(y_0)_j + C_i > \sum_{j \in N} B_{ij}(y_0)_j$, since the terms with $j \notin N$ and C_i are nonnegative and $C_i > 0$. Hence

$$\sum_{i \in N} |(y_0)_i| < \sum_{j \in N} \Big(\sum_{i \in N} B_{ij} \Big) |(y_0)_j| \leq \sum_{j \in N} |(y_0)_j|,$$

a contradiction. So $v \geq 0$. ∎

Open model. With outside demands $d \geq 0$ and A_{ij} the amount of commodity i needed per unit of commodity j (nonnegative, column sums ≤ 1), we need $x \geq 0$ with $(I - A)x = d$. If $(I - A)^{-1}$ exists and is nonnegative, $x = (I - A)^{-1}d$; this holds when $A^k \to O$, since then $(I - A)^{-1} = I + A + A^2 + \cdots$ (Section 5.3). For example, with

$$A = \begin{pmatrix} 0.30 & 0.20 & 0.30 \\ 0.10 & 0.40 & 0.10 \\ 0.30 & 0.20 & 0.30 \end{pmatrix}, \qquad (I - A)^{-1} = \begin{pmatrix} 2.0 & 1.0 & 1.0 \\ 0.5 & 2.0 & 0.5 \\ 1.0 & 1.0 & 2.0 \end{pmatrix},$$

a demand $d = (30, 20, 10)$ is met by $x = (I - A)^{-1}d = (90, 60, 70)$.

3.4 Systems of linear equations: computational aspects

Motivation. Row operations do not change the solution set. Gaussian elimination brings any augmented matrix to its unique reduced row echelon form, from which the general solution and a basis of the homogeneous solutions can be read off. The same form exposes which columns of a matrix are independent.

Definition (Equivalent systems)

Two systems are **equivalent** if they have the same solution set.

Theorem 3.13

If C is an invertible $m \times m$ matrix, $(CA)x = Cb$ is equivalent to $Ax = b$.

Proof. $As = b \Rightarrow CAs = Cb$, and $CAs = Cb \Rightarrow As = C^{-1}CAs = C^{-1}Cb = b$. ■

Corollary

If $(A' \mid b')$ comes from $(A \mid b)$ by finitely many elementary row operations, $A'x = b'$ is equivalent to $Ax = b$.

Proof. $(A' \mid b') = C(A \mid b) = (CA \mid Cb)$ with C a product of elementary, hence invertible, matrices. ■

Definition (Reduced row echelon form)

A matrix is in **reduced row echelon form** if
(a) every nonzero row precedes every zero row;
(b) the first nonzero entry of each row is the only nonzero entry in its column;
(c) the first nonzero entry of each row is 1 and lies to the right of the first nonzero entry of the preceding row.

Gaussian elimination. *Forward pass:* in the leftmost nonzero column, bring a nonzero entry to the top row (type 1), scale it to 1 (type 2), and clear the entries below it (type 3). Repeat on the submatrix below and to the right. *Backward pass:* starting from the last nonzero row, clear the entries above each leading 1. For example

$$\begin{cases} 3x_1 + 2x_2 + 3x_3 - 2x_4 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_3 - x_4 = 2 \end{cases} \quad \rightsquigarrow \quad \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{array}\right),$$

so $x_1 = 1 - x_3$, $x_2 = 2$, $x_4 = 3$, with x_3 free.

Example 1

(a) The final matrix above is in reduced row echelon form.
(b) $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ is not: the leading 1 of row 1 shares its column with a nonzero entry, violating (b), and the leading 1 of row 3 is not to the right of that of row 2, violating (c).
(c) $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is in reduced row echelon form; $\left(\begin{smallmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ is not (leading entry 2).

Theorem 3.14

Gaussian elimination transforms any matrix into reduced row echelon form.

Proof [Supplement]. After the forward pass, each nonzero row has leading entry 1 strictly to the right of the leading entry of the row above, below each leading 1 there are zeros, and zero rows are at the bottom: (a) and (c) hold. The backward pass clears the entries above each leading 1 by adding multiples of its row. This does not change the leading entries or the zeros to their left, since the row used has zeros to the left of its leading 1. So (b) holds as well. ■

Reading off the solution. Let $(A \mid b)$ be in reduced row echelon form with r nonzero rows and no row $(0 \cdots 0 \mid 1)$ (otherwise there is no solution). Solve each equation for its leading variable and set the remaining $n - r$ (free) variables equal to parameters $t_1, \ldots, t_{n-r}$. For

$$\begin{cases} 2x_1 + 3x_2 + x_3 + 4x_4 - 9x_5 = 17 \\ x_1 + x_2 + x_3 + x_4 - 3x_5 = 6 \\ x_1 + x_2 + x_3 + 2x_4 - 5x_5 = 8 \\ 2x_1 + 2x_2 + 2x_3 + 3x_4 - 8x_5 = 14 \end{cases} \rightsquigarrow \left(\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 3 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

the general solution is

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$$

Theorem 3.15

Let $Ax = b$ have r nonzero equations in n unknowns, $\operatorname{rank} A = \operatorname{rank}(A \mid b)$, and $(A \mid b)$ in reduced row echelon form. Then (a) $\operatorname{rank} A = r$;
(b) if the procedure gives $s = s_0 + t_1 u_1 + \cdots + t_{n-r} u_{n-r}$, then $\{u_1, \ldots, u_{n-r}\}$ is a basis of the homogeneous solution set K_H and s_0 is a solution.

Proof.

(a) The system is consistent, so no row is $(0 \cdots 0 \mid 1)$. Each of the r rows has a leading 1 in A, in a column that is e_i; these r columns are independent, and the column space lies in $\mathbb{F}^r$ (rows below r are absent). So $\operatorname{rank} A = r$.

(b) Taking all $t_i = 0$ shows $s_0 \in K$. By Theorem 3.9, $K_H = \{-s_0\} + K = \operatorname{span}\{u_1, \ldots, u_{n-r}\}$, and $\dim K_H = n - r$ (Theorem 3.8). A spanning set of $n - r$ vectors in an $(n-r)$-dimensional space is a basis. ∎

What the reduced row echelon form says. Let A have columns $a_1, \ldots, a_n$ and reduced row echelon form $B = CA$ (C invertible) with columns $b_j = Ca_j$. Since L_C is an isomorphism, linear relations among the b_j are exactly those among the a_j.

Theorem 3.16

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank $r > 0$ and reduced row echelon form B. Then
(a) B has exactly r nonzero rows;
(b) for each $i \le r$ there is a column $b_{j_i} = e_i$ ($j_1 < \cdots < j_r$);
(c) the columns $a_{j_1}, \ldots, a_{j_r}$ of A are linearly independent;
(d) if column k of B is $d_1 e_1 + \cdots + d_r e_r$, then $a_k = d_1 a_{j_1} + \cdots + d_r a_{j_r}$.

Proof.

(a) $\operatorname{rank} B = \operatorname{rank} A = r$ (Corollary to Theorem 3.4), and the nonzero rows of B are independent (each has a 1 where the others have 0), so there are r of them (Corollary 2 of Theorem 3.6).

(b) Let j_i be the column of the leading 1 of row i; by (b), (c) of the definition, $b_{j_i} = e_i$ and $j_1 < \cdots < j_r$.

(c) $b_{j_1}, \ldots, b_{j_r}$ are independent, and $\mathsf{L}_{C^{-1}}$ preserves independence.

(d) Every column of B lies in $\operatorname{span}\{e_1, \ldots, e_r\}$ (rows $> r$ are zero), so $b_k = \sum d_i e_i = \sum d_i b_{j_i}$; apply C^{-1}. ∎

Corollary

The reduced row echelon form of a matrix is unique.

Proof. By Theorem 3.16, j_1 is the first column of A that is nonzero, and in general j_i is the first column not in the span of the preceding columns. Every other column b_k has coordinates d_i determined by the unique relation $a_k = \sum d_i a_{j_i}$ (a_{j_i} independent). So B is determined by A. ■

Example 2

$A = \begin{pmatrix} 2&4&6&2&4 \\ 1&2&3&1&1 \\ 2&4&8&0&0 \\ 3&6&7&5&9 \end{pmatrix}$ has reduced form $B = \begin{pmatrix} 1&2&0&4&0 \\ 0&0&1&-1&0 \\ 0&0&0&0&1 \\ 0&0&0&0&0 \end{pmatrix}$. So rank $A = 3$, columns $1, 3, 5$ of A are independent, $a_2 = 2a_1$ and $a_4 = 4a_1 - a_3$.

Example 3

To reduce $S = \{2+x+2x^2+3x^3,\ 4+2x+4x^2+6x^3,\ 6+3x+8x^2+7x^3,\ 2+x+5x^3,\ 4+x+9x^3\}$ to a basis of span(S): the coordinate vectors are the columns of A in Example 2, so the first, third and fifth polynomials form a basis.

Example 4

$V = \{x \in \mathbb{R}^5 : x_1 + 7x_2 + 5x_3 - 4x_4 + 2x_5 = 0\}$ has dimension 4, with basis $\{(-7,1,0,0,0), (-5,0,1,0,0), (4,0,0,1,0), (-2,0,0,0,1)\}$. To extend the independent set $S = \{(-2,0,0,-1,-1), (1,1,-2,-1,-1), (-5,1,0,1,1)\} \subseteq V$ to a basis, row-reduce the matrix whose columns are the vectors of S followed by this basis. The pivot columns are $1, 2, 3, 6$, so $S \cup \{(4,0,0,1,0)\}$ is a basis of V.

Chapter 4. Determinants

4.1 Determinants of order 2

Motivation. For 2×2 matrices one number, $ad-bc$, decides invertibility and gives the signed area of the parallelogram spanned by the rows. It is not linear in the matrix but is linear in each row separately, and that is the property that generalises.

Definition (Determinant of order 2)

For $A=\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\in \mathrm{M}_{2\times 2}(\mathbb{F})$, $\det(A)=|A|=ad-bc$.

Example 1

For $A=\left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B=\left(\begin{smallmatrix} 3 & 2 \\ 6 & 4 \end{smallmatrix}\right)$: $\det A=-2$, $\det B=0$, but $\det(A+B)=\det\left(\begin{smallmatrix} 4 & 4 \\ 9 & 8 \end{smallmatrix}\right)=-4\neq \det A+\det B$. So det is not linear.

Theorem 4.1

$\det:\mathrm{M}_{2\times 2}(\mathbb{F})\to\mathbb{F}$ is linear in each row when the other row is fixed: for $u,v,w\in\mathbb{F}^2$, $k\in\mathbb{F}$,

$$\det\begin{pmatrix} u+kv \\ w \end{pmatrix}=\det\begin{pmatrix} u \\ w \end{pmatrix}+k\det\begin{pmatrix} v \\ w \end{pmatrix},\qquad \det\begin{pmatrix} w \\ u+kv \end{pmatrix}=\det\begin{pmatrix} w \\ u \end{pmatrix}+k\det\begin{pmatrix} w \\ v \end{pmatrix}.$$

Proof. With $u=(a_1,a_2)$, $v=(b_1,b_2)$, $w=(c_1,c_2)$: $(a_1+kb_1)c_2-(a_2+kb_2)c_1=(a_1c_2-a_2c_1)+k(b_1c_2-b_2c_1)$. The second identity is analogous. ∎

Theorem 4.2

$A\in\mathrm{M}_{2\times 2}(\mathbb{F})$ is invertible iff $\det A\neq 0$, and then $A^{-1}=\frac{1}{\det A}\left(\begin{smallmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{smallmatrix}\right)$.

Proof. If $\det A\neq 0$, multiplying out shows that the displayed matrix M satisfies $AM=MA=I$. Conversely let A be invertible, so $\operatorname{rank}A=2$ and $A_{11}\neq 0$ or $A_{21}\neq 0$. If $A_{11}\neq 0$, add $-A_{21}/A_{11}$ times row 1 to row 2, getting $\left(\begin{smallmatrix} A_{11} & A_{12} \\ 0 & A_{22}-A_{12}A_{21}/A_{11} \end{smallmatrix}\right)$, still of rank 2 (Theorem 3.4). So $A_{22}-A_{12}A_{21}/A_{11}\neq 0$, i.e. $\det A\neq 0$. If $A_{21}\neq 0$, argue the same way with the rows exchanged. ∎

4.1.1 The area of a parallelogram

For an ordered basis $\beta=\{u,v\}$ of $\mathbb{R}^2$, the **orientation** is $O\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)=\det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)/\left|\det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)\right|=\pm 1$: $+1$ if β is right-handed (rotating u counterclockwise by less than π reaches the direction of v), -1 if left-handed; set $O=1$ if $\{u,v\}$ is dependent. The **area** of the parallelogram with adjacent sides u,v satisfies

$$\operatorname{area}\begin{pmatrix} u \\ v \end{pmatrix}=O\begin{pmatrix} u \\ v \end{pmatrix}\cdot\det\begin{pmatrix} u \\ v \end{pmatrix}=\left|\det\begin{pmatrix} u \\ v \end{pmatrix}\right|.$$

Reason. $\delta(u,v)=O\cdot\text{area}$ is linear in each row (shearing $v\mapsto v+ku$ preserves area and orientation, and scaling a side scales the signed area), vanishes on equal rows, and $\delta(e_1,e_2)=1$. Such a function equals det (the $n=2$ case of Theorem 4.12).

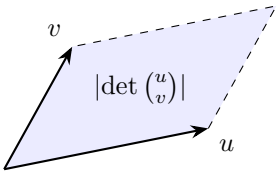


Figure. *The parallelogram spanned by u, v has area $|\det \binom{u}{v}|$; the sign of* det *records orientation.*

Example 2

For $u = (-1, 5)$, $v = (4, -2)$: $\det\begin{pmatrix} -1 & 5 \\ 4 & -2 \end{pmatrix} = 2 - 20 = -18$, so the area is 18 (and $\{u, v\}$ is left-handed).

4.2 Determinants of order n

Motivation. The $n \times n$ determinant is defined recursively by expanding along the first row. From this definition alone one proves linearity in each row and expansion along any row, which yield its behaviour under elementary row operations, the practical way to compute it.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\tilde{A}_{ij}$ is the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j.

Definition (Determinant)

For $n = 1$, $\det(A_{11}) = A_{11}$. For $n \geq 2$,

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$ is the **cofactor** of A_{ij}, and the formula is the **cofactor expansion along the first row**. For $n = 2$ it gives $A_{11}A_{22} - A_{12}A_{21}$.

Example 1

$A = \begin{pmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{pmatrix}$: $\det A = 1 \cdot (30 - 8) - 3 \cdot (18 + 8) + (-3)(-12 - 20) = 22 - 78 + 96 = 40$.

Example 2

$B = \begin{pmatrix} 0 & 1 & 3 \\ -2 & -3 & -5 \\ 4 & -4 & 4 \end{pmatrix}$: $\det B = 0 - 1 \cdot (-8 + 20) + 3 \cdot (8 + 12) = 48$.

Example 3

$C = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & 4 & -6 \end{pmatrix}$: only C_{11} and C_{14} contribute, and $\tilde{C}_{11} = A$, $\tilde{C}_{14} = B$ above, so $\det C = 2 \cdot 40 + (-1)^5 \cdot 1 \cdot 48 = 32$.

Example 4

$\det I_n = 1$, by induction: the expansion along the first row has only the term $1 \cdot \det I_{n-1}$.

Theorem 4.3

The determinant is linear in each row when the other rows are fixed: if rows $a_1, \dots, a_n$ have $a_r = u + kv$, then

$$\det(a_1, \dots, u + kv, \dots, a_n) = \det(a_1, \dots, u, \dots, a_n) + k \det(a_1, \dots, v, \dots, a_n).$$

Proof. Induction on n; $n = 1$ is clear. Let A, B, C be the matrices with row r equal to $u + kv$, u, v and the other rows equal.

Step 1 ($r = 1$). $\tilde{A}_{1j} = \tilde{B}_{1j} = \tilde{C}_{1j}$, so $\det A = \sum_j (-1)^{1+j}(u_j + kv_j) \det \tilde{A}_{1j} = \det B + k \det C$.

Step 2 ($r > 1$). $A_{1j} = B_{1j} = C_{1j}$, and $\tilde{A}_{1j}$, $\tilde{B}_{1j}$, $\tilde{C}_{1j}$ agree except in the row coming from row r, which is $u + kv$, u, v with entry j deleted. By induction $\det \tilde{A}_{1j} = \det \tilde{B}_{1j} + k \det \tilde{C}_{1j}$; multiply by $(-1)^{1+j} A_{1j}$ and sum. ∎

Corollary

A matrix with a zero row has determinant 0.

Proof. If row r is $0 = 0 + 1 \cdot 0$, Theorem 4.3 gives $\det A = \det A + \det A$. ∎

Lemma

If $n \geq 2$ and row i of $B \in \mathrm{M}_{n \times n}(\mathbb{F})$ is e_k, then $\det B = (-1)^{i+k} \det(\tilde{B}_{ik})$.

Proof. Induction on n. For $n = 2$ check directly. Let $n \geq 3$. If $i = 1$, the definition gives the claim. Let $i > 1$. For each $j \neq k$, let C_{ij} be $\tilde{B}_{1j}$ with row $i - 1$ (coming from row i of B) and its column through k removed. Row $i - 1$ of $\tilde{B}_{1j}$ is 0 if $j = k$, e_{k-1} if $j < k$, and e_k if $j > k$ (indices in $\mathbb{F}^{n-1}$). By induction,

$$\det \tilde{B}_{1j} = \begin{cases} 0 & j = k \\ (-1)^{(i-1)+(k-1)} \det C_{ij} & j < k \\ (-1)^{(i-1)+k} \det C_{ij} & j > k. \end{cases}$$

Hence $\det B = \sum_{j<k} (-1)^{1+j} B_{1j} (-1)^{i+k} \det C_{ij} + \sum_{j>k} (-1)^{1+j} B_{1j} (-1)^{i+k+1} \det C_{ij}$. On the other hand, expand $\det \tilde{B}_{ik}$ along its first row (row 1 of B without entry k). The entry B_{1j} sits in column j if $j < k$ and column $j - 1$ if $j > k$, and its minor is C_{ij}. So

$$\det \tilde{B}_{ik} = \sum_{j<k} (-1)^{1+j} B_{1j} \det C_{ij} + \sum_{j>k} (-1)^{j} B_{1j} \det C_{ij}.$$

Multiplying by $(-1)^{i+k}$ gives the expression for $\det B$. ∎

Theorem 4.4

The determinant can be expanded along any row: for each i, $\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$.

Proof. For $i = 1$ this is the definition; let $i > 1$. Row i of A is $\sum_j A_{ij} e_j$. Let B_j be A with row i replaced by e_j. By Theorem 4.3 and the Lemma, $\det A = \sum_j A_{ij} \det B_j = \sum_j A_{ij} (-1)^{i+j} \det(\tilde{B}_j)_{ij} = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$, since deleting row i removes the only difference between B_j and A. ∎

Corollary

A matrix with two identical rows has determinant 0.

Proof. Induction on n; for $n = 2$ it is clear. For $n \geq 3$, if rows $r \neq s$ coincide, expand along a row $i \notin \{r, s\}$: each $\tilde{A}_{ij}$ has two identical rows, so every term vanishes by induction. ■

Theorem 4.5

Interchanging two rows multiplies the determinant by -1.

Proof. Let rows $r < s$ of A be u, v and B have them interchanged. By Theorem 4.3 and the Corollary, $0 = \det(\dots, u+v, \dots, u+v, \dots) = \det(\dots, u, \dots, u, \dots) + \det(\dots, u, \dots, v, \dots) + \det(\dots, v, \dots, u, \dots) + \det(\dots, v, \dots, v, \dots) = 0 + \det A + \det B + 0$. ■

Theorem 4.6

Adding a multiple of one row to another leaves the determinant unchanged.

Proof. Let B be A with k·row r added to row s, and C be A with row s replaced by row r. By Theorem 4.3, $\det B = \det A + k \det C = \det A$, since $\det C = 0$ by the Corollary to Theorem 4.4. ■

Corollary

If $\operatorname{rank} A < n$, then $\det A = 0$.

Proof. The rows are dependent (Corollary 2 of Theorem 3.6), so some row $a_r = \sum_{i \neq r} c_i a_i$. Subtracting c_i·row i from row r for each $i \neq r$ produces a zero row without changing det (Theorem 4.6); apply the Corollary to Theorem 4.3. ■

Summary for row operations: type 1 multiplies det by -1; type 2 (row times k) multiplies it by k; type 3 leaves it unchanged. The determinant of an upper triangular matrix is the product of its diagonal entries (expand along the last row repeatedly), so reducing to triangular form computes det.

Example 5

For B of Example 2: swap rows $1, 2$ (sign change), clear column 1, and reach the triangular matrix $\begin{pmatrix} -2 & -3 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 24 \end{pmatrix}$. So $\det B = -(-2)(1)(24) = 48$.

Example 6

For $\begin{pmatrix} 2 & 3 & 1 & 1 \\ -2 & -2 & 2 & -1 \\ 4 & 7 & 2 & 3 \\ -6 & -9 & 5 & 1 \end{pmatrix}$, type 3 operations give an upper triangular matrix with diagonal product -40, the determinant.

4.3 Properties of determinants

Motivation. The determinant is multiplicative, detects invertibility, ignores transposition, solves square systems (Cramer's rule), and measures n-dimensional volume.

From the row-operation rules: $\det E = -1$, k, 1 for elementary E of types $1, 2, 3$, and $\det(EA) = \det E \cdot \det A$ for elementary E.

Theorem 4.7

$\det(AB) = \det(A)\det(B)$ for $A, B \in \mathrm{M}_{n\times n}(\mathbb{F})$.

Proof.

Step 1 ($\operatorname{rank} A < n$*).* $\operatorname{rank}(AB) \le \operatorname{rank} A < n$ (Theorem 3.7), so both sides are 0 (Corollary to Theorem 4.6).

Step 2 ($\operatorname{rank} A = n$*).* $A = E_m \cdots E_1$ is a product of elementary matrices (Corollary 3 of Theorem 3.6). Applying $\det(EM) = \det E \det M$ repeatedly, $\det(AB) = \det E_m \cdots \det E_1 \det B$ and $\det A = \det E_m \cdots \det E_1$. ■

Corollary

A is invertible iff $\det A \neq 0$, and then $\det(A^{-1}) = \frac{1}{\det A}$.

Proof. If A is not invertible, $\operatorname{rank} A < n$ and $\det A = 0$. If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$. ■

Theorem 4.8

$\det(A^t) = \det(A)$.

Proof. If A is not invertible, neither is A^t ($\operatorname{rank} A^t = \operatorname{rank} A$), and both determinants are 0. Otherwise $A = E_m \cdots E_1$, so $A^t = E_1^t \cdots E_m^t$. Each E_i^t is elementary of the same type with $\det E_i^t = \det E_i$ (types 1, 2 are symmetric; a type 3 transpose is again type 3). By Theorem 4.7, $\det A^t = \prod \det E_i^t = \prod \det E_i = \det A$. ■

Consequently all row statements hold for columns: expansion along any column, and the effect of column operations.

Theorem 4.9 (Cramer's rule)

If $Ax = b$ is an $n \times n$ system with $\det A \neq 0$, it has a unique solution, and

$$x_k = \frac{\det M_k}{\det A} \qquad (1 \le k \le n),$$

where M_k is A with column k replaced by b.

Proof. Uniqueness: A is invertible (Theorem 3.10). Let x be the solution and X_k be I_n with column k replaced by x. Then $AX_k = M_k$ (column $j \neq k$ of AX_k is $Ae_j = a_j$, column k is $Ax = b$). Expanding along row k, $\det X_k = x_k$. So $\det M_k = \det A \cdot x_k$. ■

Example 1

$x_1 + 2x_2 + 3x_3 = 2$, $x_1 + x_3 = 3$, $x_1 + x_2 - x_3 = 1$: $\det A = 6$, $\det M_1 = 15$, $\det M_2 = -6$, $\det M_3 = 3$, so $(x_1, x_2, x_3) = (\frac{5}{2}, -1, \frac{1}{2})$.

Volume. For rows $a_1, \ldots, a_n \in \mathbb{R}^n$ of A, $|\det A|$ is the n-dimensional volume of the parallelepiped with adjacent sides $a_1, \ldots, a_n$ (this follows as in Section 4.1.1, the multilinear-alternating argument of Section 4.5).

Example 2

$a_1 = (1,-2,1)$, $a_2 = (1,0,-1)$, $a_3 = (1,1,1)$ span a parallelepiped of volume $\left|\det\begin{pmatrix}1&-2&1\\1&0&-1\\1&1&1\end{pmatrix}\right| = 6$. (These vectors are mutually orthogonal, of lengths $\sqrt{6}, \sqrt{2}, \sqrt{3}$, and $\sqrt{6}\sqrt{2}\sqrt{3} = 6$.)

For an ordered basis β of $\mathbb{R}^n$ with vectors as rows of A, β is **right-handed** if $\det A > 0$ and **left-handed** if $\det A < 0$. A change of coordinates with $\det Q > 0$ preserves handedness.

4.4 Summary: important facts about determinants

Motivation. The facts needed later, collected in one place.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$:

1. $\det A = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any i, and $= \sum_i (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any j (Theorems 4.4, 4.8).

2. Interchanging two rows (columns) changes the sign; multiplying a row (column) by k multiplies det by k; adding a multiple of a row (column) to another leaves it unchanged.

3. det of an upper or lower triangular matrix is the product of the diagonal entries; $\det I = 1$.

4. Two identical rows (columns), or a zero row, give det $= 0$.

5. $\det(AB) = \det A \det B$; $\det A^t = \det A$.

6. A is invertible iff $\det A \neq 0$, and then $\det A^{-1} = (\det A)^{-1}$.

7. Similar matrices have equal determinants: $\det(Q^{-1}AQ) = \det Q^{-1} \det A \det Q = \det A$. Hence the **determinant of a linear operator** $\det \mathsf{T} = \det[\mathsf{T}]_\beta$ is independent of β.

8. For $A = \left(\begin{smallmatrix} B & C \\ O & D \end{smallmatrix}\right)$ with square B, D, $\det A = \det B \det D$.

Proof of 3 and 8 [Supplement].

3. For upper triangular A expand along the first column: $\det A = A_{11} \det \tilde{A}_{11}$, and $\tilde{A}_{11}$ is upper triangular; induct. Lower triangular: transpose.

8. Induction on the size k of B, expanding along the first column. For $k = 1$, $\det A = B_{11} \det D$. For $k > 1$, only rows $i \leq k$ contribute, and $\tilde{A}_{i1} = \left(\begin{smallmatrix} \tilde{B}_{i1} & C' \\ O & D \end{smallmatrix}\right)$, so $\det A = \sum_{i\leq k} (-1)^{i+1} B_{i1} \det \tilde{B}_{i1} \det D = \det B \det D$. ∎

4.5 A characterization of the determinant *

Motivation. Linearity in each row, vanishing on matrices with two equal adjacent rows, and the value 1 at I determine the determinant uniquely. This explains why det measures signed volume and gives a second proof of multiplicativity.

Definition (n-linear; alternating)

$\delta : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is **n-linear** if it is linear in each row when the others are fixed. It is **alternating** if $\delta(A) = 0$ whenever two adjacent rows of A are identical.

Example 1

$\delta(A) = 0$ is n-linear.

Example 2

$\delta_j(A) = A_{1j}A_{2j}\cdots A_{nj}$ (product of column j) is n-linear: it is linear in A_{rj}, i.e. in row r.

Example 3

$\delta(A) = A_{11}A_{22}\cdots A_{nn}$ is n-linear.

Example 4

tr is not n-linear for $n \geq 2$: $\mathrm{tr}(2I) = 2n \neq 2\mathrm{tr}(I)$, whereas an n-linear function would scale $2I$ (each of n rows doubled) by 2^n.

det is n-linear (Theorem 4.3) and alternating (Corollary to Theorem 4.4).

Theorem 4.10

Let δ be alternating and n-linear.
(a) Interchanging two rows of A changes $\delta(A)$ to $-\delta(A)$.
(b) If A has two identical rows, $\delta(A) = 0$.

Proof.

(a) For adjacent rows $r, r+1$ with entries u, v, the computation of Theorem 4.5 works because only adjacent equal rows are used: $0 = \delta(\dots, u+v, u+v, \dots) = \delta(A) + \delta(B)$. A swap of rows $r < s$ is a composition of $2(s-r)-1$ adjacent swaps (move row r down to position s, then row s up to position r), an odd number. So δ changes sign.

(b) Swap rows to make the identical rows adjacent; by (a) this changes δ only by a sign, and the result is 0. ■

Corollary 1

If B is A with a multiple of a row added to another row, then $\delta(B) = \delta(A)$.

Proof. As for Theorem 4.6, using Theorem 4.10(b). ■

Corollary 2

If $\mathrm{rank}\, M < n$, then $\delta(M) = 0$.

Proof. As for the Corollary to Theorem 4.6: row operations of type 3 (Corollary 1) produce a zero row, and δ of a matrix with a zero row is 0 by linearity. ■

Corollary 3

For elementary E_1, E_2, E_3 of types $1, 2, 3$, with E_2 multiplying a row of I by k: $\delta(E_1) = -\delta(I)$, $\delta(E_2) = k\delta(I)$, $\delta(E_3) = \delta(I)$.

Proof. Theorem 4.10(a), linearity in the scaled row, and Corollary 1, applied to I. ■

Theorem 4.11

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(AB) = \delta(A)\delta(B)$.

Proof [Supplement]. By Theorem 4.10, linearity and Corollary 1, $\delta(EB) = \delta(E)\delta(B)$ for elementary E (with $\delta(E) = -1, k, 1$ by Corollary 3). Now repeat the proof of Theorem 4.7, using Corollary 2 when $\operatorname{rank} A < n$. ■

Theorem 4.12

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(A) = \det(A)$ for all A.

Proof. If $\operatorname{rank} A < n$, both vanish (Corollary 2 and the Corollary to Theorem 4.6). Otherwise $A = E_m \cdots E_1$, and by Theorem 4.11 and Corollary 3, $\delta(A) = \prod \delta(E_i) = \prod \det(E_i) = \det A$. ■

So det is the *unique* alternating n-linear function with value 1 at I.

Chapter 5. Diagonalization

5.1 Eigenvalues and eigenvectors

Motivation. An operator is easiest to understand in a basis where its matrix is diagonal. Such a basis consists of eigenvectors, vectors that T only stretches, and the stretching factors are the zeros of the characteristic polynomial $\det(A - tI)$.

Definition (Diagonalizable)

A linear operator T on finite-dimensional V is **diagonalizable** if $[\mathsf{T}]_\beta$ is diagonal for some ordered basis β. A square matrix A is diagonalizable if L_A is, i.e. if A is similar to a diagonal matrix.

Definition (Eigenvector, eigenvalue)

A nonzero $v \in V$ is an **eigenvector** of T if $\mathsf{T}(v) = \lambda v$ for some scalar λ, the corresponding **eigenvalue**. For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, eigenvectors and eigenvalues are those of L_A: $Av = \lambda v$, $v \neq 0$.

Theorem 5.1

T is diagonalizable iff V has an ordered basis β of eigenvectors of T. If $\beta = \{v_1, \ldots, v_n\}$ is such a basis and $D = [\mathsf{T}]_\beta$, then D is diagonal and D_{jj} is the eigenvalue of v_j.

Proof. $[\mathsf{T}]_\beta$ is diagonal iff each column $[\mathsf{T}(v_j)]_\beta$ is $D_{jj}e_j$, i.e. $\mathsf{T}(v_j) = D_{jj}v_j$ for all j. Basis vectors are nonzero, so this says each v_j is an eigenvector with eigenvalue D_{jj}. ■

To diagonalize: find a basis of eigenvectors γ; then $Q^{-1}AQ$ is diagonal, where the columns of Q are the vectors of γ (Corollary to Theorem 2.23).

Example 1

$A = \left(\begin{smallmatrix} 1 & 3 \\ 4 & 2 \end{smallmatrix}\right)$: $A(1,-1)^t = -2(1,-1)^t$ and $A(3,4)^t = 5(3,4)^t$. With $Q = \left(\begin{smallmatrix} 1 & 3 \\ -1 & 4 \end{smallmatrix}\right)$, $Q^{-1}AQ = \left(\begin{smallmatrix} -2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$.

Example 2

The rotation of $\mathbb{R}^2$ by $\pi/2$ has no eigenvectors (no nonzero vector is mapped to a multiple of itself), so it is not diagonalizable.

Example 3

On $C^\infty(\mathbb{R})$, $\mathsf{T}(f) = f'$ has $\mathsf{T}(f) = \lambda f$ iff $f = ce^{\lambda t}$ (as in Theorem 2.30): every real λ is an eigenvalue, with eigenvectors $ce^{\lambda t}$, $c \neq 0$.

Theorem 5.2

λ is an eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ iff $\det(A - \lambda I_n) = 0$.

Proof. λ is an eigenvalue iff $(A-\lambda I)v = 0$ for some $v \neq 0$, iff $A-\lambda I$ is not invertible, iff $\det(A-\lambda I) = 0$ (Corollary to Theorem 4.7). ■

Definition (Characteristic polynomial)

$f(t) = \det(A - tI_n)$ is the **characteristic polynomial** of A. For an operator T on n-dimensional V, it is that of $[\mathsf{T}]_\beta$, independent of β since similar matrices have the same characteristic polynomial: $\det(Q^{-1}AQ - tI) = \det(Q^{-1}(A-tI)Q) = \det(A-tI)$. Write $f(t) = \det(\mathsf{T} - t\mathsf{I})$.

Example 4

$A = \left(\begin{smallmatrix} 1 & 1 \\ 4 & 1 \end{smallmatrix}\right)$: $f(t) = (1-t)^2 - 4 = (t-3)(t+1)$, so the eigenvalues are 3 and -1.

Example 5

$\mathsf{T}(f) = f + (x+1)f'$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, so $f(t) = -(t-1)(t-2)(t-3)$ and the eigenvalues are $1, 2, 3$.

Theorem 5.3

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$: (a) $f(t)$ has degree n and leading coefficient $(-1)^n$;
(b) A has at most n distinct eigenvalues.

Proof [Supplement].

(a) Induction on n, expanding $\det(A-tI)$ along the first row: the term $(A_{11}-t)\det(\widetilde{(A-tI)}_{11})$ has degree n and leading coefficient $(-1)\cdot(-1)^{n-1}$ by induction, while the other terms contain a minor missing two diagonal entries, of degree $\leq n-2$.

(b) A nonzero polynomial of degree n has at most n zeros (Section 1.6.1). ■

Theorem 5.4

If λ is an eigenvalue of T, then v is an eigenvector for λ iff $v \neq 0$ and $v \in \mathrm{N}(\mathsf{T} - \lambda\mathsf{I})$.

Proof. $\mathsf{T}(v) = \lambda v$ iff $(\mathsf{T} - \lambda\mathsf{I})(v) = 0$. ■

Example 6

For A of Example 4: $\mathrm{N}(A-3I) = \mathrm{span}\{(1,2)^t\}$ and $\mathrm{N}(A+I) = \mathrm{span}\{(1,-2)^t\}$. So $\beta = \{(1,2),(1,-2)\}$ is a basis of eigenvectors and $Q^{-1}AQ = \mathrm{diag}(3,-1)$ for $Q = \left(\begin{smallmatrix} 1 & 1 \\ 2 & -2 \end{smallmatrix}\right)$.

Example 7

For T of Example 5, solving $(A-\lambda I)x = 0$ gives eigenvectors 1, $1+x$, $1+2x+x^2 = (1+x)^2$ for $\lambda = 1, 2, 3$. In the basis $\{1, 1+x, (1+x)^2\}$, $[\mathsf{T}] = \mathrm{diag}(1,2,3)$.

In general one computes eigenvectors of T through coordinates: v is an eigenvector of T iff $[v]_\beta$ is one of $[\mathsf{T}]_\beta$.

5.2 Diagonalizability

Motivation. Eigenvectors for distinct eigenvalues are independent, so n distinct eigenvalues guarantee a diagonal form. In general, diagonalizability means that the characteristic polynomial splits and that each eigenspace is as large as the multiplicity allows, i.e. V is the direct sum of the eigenspaces.

Theorem 5.5

Eigenvectors $v_1, \dots, v_k$ of T for distinct eigenvalues $\lambda_1, \dots, \lambda_k$ are linearly independent.

Proof. Induction on k; $k = 1$ holds as $v_1 \neq 0$. If $\sum_{i=1}^{k} a_i v_i = 0$, apply $\mathsf{T} - \lambda_k \mathsf{I}$: $\sum_{i<k} a_i(\lambda_i - \lambda_k)v_i = 0$. By induction $a_i(\lambda_i - \lambda_k) = 0$, so $a_i = 0$ for $i < k$; then $a_k v_k = 0$ gives $a_k = 0$. ■

Corollary

If $\dim V = n$ and T has n distinct eigenvalues, T is diagonalizable.

Proof. One eigenvector per eigenvalue gives n independent vectors (Theorem 5.5), a basis (Corollary 2 of Theorem 1.10); apply Theorem 5.1. ■

Example 1

$A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues $0, 2$, hence is diagonalizable. The converse of the Corollary fails: I is diagonal with one eigenvalue.

Definition (Splitting)

$f \in \mathsf{P}(\mathbb{F})$ **splits over** $\mathbb{F}$ if $f(t) = c(t - a_1) \cdots (t - a_n)$ with $c, a_i \in \mathbb{F}$. E.g. $(t^2 + 1)(t - 2)$ splits over $\mathbb{C}$ but not over $\mathbb{R}$.

Theorem 5.6

The characteristic polynomial of a diagonalizable operator splits.

Proof. If $[\mathsf{T}]_\beta = D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, then $f(t) = \det(D - tI) = (\lambda_1 - t) \cdots (\lambda_n - t) = (-1)^n(t - \lambda_1) \cdots (t - \lambda_n)$. ■

Definition (Multiplicity; eigenspace)

The (algebraic) **multiplicity** of an eigenvalue λ is the largest k with $(t - \lambda)^k \mid f(t)$. The **eigenspace** is $E_\lambda = \{x : \mathsf{T}(x) = \lambda x\} = \mathrm{N}(\mathsf{T} - \lambda \mathsf{I})$.

Example 2

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{smallmatrix}\right)$ has $f(t) = -(t - 3)^2(t - 4)$: $\lambda = 3$ has multiplicity 2, $\lambda = 4$ multiplicity 1.

Theorem 5.7

If λ is an eigenvalue of multiplicity m of T on finite-dimensional V, then $1 \leq \dim E_\lambda \leq m$.

Proof. Let $\{v_1, \dots, v_p\}$ be a basis of E_λ ($p \geq 1$), extended to an ordered basis β of V. Then $A = [\mathsf{T}]_\beta = \begin{pmatrix} \lambda I_p & B \\ O & C \end{pmatrix}$, so by fact 8 of Section 4.4, $f(t) = \det(A - tI) = (\lambda - t)^p \det(C - tI)$. Hence $(t - \lambda)^p \mid f$, and $p \leq m$. ■

Example 3

$\mathsf{T}(f) = f'$ on $\mathsf{P}_2(\mathbb{R})$: $[\mathsf{T}] = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$, $f(t) = -t^3$, and E_0 is the constants, of dimension $1 < 3$. T is not diagonalizable.

Example 4

$\mathsf{T}(a_1, a_2, a_3) = (4a_1 + a_3,\ 2a_1 + 3a_2 + 2a_3,\ a_1 + 4a_3)$ has $f(t) = -(t - 5)(t - 3)^2$, $E_5 = \operatorname{span}\{(1, 2, 1)\}$ and $E_3 = \operatorname{span}\{(0, 1, 0), (-1, 0, 1)\}$. The union of these bases is a basis of eigenvectors.

Lemma

Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T and $v_i \in E_{\lambda_i}$. If $v_1 + \cdots + v_k = 0$, then all $v_i = 0$.

Proof. Otherwise the nonzero v_i are eigenvectors for distinct eigenvalues with a nontrivial relation (all coefficients 1), contradicting Theorem 5.5. ■

Theorem 5.8

If $\lambda_1, \dots, \lambda_k$ are distinct eigenvalues of T and each $S_i \subseteq E_{\lambda_i}$ is finite and linearly independent, then $S = S_1 \cup \cdots \cup S_k$ is linearly independent.

Proof. Let $S_i = \{v_{i1}, \dots, v_{in_i}\}$ and $\sum_{i,j} a_{ij} v_{ij} = 0$. Put $w_i = \sum_j a_{ij} v_{ij} \in E_{\lambda_i}$. Then $\sum w_i = 0$, so each $w_i = 0$ by the Lemma, and independence of S_i gives $a_{ij} = 0$. ■

Theorem 5.9

Let T on finite-dimensional V have splitting characteristic polynomial and distinct eigenvalues $\lambda_1, \dots, \lambda_k$ with multiplicities m_i.
(a) T is diagonalizable iff $m_i = \dim E_{\lambda_i}$ for all i.
(b) If T is diagonalizable and β_i is an ordered basis of E_{λ_i}, then $\beta = \beta_1 \cup \cdots \cup \beta_k$ is an ordered basis of V consisting of eigenvectors.

Proof. Let $n = \dim V$, $d_i = \dim E_{\lambda_i}$. As f splits, $\sum m_i = n$.

Step 1 (only if). Let β be a basis of eigenvectors and $\beta_i' = \beta \cap E_{\lambda_i}$, with n_i vectors. Then $n_i \leq d_i \leq m_i$, and $\sum n_i = n$ since every vector of β is in some E_{λ_i}. So $n = \sum n_i \leq \sum d_i \leq \sum m_i = n$, forcing $d_i = m_i$.

Step 2 (if) and (b). If $d_i = m_i$, then β in (b) is independent by Theorem 5.8 with $\sum d_i = \sum m_i = n$ vectors, hence a basis of eigenvectors. Under the hypothesis of (b), Step 1 shows $d_i = m_i$, and the same argument applies. ■

Test for diagonalization. T is diagonalizable iff (1) its characteristic polynomial splits and (2) for each eigenvalue, $m_i = \dim E_{\lambda_i} = n - \operatorname{rank}(\mathsf{T} - \lambda_i \mathsf{I})$.

Example 5

$A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right)$: $\lambda = 3$ has multiplicity 2 but $\operatorname{rank}(A - 3I) = 2$, so $\dim E_3 = 1$; A is not diagonalizable.

Example 6

$\mathsf{T}(f) = f(1) + f'(0)x + (f'(0) + f''(0))x^2$ on $\mathsf{P}_2(\mathbb{R})$: in the standard basis $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, $f(t) = -(t-1)^2(t-2)$, $E_1 = \operatorname{span}\{1, -x + x^2\}$ (dimension 2), $E_2 = \operatorname{span}\{1 + x^2\}$. So T is diagonalizable with basis $\{1, -x + x^2, 1 + x^2\}$.

Example 7

$A = \left(\begin{smallmatrix} 0 & -2 \\ 1 & 3 \end{smallmatrix}\right)$ has eigenvalues $1, 2$ with eigenvectors $(-2, 1)^t, (-1, 1)^t$. With $Q = \left(\begin{smallmatrix} -2 & -1 \\ 1 & 1 \end{smallmatrix}\right)$, $A = QDQ^{-1}$, $D = \operatorname{diag}(1, 2)$, so

$$A^n = QD^nQ^{-1} = \begin{pmatrix} 2 - 2^n & 2 - 2^{n+1} \\ -1 + 2^n & -1 + 2^{n+1} \end{pmatrix}.$$

5.2.1 Systems of differential equations

Consider $x' = Ax$ with $A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{smallmatrix}\right)$, $x = (x_1, x_2, x_3)^t$ real functions. A has eigenvalues 2 (eigenvectors $(-1, 1, 0)$, $(-1, 0, 1)$) and 4 (eigenvector $(-1, -2, 1)$), so $Q^{-1}AQ = D = \operatorname{diag}(2, 2, 4)$ with Q having these columns. For $y = Q^{-1}x$, $y' = Q^{-1}x' = Dy$, i.e. $y_1' = 2y_1$, $y_2' = 2y_2$, $y_3' = 4y_3$, so $y_i = c_i e^{\lambda_i t}$. Hence

$$x = Qy = e^{2t}\left(c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right) + c_3 e^{4t} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}.$$

5.2.2 Direct sums

Definition (Sum of subspaces)

$W_1 + \cdots + W_k = \sum_i W_i = \{v_1 + \cdots + v_k : v_i \in W_i\}$, a subspace.

Example 8

In $\mathbb{R}^3$, the xy-plane W_1 and the yz-plane W_2 satisfy $\mathbb{R}^3 = W_1 + W_2$: $(a, b, c) = (a, 0, 0) + (0, b, c)$. The representation is not unique.

Definition (Direct sum)

$V = W_1 \oplus \cdots \oplus W_k$ if $V = \sum_i W_i$ and $W_j \cap \sum_{i \neq j} W_i = \{0\}$ for each j.

Example 9

$\mathbb{R}^4 = W_1 \oplus W_2 \oplus W_3$ for $W_1 = \{(a, b, 0, 0)\}$, $W_2 = \{(0, 0, c, 0)\}$, $W_3 = \{(0, 0, 0, d)\}$.

Theorem 5.10

For subspaces $W_1, \dots, W_k$ of finite-dimensional V, the following are equivalent:
(a) $V = W_1 \oplus \cdots \oplus W_k$;
(b) $V = \sum W_i$, and $v_1 + \cdots + v_k = 0$ with $v_i \in W_i$ implies all $v_i = 0$;
(c) each $v \in V$ is uniquely $v = v_1 + \cdots + v_k$ with $v_i \in W_i$;
(d) if γ_i is an ordered basis of W_i for each i, then $\gamma_1 \cup \cdots \cup \gamma_k$ is an ordered basis of V;
(e) for each i there is an ordered basis γ_i of W_i with $\gamma_1 \cup \cdots \cup \gamma_k$ an ordered basis of V.

Proof.

Step (a)⇒(b). If $\sum v_i = 0$, then $-v_j = \sum_{i \neq j} v_i \in W_j \cap \sum_{i \neq j} W_i = \{0\}$.

Step (b)⇒(c). If $v = \sum v_i = \sum w_i$, then $\sum (v_i - w_i) = 0$, so $v_i = w_i$.

Step (c)⇒(d). The union generates V since $V = \sum W_i$. If a linear combination of all the basis vectors is 0, group it as $\sum_i u_i = 0$ with $u_i \in W_i$ a combination of γ_i; uniqueness for $0 = \sum 0$ gives $u_i = 0$, and independence of γ_i gives zero coefficients.

Step (d)⇒(e). Bases exist.

Step (e)⇒(a). $V = \operatorname{span}(\bigcup \gamma_i) \subseteq \sum W_i$. If $v \in W_j \cap \sum_{i \neq j} W_i$, write v as a combination of γ_j and also of $\bigcup_{i \neq j} \gamma_i$; subtracting gives a relation among the vectors of the basis $\bigcup \gamma_i$, so all coefficients vanish and $v = 0$. ■

Theorem 5.11

T on finite-dimensional V is diagonalizable iff V is the direct sum of the eigenspaces of T.

Proof. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues. If T is diagonalizable, with γ_i a basis of E_{λ_i}, then $\bigcup \gamma_i$ is a basis of V (Theorem 5.9(b)), so $V = \bigoplus E_{\lambda_i}$ by Theorem 5.10(e)⇒(a). Conversely, if $V = \bigoplus E_{\lambda_i}$, then by (a)⇒(d) the union of bases of the E_{λ_i} is a basis of eigenvectors. ■

Example 10

$\mathsf{T}(a, b, c, d) = (a, b, 2c, 3d)$ on $\mathbb{R}^4$: $E_1 = W_1$, $E_2 = W_2$, $E_3 = W_3$ of Example 9, and $\mathbb{R}^4 = E_1 \oplus E_2 \oplus E_3$.

5.3 Matrix limits and Markov chains *

Motivation. Powers A^m describe repeated transitions, as in population movements or Markov chains. Whether A^m converges is decided by the eigenvalues. For a regular transition matrix the limit exists and every column of it is the unique stationary probability vector.

Definition (Matrix limit)

$A_m \to L$ ($n \times p$, complex entries) if $(A_m)_{ij} \to L_{ij}$ for all i, j; write $\lim_m A_m = L$.

Example 1

$$A_m = \begin{pmatrix} 1 - \frac{1}{m} & (\frac{3}{4})^m \\ \frac{3m^2}{m^2+1} + i\frac{2m+1}{m-1} & (1+\frac{1}{m})^m \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 3+2i & e \end{pmatrix}.$$

Theorem 5.12

If $A_m \to L$ $(n \times p)$, then $PA_m \to PL$ and $A_m Q \to LQ$ for any $P \in \mathrm{M}_{r\times n}(\mathbb{C})$, $Q \in \mathrm{M}_{p\times s}(\mathbb{C})$.

Proof. $(PA_m)_{ij} = \sum_k P_{ik}(A_m)_{kj} \to \sum_k P_{ik}L_{kj}$, a finite sum of convergent sequences; similarly for $A_m Q$. ∎

Corollary

If $A^m \to L$ and Q is invertible, then $(QAQ^{-1})^m \to QLQ^{-1}$.

Proof. $(QAQ^{-1})^m = QA^mQ^{-1}$; apply Theorem 5.12 twice. ∎

Let $S = \{\lambda \in \mathbb{C} : |\lambda| < 1\} \cup \{1\}$. For $\lambda \in \mathbb{C}$, λ^m converges iff $\lambda \in S$.

Theorem 5.13

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\lim A^m$ exists iff (a) every eigenvalue of A lies in S, and (b) if 1 is an eigenvalue, $\dim E_1$ equals the multiplicity of 1.

Proof [Supplement] (the book gives part of it and cites the rest; we use Theorems 5.23 and 5.24 of Section 5.4).

Step 1 (generalized eigenspaces). Over $\mathbb{C}$, $f(t) = \pm\prod_i (t-\lambda_i)^{m_i}$. Let $K_i = \mathrm{N}((A-\lambda_i I)^{m_i})$; it is L_A-invariant. The polynomials $g_i = \prod_{j\neq i}(t-\lambda_j)^{m_j}$ have no common zero, so $\sum_i p_i g_i = 1$ for some polynomials p_i (Appendix E). Put $P_i = p_i(A)g_i(A)$. Then $\sum P_i = I$, and $(A-\lambda_i I)^{m_i}P_i = \pm p_i(A)f(A) = O$ (Cayley–Hamilton), so $\mathrm{R}\,P_i \subseteq K_i$ and $\mathbb{C}^n = \sum K_i$. If $\sum v_i = 0$ with $v_i \in K_i$, apply $p_j(A)g_j(A)$: it kills v_i for $i \neq j$ (because $(t-\lambda_i)^{m_i} \mid g_j$), so $P_j v_j = 0$; since $v_j = \sum_i P_i v_j = P_j v_j$, $v_j = 0$. So $\mathbb{C}^n = \bigoplus K_i$ (Theorem 5.10).

Step 2 ($\dim K_i = m_i$). On K_i, $A - \lambda_i I$ is nilpotent, so λ_i is the only eigenvalue of $A|_{K_i}$. By Theorem 5.24, f is the product of the characteristic polynomials of the $A|_{K_i}$, so $\dim K_i = m_i$.

Step 3 (sufficiency). On K_i write $A = \lambda_i I + N$ with $N^{m_i} = O$. Then $A^m = \sum_{k<m_i}\binom{m}{k}\lambda_i^{m-k}N^k$. If $|\lambda_i| < 1$, then $\binom{m}{k}|\lambda_i|^{m-k} \to 0$ for each fixed k, so $A^m \to O$ on K_i. If $\lambda_i = 1$, (b) and $E_1 \subseteq K_1$, $\dim K_1 = m_1 = \dim E_1$ give $K_1 = E_1$, where $A = I$. Choosing a basis adapted to $\bigoplus K_i$, A is similar to a block-diagonal matrix whose powers converge; apply the Corollary to Theorem 5.12.

Step 4 (necessity of (a)). If $Av = \lambda v$, $v \neq 0$, and $L = \lim A^m$, then $Lv = \lim A^m v = \lim \lambda^m v$ (Theorem 5.12), so λ^m converges and $\lambda \in S$.

Step 5 (necessity of (b)). If $\dim E_1 < m_1 = \dim K_1$, then $A - I$ is nilpotent but nonzero on K_1, so some $w \in K_1$ has $u = (A-I)w \neq 0$ and $(A-I)u = 0$. Then $A^m w = w + mu$ diverges. ∎

Theorem 5.14

If every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in S and A is diagonalizable, then $\lim A^m$ exists.

Proof. $A = QDQ^{-1}$ with $D = \mathrm{diag}(\lambda_i)$, $\lambda_i \in S$, so $D^m = \mathrm{diag}(\lambda_i^m)$ converges, and $A^m = QD^mQ^{-1}$ converges by the Corollary to Theorem 5.12. ∎

Markov chains. A **transition matrix** (stochastic matrix) has nonnegative real entries and columns summing to 1: A_{ij} is the probability of moving from state j to state i in one step. A **probability vector** has nonnegative coordinates summing to 1. If P gives the initial distribution, A^mP gives the distribution after m steps, and $(A^m)_{ij}$ is the probability of going from j to i in m steps. Let $u = (1,\dots,1)^t$.

Theorem 5.15

Let M be $n \times n$ with nonnegative real entries and $v \in \mathbb{R}^n$ with nonnegative coordinates. Then
(a) M is a transition matrix iff $M^t u = u$;
(b) v is a probability vector iff $u^t v = (1)$.

Proof. $(M^t u)_j$ is the j-th column sum of M, and $u^t v$ is the coordinate sum of v. ■

Corollary

(a) Products of transition matrices are transition matrices; in particular so are their powers.
(b) A transition matrix times a probability vector is a probability vector.

Proof. Nonnegativity is preserved. (a) $(MN)^t u = N^t M^t u = N^t u = u$.
(b) $u^t(Mv) = (M^t u)^t v = u^t v = 1$. ■

Definition (Regular)

A transition matrix is **regular** if some power of it has only positive entries.

Example 2

$\left(\begin{smallmatrix} 0.90 & 0.02 \\ 0.10 & 0.98 \end{smallmatrix}\right)$ (city/suburb migration) is regular. A matrix with an absorbing state, i.e. a column e_1, is not, since the first column of every power is e_1.

Definition (Gerschgorin disks)

For $A \in \mathrm{M}_{n\times n}(\mathbb{C})$, $\rho_i(A) = \sum_j |A_{ij}|$ (row sums), $\nu_j(A) = \sum_i |A_{ij}|$ (column sums), $\rho(A) = \max_i \rho_i(A)$, $\nu(A) = \max_j \nu_j(A)$. The i-th **Gerschgorin disk** C_i has center A_{ii} and radius $r_i = \rho_i(A) - |A_{ii}|$.

Example 3

For $A = \left(\begin{smallmatrix} 1 & 2 & -3 \\ 0 & 4 & 1 \\ 2 & -1 & 5 \end{smallmatrix}\right)$: $\rho_i = 6, 5, 8$ and $\nu_j = 3, 7, 9$, so $\rho(A) = 8$, $\nu(A) = 9$.

Theorem 5.16 (Gerschgorin's disk theorem)

Every eigenvalue of $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ lies in some Gerschgorin disk.

Proof. Let $Av = \lambda v$, $v \neq 0$, and choose k with $|v_k|$ maximal. From $\lambda v_k = \sum_j A_{kj} v_j$, $|\lambda - A_{kk}||v_k| = |\sum_{j\neq k} A_{kj} v_j| \leq \sum_{j\neq k} |A_{kj}||v_k| = r_k |v_k|$, so $|\lambda - A_{kk}| \leq r_k$. ■

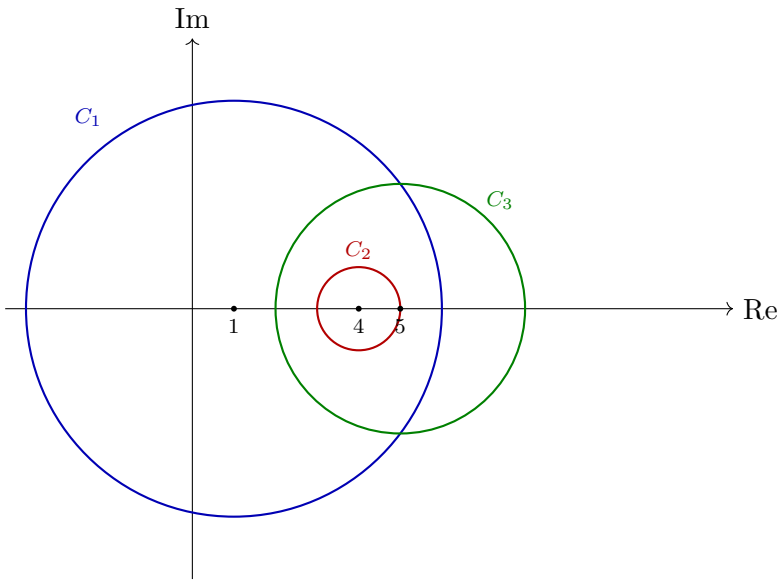


Figure. *Gerschgorin disks of the matrix of Example 3: centers* $1, 4, 5$*, radii* $5, 1, 3$*. All eigenvalues lie in their union.*

Corollary 1

$|\lambda| \le \rho(A)$ for every eigenvalue λ of A.

Proof. $|\lambda| \le |\lambda - A_{kk}| + |A_{kk}| \le r_k + |A_{kk}| = \rho_k(A) \le \rho(A)$. ■

Corollary 2

$|\lambda| \le \min\{\rho(A), \nu(A)\}$.

Proof. λ is an eigenvalue of A^t (same characteristic polynomial, Theorem 4.8), and $\rho(A^t) = \nu(A)$; apply Corollary 1 to both. ■

Corollary 3

Every eigenvalue λ of a transition matrix satisfies $|\lambda| \le 1$.

Proof. $\nu(A) = 1$; apply Corollary 2. ■

Theorem 5.17

Every transition matrix has 1 as an eigenvalue.

Proof. $A^t u = u$ (Theorem 5.15), so 1 is an eigenvalue of A^t, hence of A. ■

Theorem 5.18

Let $A \in \mathrm{M}_{n\times n}(\mathbb{C})$ have only positive entries, and let λ be an eigenvalue with $|\lambda| = \rho(A)$. Then $\lambda = \rho(A)$ and $\{u\}$ is a basis of E_λ.

Proof. Let $Av = \lambda v$, $v \neq 0$, $b = \max_j |v_j| = |v_k|$. Then

$$\rho(A)b = |\lambda||v_k| = \Big|\sum_j A_{kj}v_j\Big| \le \sum_j A_{kj}|v_j| \le \sum_j A_{kj}b = \rho_k(A)b \le \rho(A)b.$$

So equality holds throughout. Since all $A_{kj} > 0$, $|v_j| = b$ for all j, and the numbers $A_{kj}v_j$ all have the same argument (equality in the triangle inequality), so all v_j equal some $z \neq 0$: $v = zu$. Then $\lambda zu = Azu = z(\rho_1(A), \ldots, \rho_n(A))^t$, so every $\rho_i(A) = \lambda$, i.e. $\lambda = \rho(A)$. Every eigenvector for λ is a multiple of u. ■

Corollary 1

If A has positive entries and $|\lambda| = \nu(A)$ for an eigenvalue λ, then $\lambda = \nu(A)$ and $\dim E_\lambda = 1$.

Proof. Apply Theorem 5.18 to A^t ($\rho(A^t) = \nu(A)$), and note $\operatorname{rank}(A - \lambda I) = \operatorname{rank}(A^t - \lambda I)$. ■

Corollary 2

If A is a transition matrix with positive entries, every eigenvalue $\lambda \neq 1$ has $|\lambda| < 1$, and $\dim E_1 = 1$.

Proof. $\nu(A) = 1$. By Corollary 3 of Theorem 5.16, $|\lambda| \leq 1$, and if $|\lambda| = 1$ then $\lambda = 1$ by Corollary 1. Also $\dim E_1 = 1$ by Corollary 1. ■

Theorem 5.19

Let A be a regular transition matrix and λ an eigenvalue. Then (a) $|\lambda| \leq 1$;
(b) if $|\lambda| = 1$, then $\lambda = 1$ and $\dim E_\lambda = 1$.

Proof.

(a) Corollary 3 of Theorem 5.16.

(b) Let A^k be positive; then so is $A^{k+1} = A \cdot A^k$ (each row of A has a positive entry, since otherwise $A^k = A \cdot A^{k-1}$ would have a zero row). If $Av = \lambda v$, $|\lambda| = 1$, then λ^k and λ^{k+1} are eigenvalues of modulus 1 of positive transition matrices, so both equal 1 (Corollary 2 above), and $\lambda = \lambda^{k+1}/\lambda^k = 1$. Finally $E_1(A) \subseteq E_1(A^k)$, which has dimension 1. ■

Corollary

A diagonalizable regular transition matrix A has a limit $\lim A^m$.

Proof. Theorems 5.19 and 5.14. ■

Theorem 5.20

Let A be an $n \times n$ regular transition matrix. Then
(a) 1 has multiplicity 1 as an eigenvalue;
(b) $L = \lim A^m$ exists;
(c) L is a transition matrix;
(d) $AL = LA = L$;
(e) all columns of L equal the unique probability vector v with $Av = v$;
(f) $\lim A^m w = v$ for every probability vector w.

Proof.

(a) [Supplement] (The book defers this to Chapter 7.) Transition matrices have entries in $[0, 1]$, so $\{A^m w\}$ is bounded for each $w \in \mathbb{C}^n$. Suppose $(A - I)^2 w = 0$. Then $u' = (A - I)w \in E_1$ and $A^m w = w + mu'$, so boundedness forces $u' = 0$. Thus $\mathrm{N}((A - I)^2) = \mathrm{N}(A - I)$. Then $\mathrm{N}(A - I) \cap \mathrm{R}(A - I) = \{0\}$: if $x = (A - I)y$ and $(A - I)x = 0$, then $y \in \mathrm{N}(A - I)$ and $x = 0$. By

the dimension theorem $\mathbb{C}^n = \mathrm{N}(A-I) \oplus \mathrm{R}(A-I)$, both A-invariant, and 1 is not an eigenvalue of A on $\mathrm{R}(A-I)$. By Theorem 5.24 the multiplicity of 1 is $\dim \mathrm{N}(A-I) = 1$ (Theorem 5.19).

(b) By (a) and Theorem 5.19, the conditions of Theorem 5.13 hold.

(c) A^m is a transition matrix, so $u^t L = \lim u^t A^m = u^t$; entries stay nonnegative.

(d) $AL = \lim A^{m+1} = L$ by Theorem 5.12, and likewise $LA = L$.

(e) By (d) each column of L lies in E_1, and by (c) it is a probability vector. As $\dim E_1 = 1$, there is exactly one probability vector v in E_1.

(f) $y = \lim A^m w = Lw$ is a probability vector with $Ay = ALw = Lw = y$, so $y = v$ by (e). ∎

v is the **fixed probability vector** (**stationary vector**) of A.

Example 4

Preferences (bread, wine, "thou") with initial distribution $P = (0.50, 0.30, 0.20)^t$ and transition matrix $A = \begin{pmatrix} 0.40 & 0.20 & 0.20 \\ 0.10 & 0.70 & 0.20 \\ 0.50 & 0.10 & 0.60 \end{pmatrix}$. A is positive, with eigenvalues $1, 0.5, 0.2$, and $AP = (0.30, 0.30, 0.40)^t$. The stationary vector is $v = (0.25, 0.35, 0.40)^t$: in the long run 25%, 35%, 40%, whatever the initial distribution.

Example 5

Crop rotation (corn, soybeans, wheat; half of each crop's land switches to each other crop): $A = \frac{1}{2}\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, $P = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})^t$ from $300, 200, 100$ acres. A^2 is positive, the eigenvalues are $1, -\frac{1}{2}, -\frac{1}{2}$, and the stationary vector is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^t$. Eventually $600 \cdot \frac{1}{3} = 200$ acres go to each crop.

An **absorbing** state (one never left) leads to non-regular chains; the limit can still exist by Theorem 5.13.

5.4 Invariant subspaces and the Cayley–Hamilton theorem

Motivation. Even non-diagonalizable operators can be broken into pieces on invariant subspaces. Cyclic subspaces $\mathrm{span}\{v, \mathsf{T}v, \mathsf{T}^2 v, \dots\}$ have computable characteristic polynomials, which yields the Cayley–Hamilton theorem: every operator is annihilated by its characteristic polynomial.

Definition (Invariant subspace)

$W \subseteq V$ is **T-invariant** if $\mathsf{T}(W) \subseteq W$. Then $\mathsf{T}_W : W \to W$ denotes the restriction.

Example 1

$\{0\}$, V, $\mathrm{R}(\mathsf{T})$, $\mathrm{N}(\mathsf{T})$ and every eigenspace E_λ are T-invariant.

Example 2

For $\mathsf{T}(a,b,c) = (a+b, b+c, 0)$, the xy-plane $\{(x,y,0)\}$ and the x-axis $\{(x,0,0)\}$ are T-invariant.

Definition (Cyclic subspace)

For $x \in V$, the **T-cyclic subspace generated by** x is $W = \operatorname{span}\{x, \mathsf{T}(x), \mathsf{T}^2(x), \dots\}$, the smallest T-invariant subspace containing x.

Example 3

For $\mathsf{T}(a,b,c) = (-b+c, a+c, 3c)$: $\mathsf{T}(e_1) = e_2$, $\mathsf{T}(e_2) = -e_1$, so the cyclic subspace generated by e_1 is $\operatorname{span}\{e_1, e_2\}$.

Example 4

$\mathsf{T}(f) = f'$ on $\mathsf{P}(\mathbb{R})$: the cyclic subspace generated by x^2 is $\operatorname{span}\{x^2, 2x, 2\} = \mathsf{P}_2(\mathbb{R})$.

Theorem 5.21

If W is T-invariant (V finite-dimensional), the characteristic polynomial of T_W divides that of T.

Proof. Extend a basis γ of W (k vectors) to a basis β of V. Then $[\mathsf{T}]_\beta = \begin{pmatrix} B_1 & B_2 \\ O & B_3 \end{pmatrix}$ with $B_1 = [\mathsf{T}_W]_\gamma$, and $f(t) = \det(B_1 - tI)\det(B_3 - tI)$ (Section 4.4, fact 8). ■

Example 5

$\mathsf{T}(a,b,c,d) = (a+b+2c-d,\ b+d,\ 2c-d,\ c+d)$ and $W = \operatorname{span}\{e_1, e_2\}$, which is invariant. $[\mathsf{T}_W] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has characteristic polynomial $(t-1)^2$, which divides $f(t) = (t-1)^2(t^2-3t+3)$.

Theorem 5.22

Let W be the T-cyclic subspace generated by $v \neq 0$ and $k = \dim W$ (V finite-dimensional). Then
(a) $\{v, \mathsf{T}(v), \dots, \mathsf{T}^{k-1}(v)\}$ is a basis of W;
(b) if $a_0 v + a_1 \mathsf{T}(v) + \cdots + a_{k-1}\mathsf{T}^{k-1}(v) + \mathsf{T}^k(v) = 0$, the characteristic polynomial of T_W is $(-1)^k(a_0 + a_1 t + \cdots + a_{k-1}t^{k-1} + t^k)$.

Proof.

(a) Let j be the largest integer with $\beta = \{v, \dots, \mathsf{T}^{j-1}(v)\}$ independent ($j \geq 1$ as $v \neq 0$). Then $\mathsf{T}^j(v) \in Z = \operatorname{span}(\beta)$ by Theorem 1.7. Z is T-invariant: T maps $\mathsf{T}^i(v)$ to $\mathsf{T}^{i+1}(v) \in Z$ for $i < j$. So $Z \supseteq W$ (the smallest invariant subspace containing v), $Z = W$, and $j = k$.

(b) In the basis β of (a), $[\mathsf{T}_W]_\beta$ has $e_2, \dots, e_k$ as its first $k-1$ columns and $(-a_0, \dots, -a_{k-1})^t$ as its last column. Expanding $\det([\mathsf{T}_W]_\beta - tI)$ along the last column (or by induction on k) gives $(-1)^k(a_0 + a_1 t + \cdots + t^k)$. ■

Example 6

For T of Example 3 and $W = \operatorname{span}\{e_1, e_2\}$: $\mathsf{T}^2(e_1) = -e_1$, so $e_1 + \mathsf{T}^2(e_1) = 0$ and the characteristic polynomial of T_W is $t^2 + 1$.

Theorem 5.23 (Cayley–Hamilton)

If $f(t)$ is the characteristic polynomial of T on finite-dimensional V, then $f(\mathsf{T}) = \mathsf{T}_0$.

Proof. We show $f(\mathsf{T})(v) = 0$ for every $v \neq 0$. Let W be the T-cyclic subspace generated by v, $k = \dim W$, and a_i as in Theorem 5.22(b). Then $g(t) = (-1)^k(a_0 + \cdots + t^k)$ is the characteristic polynomial of T_W, and $g(\mathsf{T})(v) = (-1)^k(a_0 v + \cdots + \mathsf{T}^k(v)) = 0$. By Theorem 5.21, $f = qg$, so $f(\mathsf{T})(v) = q(\mathsf{T})(g(\mathsf{T})(v)) = 0$. ■

Example 7

$\mathsf{T}(a,b) = (a+2b, -2a+b)$, $A = \left(\begin{smallmatrix} 1 & 2 \\ -2 & 1 \end{smallmatrix}\right)$, $f(t) = t^2 - 2t + 5$: $A^2 - 2A + 5I = \left(\begin{smallmatrix} -3 & 4 \\ -4 & -3 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 4 \\ -4 & 2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right) = O$.

Corollary (Cayley–Hamilton for matrices)

If f is the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, then $f(A) = O$.

Proof. $f(\mathsf{L}_A) = \mathsf{L}_{f(A)}$ is the zero map by Theorem 5.23; apply Theorem 2.15(b). ■

5.4.1 Invariant subspaces and direct sums

Theorem 5.24

If $V = W_1 \oplus \cdots \oplus W_k$ with each W_i T-invariant and f_i the characteristic polynomial of T_{W_i}, then $f_1 \cdots f_k$ is the characteristic polynomial of T.

Proof. Induction on k; $k = 1$ is trivial. For $k = 2$, take bases β_1, β_2 of W_1, W_2; $\beta = \beta_1 \cup \beta_2$ is a basis (Theorem 5.10) and $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} [\mathsf{T}_{W_1}] & O \\ O & [\mathsf{T}_{W_2}] \end{smallmatrix}\right)$, whose characteristic polynomial is $f_1 f_2$ (Section 4.4, fact 8). For $k > 2$, $W = W_1 \oplus \cdots \oplus W_{k-1}$ is T-invariant with $V = W \oplus W_k$; apply the case $k = 2$ and induction to T_W. ■

Example 8

$\mathsf{T}(a,b,c,d) = (2a-b, a+b, c-d, c+d)$, $W_1 = \mathrm{span}\{e_1, e_2\}$, $W_2 = \mathrm{span}\{e_3, e_4\}$: $f_1 = t^2 - 3t + 3$, $f_2 = t^2 - 2t + 2$, and $f = f_1 f_2$.

Definition (Direct sum of matrices)

For $B_1 \in \mathrm{M}_{m\times m}(\mathbb{F})$, $B_2 \in \mathrm{M}_{n\times n}(\mathbb{F})$, $B_1 \oplus B_2 = \left(\begin{smallmatrix} B_1 & O \\ O & B_2 \end{smallmatrix}\right)$; inductively $B_1 \oplus \cdots \oplus B_k = (B_1 \oplus \cdots \oplus B_{k-1}) \oplus B_k$, block diagonal.

Example 9

$\left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right) \oplus (3) \oplus \left(\begin{smallmatrix} 1 & 2 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ is the 6×6 block-diagonal matrix with these blocks on the diagonal.

Theorem 5.25

If $V = W_1 \oplus \cdots \oplus W_k$ with T-invariant W_i, ordered bases β_i of W_i, $\beta = \beta_1 \cup \cdots \cup \beta_k$, $A = [\mathsf{T}]_\beta$ and $B_i = [\mathsf{T}_{W_i}]_{\beta_i}$, then $A = B_1 \oplus \cdots \oplus B_k$.

Proof. Induction on k. For $k = 2$: if $v \in \beta_1$, then $\mathsf{T}(v) \in W_1$ is a combination of β_1 alone, with coordinates the corresponding column of B_1; similarly for β_2. So $A = B_1 \oplus B_2$. For $k > 2$ apply this to $W = W_1 \oplus \cdots \oplus W_{k-1}$ and W_k, and use induction on T_W. ■

Chapter 6. Inner Product Spaces

6.1 Inner products and norms

Motivation. Lengths and angles need more than vector-space structure. An inner product, modelled on the dot product, supplies them over $\mathbb{R}$ and $\mathbb{C}$, together with the Cauchy–Schwarz and triangle inequalities, and with the notion of orthogonality that drives the rest of the chapter.

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\bar{c}$ is the complex conjugate.

Definition (Inner product)

An **inner product** on V assigns to $x, y \in V$ a scalar $\langle x, y\rangle \in \mathbb{F}$ such that for all $x, y, z \in V$, $c \in \mathbb{F}$:
(a) $\langle x + z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$;
(b) $\langle cx, y\rangle = c\langle x, y\rangle$;
(c) $\overline{\langle x, y\rangle} = \langle y, x\rangle$;
(d) $\langle x, x\rangle > 0$ if $x \neq 0$. V with an inner product is an **inner product space**: **complex** if $\mathbb{F} = \mathbb{C}$, **real** if $\mathbb{F} = \mathbb{R}$.

By (a), (b), $\langle \sum a_i x_i, y\rangle = \sum a_i \langle x_i, y\rangle$. Over $\mathbb{R}$, (c) reads $\langle x, y\rangle = \langle y, x\rangle$.

Example 1

The **standard inner product** on $\mathbb{F}^n$: $\langle x, y\rangle = \sum_i a_i \bar{b}_i$ for $x = (a_i)$, $y = (b_i)$; for $\mathbb{F} = \mathbb{R}$ it is the dot product $x \cdot y$.

Example 2

If $\langle \cdot, \cdot \rangle$ is an inner product and $r > 0$, then $r\langle x, y\rangle$ is one; for $r \leq 0$, (d) fails.

Example 3

On $C([0,1])$ (real), $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product; (d) holds because $f^2 \geq 0$ is continuous and not identically 0 when $f \neq 0$.

Definition (Conjugate transpose)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $A^* \in \mathrm{M}_{n\times m}(\mathbb{F})$ has $(A^*)_{ij} = \overline{A_{ji}}$ (the **adjoint**); for real A, $A^* = A^t$.

Example 4

$\begin{pmatrix} i & 1+2i \\ 2 & 3+4i \end{pmatrix}^* = \begin{pmatrix} -i & 2 \\ 1-2i & 3-4i \end{pmatrix}$.

Example 5

On $\mathrm{M}_{n\times n}(\mathbb{F})$, $\langle A, B\rangle = \mathrm{tr}(B^*A)$ is an inner product (the **Frobenius inner product**): $\langle A, A\rangle = \sum_{i,k} |A_{ki}|^2 > 0$ for $A \neq O$.

The space H of continuous complex functions on $[0,2\pi]$ with $\langle f,g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(t)\overline{g(t)}\,dt$ is an inner product space.

Theorem 6.1

For $x,y,z \in V$, $c \in \mathbb{F}$:
(a) $\langle x, y+z\rangle = \langle x,y\rangle + \langle x,z\rangle$;
(b) $\langle x, cy\rangle = \bar{c}\langle x,y\rangle$;
(c) $\langle x,0\rangle = \langle 0,x\rangle = 0$;
(d) $\langle x,x\rangle = 0$ iff $x = 0$;
(e) if $\langle x,y\rangle = \langle x,z\rangle$ for all x, then $y = z$.

Proof.

(a) $\langle x, y+z\rangle = \overline{\langle y+z, x\rangle} = \overline{\langle y,x\rangle} + \overline{\langle z,x\rangle} = \langle x,y\rangle + \langle x,z\rangle$.

(b) $\langle x, cy\rangle = \overline{c\langle y,x\rangle} = \bar{c}\langle x,y\rangle$.

(c) $\langle 0,x\rangle = \langle 0\cdot 0, x\rangle = 0$, then conjugate.

(d) From (c) and axiom (d).

(e) $\langle x, y-z\rangle = 0$ for all x; take $x = y-z$ and use (d). ∎

An inner product is linear in the first and **conjugate linear** in the second variable.

Definition (Norm)

$\|x\| = \sqrt{\langle x,x\rangle}$ is the **norm** (length) of x.

Example 6

On $\mathbb{F}^n$, $\|(a_1,\dots,a_n)\| = \left(\sum |a_i|^2\right)^{1/2}$, the Euclidean length.

Theorem 6.2

For $x,y \in V$, $c \in \mathbb{F}$:
(a) $\|cx\| = |c|\|x\|$;
(b) $\|x\| = 0$ iff $x = 0$; always $\|x\| \geq 0$;
(c) (Cauchy–Schwarz) $|\langle x,y\rangle| \leq \|x\|\|y\|$;
(d) (triangle inequality) $\|x+y\| \leq \|x\| + \|y\|$.

Proof.

(a) $\|cx\|^2 = c\bar{c}\langle x,x\rangle$.

(b) Theorem 6.1(d).

(c) Trivial if $y = 0$. Otherwise, for $c = \langle x,y\rangle/\langle y,y\rangle$,

$$0 \leq \|x - cy\|^2 = \langle x,x\rangle - \bar{c}\langle x,y\rangle - c\langle y,x\rangle + c\bar{c}\langle y,y\rangle = \|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2}.$$

(d) $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x,y\rangle + \|y\|^2 \leq \|x\|^2 + 2|\langle x,y\rangle| + \|y\|^2 \leq (\|x\| + \|y\|)^2$ by (c). ∎

Example 7

On $\mathbb{F}^n$ with the standard inner product, (c) and (d) read

$$\left|\sum a_i\bar{b}_i\right| \le \left(\sum |a_i|^2\right)^{1/2}\left(\sum |b_i|^2\right)^{1/2}, \qquad \left(\sum |a_i+b_i|^2\right)^{1/2} \le \left(\sum |a_i|^2\right)^{1/2} + \left(\sum |b_i|^2\right)^{1/2}.$$

In $\mathbb{R}^2, \mathbb{R}^3$, $\langle x, y\rangle = \|x\|\|y\|\cos\theta$, θ the angle between x and y.

Definition (Orthogonality)

x, y are **orthogonal** (perpendicular) if $\langle x, y\rangle = 0$. $S \subseteq V$ is **orthogonal** if distinct vectors of S are orthogonal, and **orthonormal** if moreover all vectors are **unit vectors** ($\|x\| = 1$). Replacing $x \neq 0$ by $x/\|x\|$ is **normalizing**.

Example 8

$\{(1,1,0),(1,-1,1),(-1,1,2)\}$ is orthogonal in $\mathbb{F}^3$; normalizing gives the orthonormal set $\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{3}}(1,-1,1), \frac{1}{\sqrt{6}}(-1,1,2)\}$.

Example 9

In H, $f_n(t) = e^{int}$ ($n \in \mathbb{Z}$) form an orthonormal set: for $m \neq n$, $\langle f_m, f_n\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)t}\,dt = \frac{1}{2\pi i(m-n)} e^{i(m-n)t}\Big|_0^{2\pi} = 0$, and $\langle f_n, f_n\rangle = 1$.

6.2 The Gram–Schmidt orthogonalization process and orthogonal complements

Motivation. In an orthonormal basis coordinates are just inner products. Gram–Schmidt converts any basis into an orthonormal one, and orthogonal projection onto a subspace gives the closest point. This yields orthogonal complements and best approximations.

Definition (Orthonormal basis)

An **orthonormal basis** is an ordered basis that is orthonormal.

Example 1

The standard ordered basis of $\mathbb{F}^n$ is orthonormal.

Example 2

$\{\frac{1}{\sqrt{5}}(1,2), \frac{1}{\sqrt{5}}(2,-1)\}$ is an orthonormal basis of $\mathbb{R}^2$.

Theorem 6.3

If $S = \{v_1, \ldots, v_k\}$ is orthogonal with nonzero vectors and $y \in \operatorname{span}(S)$, then $y = \sum_{i=1}^{k} \frac{\langle y, v_i\rangle}{\|v_i\|^2} v_i$.

Proof. If $y = \sum a_i v_i$, then $\langle y, v_j\rangle = \sum a_i \langle v_i, v_j\rangle = a_j \|v_j\|^2$. ∎

Corollary 1

If S is orthonormal and $y \in \operatorname{span}(S)$, then $y = \sum \langle y, v_i\rangle v_i$.

Proof. $\|v_i\| = 1$ in Theorem 6.3. ∎

Corollary 2

An orthogonal set of nonzero vectors is linearly independent.

Proof. If $\sum a_i v_i = 0$, Theorem 6.3 with $y = 0$ gives $a_i = \langle 0, v_i\rangle / \|v_i\|^2 = 0$. ∎

Example 3

By Corollary 2 the orthonormal set $\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{3}}(1,-1,1), \frac{1}{\sqrt{6}}(-1,1,2)\}$ of Example 8 of Section 6.1 is a basis of $\mathbb{R}^3$, and for $x = (2,1,3)$ the coefficients are $\frac{3}{\sqrt{2}}$, $\frac{4}{\sqrt{3}}$, $\frac{5}{\sqrt{6}}$. Also H is infinite-dimensional (Example 9 of Section 6.1).

Theorem 6.4 (Gram–Schmidt)

Let $S = \{w_1, \dots, w_n\}$ be linearly independent. Put $v_1 = w_1$ and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j\rangle}{\|v_j\|^2} v_j \qquad (2 \le k \le n). \tag{6.1}$$

Then $S' = \{v_1, \dots, v_n\}$ is an orthogonal set of nonzero vectors with $\operatorname{span}(S') = \operatorname{span}(S)$.

Proof. Induction on n; let $S_k = \{w_1, \dots, w_k\}$, $S'_k = \{v_1, \dots, v_k\}$, and assume S'_{k-1} is orthogonal, of nonzero vectors, with $\operatorname{span} S'_{k-1} = \operatorname{span} S_{k-1}$. If $v_k = 0$, then $w_k \in \operatorname{span} S'_{k-1} = \operatorname{span} S_{k-1}$, contradicting independence. For $i < k$, by (6.1) and orthogonality, $\langle v_k, v_i\rangle = \langle w_k, v_i\rangle - \frac{\langle w_k, v_i\rangle}{\|v_i\|^2}\|v_i\|^2 = 0$. Finally $\operatorname{span} S'_k \subseteq \operatorname{span} S_k$ by (6.1), and both have dimension k (S'_k is independent by Corollary 2), so they are equal. ∎

Example 4

In $\mathbb{R}^4$, $w_1 = (1,0,1,0)$, $w_2 = (1,1,1,1)$, $w_3 = (0,1,2,1)$: $v_1 = w_1$, $v_2 = w_2 - \frac{2}{2}v_1 = (0,1,0,1)$, $v_3 = w_3 - \frac{2}{2}v_1 - \frac{2}{2}v_2 = (-1,0,1,0)$. Normalizing gives the orthonormal basis $\frac{1}{\sqrt{2}}(1,0,1,0)$, $\frac{1}{\sqrt{2}}(0,1,0,1)$, $\frac{1}{\sqrt{2}}(-1,0,1,0)$ of $\operatorname{span}\{w_1, w_2, w_3\}$.

Example 5

On $\mathsf{P}(\mathbb{R})$ with $\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt$, Gram–Schmidt applied to $1, x, x^2$ gives 1, x, $x^2 - \frac{1}{3}$, and normalizing,

$$u_1 = \frac{1}{\sqrt{2}}, \qquad u_2 = \sqrt{\tfrac{3}{2}}\,x, \qquad u_3 = \sqrt{\tfrac{5}{8}}\,(3x^2 - 1).$$

Continuing yields (multiples of) the **Legendre polynomials**.

Theorem 6.5

A nonzero finite-dimensional inner product space V has an orthonormal basis $\beta = \{v_1, \ldots, v_n\}$, and $x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i$ for every $x \in V$.

Proof. Apply Theorem 6.4 to any basis and normalize (Corollary 2); the expansion is Corollary 1 of Theorem 6.3. ∎

Example 6

For $f = 1 + 2x + 3x^2$ in the basis of Example 5: $\langle f, u_1 \rangle = 2\sqrt{2}$, $\langle f, u_2 \rangle = \frac{2\sqrt{6}}{3}$, $\langle f, u_3 \rangle = \frac{2\sqrt{10}}{5}$, so $f = 2\sqrt{2}u_1 + \frac{2\sqrt{6}}{3}u_2 + \frac{2\sqrt{10}}{5}u_3$.

Corollary

If $\beta = \{v_1, \ldots, v_n\}$ is an orthonormal basis and $A = [\mathsf{T}]_\beta$, then $A_{ij} = \langle \mathsf{T}(v_j), v_i \rangle$.

Proof. $\mathsf{T}(v_j) = \sum_i \langle \mathsf{T}(v_j), v_i \rangle v_i$ by Theorem 6.5. ∎

Definition (Fourier coefficients)

For an orthonormal subset β (possibly infinite), the **Fourier coefficients** of x are the $\langle x, y \rangle$, $y \in \beta$.

Example 7

For $f(t) = t$ in H and $f_n = e^{int}$: $\langle f, f_n \rangle = \frac{-1}{in} = \frac{i}{n}$ for $n \neq 0$ and $\langle f, 1 \rangle = \pi$. By Bessel's inequality ($\|x\|^2 \geq \sum |\langle x, v_i \rangle|^2$ for orthonormal $\{v_i\}$, which follows from Theorem 6.6 below) and $\|f\|^2 = \frac{4}{3}\pi^2$, $\frac{4}{3}\pi^2 \geq 2\sum_{n=1}^{k} \frac{1}{n^2} + \pi^2$, so $\sum_{n \geq 1} \frac{1}{n^2} \leq \frac{\pi^2}{6}$ (in fact equality holds).

Definition (Orthogonal complement)

For nonempty $S \subseteq V$, $S^\perp = \{x \in V : \langle x, y \rangle = 0 \text{ for all } y \in S\}$, a subspace.

Example 8

$\{0\}^\perp = V$ and $V^\perp = \{0\}$.

Example 9

In $\mathbb{R}^3$, $\{e_3\}^\perp$ is the xy-plane.

Theorem 6.6

Let W be a finite-dimensional subspace of V and $y \in V$. There are unique $u \in W$, $z \in W^\perp$ with $y = u + z$. If $\{v_1, \ldots, v_k\}$ is an orthonormal basis of W, then $u = \sum_{i=1}^{k} \langle y, v_i \rangle v_i$.

Proof. Put $u = \sum \langle y, v_i \rangle v_i$, $z = y - u$. Then $\langle z, v_j \rangle = \langle y, v_j \rangle - \langle y, v_j \rangle = 0$, so $z \in W^\perp$. If also $y = u' + z'$, then $u - u' = z' - z \in W \cap W^\perp = \{0\}$ (a vector orthogonal to itself is 0). ∎

Corollary

u is the unique vector of W closest to y: $\|y-x\| \geq \|y-u\|$ for $x \in W$, with equality iff $x = u$.

Proof. $y - x = z + (u - x)$ with $u - x \in W$ orthogonal to z, so $\|y-x\|^2 = \|z\|^2 + \|u-x\|^2 \geq \|z\|^2 = \|y-u\|^2$ (Pythagoras: $\|a+b\|^2 = \|a\|^2 + \|b\|^2$ if $\langle a, b\rangle = 0$), with equality iff $u = x$. ■

u is the **orthogonal projection** of y on W.

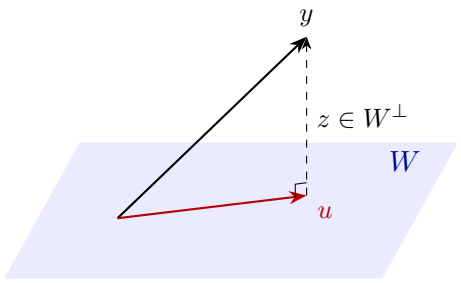


Figure. *Orthogonal projection: $y = u + z$ with $u \in W$, $z \perp W$; u is the point of W closest to y.*

Example 10

In $\mathsf{P}_3(\mathbb{R})$ with $\langle f, g\rangle = \int_{-1}^{1} fg$, the orthogonal projection of x^3 on $\mathsf{P}_2(\mathbb{R})$ is $\sum \langle x^3, u_i\rangle u_i = \frac{\sqrt{6}}{5} \cdot \sqrt{\frac{3}{2}}x = \frac{3}{5}x$ ($\langle x^3, u_1\rangle = \langle x^3, u_3\rangle = 0$).

Theorem 6.7

Let $S = \{v_1, \ldots, v_k\}$ be orthonormal in n-dimensional V.
(a) S extends to an orthonormal basis $\{v_1, \ldots, v_n\}$ of V.
(b) If $W = \operatorname{span}(S)$, then $\{v_{k+1}, \ldots, v_n\}$ is an orthonormal basis of $W^{\perp}$.
(c) For any subspace W, $\dim V = \dim W + \dim W^{\perp}$.

Proof.

(a) Extend S to a basis $\{v_1, \ldots, v_k, w_{k+1}, \ldots, w_n\}$ and apply Gram–Schmidt. The first k vectors are unchanged (they are already orthonormal), then normalize the rest.

(b) $S_1 = \{v_{k+1}, \ldots, v_n\} \subseteq W^{\perp}$ is orthonormal, hence independent. If $x \in W^{\perp}$, then $x = \sum_{i=1}^{n} \langle x, v_i\rangle v_i = \sum_{i>k} \langle x, v_i\rangle v_i$.

(c) Apply (a), (b) to an orthonormal basis of W. ■

Example 11

For $W = \operatorname{span}\{e_1, e_2\} \subseteq \mathbb{F}^3$, $(a, b, c) \in W^{\perp}$ iff $a = b = 0$, so $W^{\perp} = \operatorname{span}\{e_3\}$.

6.3 The adjoint of a linear operator

Motivation. Every linear functional on a finite-dimensional inner product space is $\langle \cdot, y\rangle$ for a unique y. This produces the adjoint T^*, characterised by $\langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}^* y\rangle$ and represented by the conjugate transpose. Its first payoffs are least squares and minimal solutions.

Theorem 6.8

If V is finite-dimensional and $g : V \to \mathbb{F}$ is linear, there is a unique $y \in V$ with $g(x) = \langle x, y\rangle$ for all x.

Proof. With an orthonormal basis $\{v_1, \ldots, v_n\}$ put $y = \sum_i \overline{g(v_i)} v_i$. Then $h(x) = \langle x, y\rangle$ is linear with $h(v_j) = \langle v_j, y\rangle = g(v_j)$, so $h = g$ (Corollary to Theorem 2.6). Uniqueness: Theorem 6.1(e). ■

Example 1

$g(a_1, a_2) = 2a_1 + a_2$ on $\mathbb{R}^2$: $y = 2e_1 + e_2 = (2, 1)$.

Theorem 6.9

For a linear operator T on finite-dimensional V there is a unique $\mathsf{T}^* : V \to V$ with $\langle \mathsf{T}(x), y\rangle = \langle x, \mathsf{T}^*(y)\rangle$ for all x, y; T^* is linear.

Proof. For fixed y, $g(x) = \langle \mathsf{T}(x), y\rangle$ is linear, so $g(x) = \langle x, y'\rangle$ for a unique y' (Theorem 6.8); set $\mathsf{T}^*(y) = y'$. Linearity: $\langle x, \mathsf{T}^*(cy_1 + y_2)\rangle = \langle \mathsf{T}x, cy_1 + y_2\rangle = \bar{c}\langle \mathsf{T}x, y_1\rangle + \langle \mathsf{T}x, y_2\rangle = \langle x, c\mathsf{T}^* y_1 + \mathsf{T}^* y_2\rangle$, so $\mathsf{T}^*(cy_1 + y_2) = c\mathsf{T}^*(y_1) + \mathsf{T}^*(y_2)$ by Theorem 6.1(e). Uniqueness: Theorem 6.1(e). ■

T^* is the **adjoint** of T; also $\langle x, \mathsf{T}(y)\rangle = \langle \mathsf{T}^*(x), y\rangle$. (In infinite dimensions an adjoint need not exist.)

Theorem 6.10

If β is an orthonormal basis of finite-dimensional V, then $[\mathsf{T}^*]_\beta = ([\mathsf{T}]_\beta)^*$.

Proof. By the Corollary to Theorem 6.5, $([\mathsf{T}^*]_\beta)_{ij} = \langle \mathsf{T}^*(v_j), v_i\rangle = \overline{\langle v_i, \mathsf{T}^*(v_j)\rangle} = \overline{\langle \mathsf{T}(v_i), v_j\rangle} = \overline{([\mathsf{T}]_\beta)_{ji}}$. ■

Corollary

$\mathsf{L}_{A^*} = (\mathsf{L}_A)^*$ for $A \in \mathrm{M}_{n\times n}(\mathbb{F})$.

Proof. With the standard (orthonormal) basis β, $[(\mathsf{L}_A)^*]_\beta = A^* = [\mathsf{L}_{A^*}]_\beta$. ■

Example 2

$\mathsf{T}(a_1, a_2) = (2ia_1 + 3a_2, a_1 - a_2)$ on $\mathbb{C}^2$: $[\mathsf{T}] = \left(\begin{smallmatrix} 2i & 3 \\ 1 & -1 \end{smallmatrix}\right)$, so $[\mathsf{T}^*] = \left(\begin{smallmatrix} -2i & 1 \\ 3 & -1 \end{smallmatrix}\right)$ and $\mathsf{T}^*(a_1, a_2) = (-2ia_1 + a_2, 3a_1 - a_2)$.

Theorem 6.11

For linear operators T, U on V having adjoints:
(a) $(\mathsf{T} + \mathsf{U})^* = \mathsf{T}^* + \mathsf{U}^*$;
(b) $(c\mathsf{T})^* = \bar{c}\mathsf{T}^*$;
(c) $(\mathsf{TU})^* = \mathsf{U}^*\mathsf{T}^*$;
(d) $\mathsf{T}^{**} = \mathsf{T}$;
(e) $\mathsf{I}^* = \mathsf{I}$.

Proof. Each follows from uniqueness in Theorem 6.9 and Theorem 6.1(e). (a) $\langle (\mathsf{T} + \mathsf{U})x, y\rangle = \langle x, \mathsf{T}^* y + \mathsf{U}^* y\rangle$.

(b) $\langle c\mathsf{T}x, y\rangle = \langle x, \bar{c}\mathsf{T}^*y\rangle$.

(c) $\langle \mathsf{T}\mathsf{U}x, y\rangle = \langle \mathsf{U}x, \mathsf{T}^*y\rangle = \langle x, \mathsf{U}^*\mathsf{T}^*y\rangle$.

(d) $\langle \mathsf{T}^*x, y\rangle = \overline{\langle y, \mathsf{T}^*x\rangle} = \overline{\langle \mathsf{T}y, x\rangle} = \langle x, \mathsf{T}y\rangle$.

(e) Clear. ■

Corollary

For $n \times n$ matrices: $(A+B)^* = A^* + B^*$, $(cA)^* = \bar{c}A^*$, $(AB)^* = B^*A^*$, $A^{**} = A$, $I^* = I$.

Proof. Apply Theorem 6.11 to $\mathsf{L}_A, \mathsf{L}_B$ and use the Corollary to Theorem 6.10. (Or compute directly; this works for rectangular matrices too.) ■

6.3.1 Least squares approximation

Given data $(t_1, y_1), \ldots, (t_m, y_m)$, the **least squares line** $y = ct+d$ minimises $E = \sum_i (y_i - ct_i - d)^2 = \|y - Ax\|^2$ with $A = \begin{pmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix}$, $x = \binom{c}{d}$. Write $\langle \cdot, \cdot \rangle_n$ for the standard inner product on $\mathbb{F}^n$.

Lemma 1

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $x \in \mathbb{F}^n$, $y \in \mathbb{F}^m$: $\langle Ax, y\rangle_m = \langle x, A^*y\rangle_n$.

Proof. $\langle Ax, y\rangle_m = y^*Ax = (A^*y)^*x = \langle x, A^*y\rangle_n$, viewing 1×1 matrices as scalars. ■

Lemma 2

$\mathrm{rank}(A^*A) = \mathrm{rank}(A)$ for $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.

Proof. If $A^*Ax = 0$, then $0 = \langle A^*Ax, x\rangle_n = \langle Ax, Ax\rangle_m$, so $Ax = 0$. Hence $\mathrm{N}(A^*A) = \mathrm{N}(A)$, and the dimension theorem gives equal ranks. ■

Corollary

If $\mathrm{rank}\, A = n$, then A^*A is invertible.

Proof. A^*A is $n \times n$ of rank n by Lemma 2. ■

Theorem 6.12

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $y \in \mathbb{F}^m$, there is $x_0 \in \mathbb{F}^n$ with $(A^*A)x_0 = A^*y$ and $\|Ax_0 - y\| \le \|Ax - y\|$ for all x. If $\mathrm{rank}\, A = n$, then $x_0 = (A^*A)^{-1}A^*y$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_A)$. By the Corollary to Theorem 6.6 there is a closest vector $Ax_0 \in W$ to y, and $Ax_0 - y \in W^\perp$. So $\langle Ax, Ax_0 - y\rangle_m = 0$ for all x, i.e. $\langle x, A^*(Ax_0 - y)\rangle_n = 0$ (Lemma 1), i.e. $A^*Ax_0 = A^*y$. If $\mathrm{rank}\, A = n$, invert A^*A (Corollary to Lemma 2). ■

For data $(1,2), (2,3), (3,5), (4,7)$: $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*y = (51, 17)^t$, so $x_0 = \frac{1}{20}\left(\begin{smallmatrix} 4 & -10 \\ -10 & 30 \end{smallmatrix}\right)\binom{51}{17} = \binom{1.7}{0}$. The least squares line is $y = 1.7t$, with error $E = 0.3$. Polynomial fits work the same way with columns t_i^k.

6.3.2 Minimal solutions to systems of linear equations

A solution s of $Ax = b$ is **minimal** if $\|s\| \le \|u\|$ for every solution u.

Theorem 6.13

Let $Ax = b$ be consistent, $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.
(a) There is exactly one minimal solution s, and $s \in \mathrm{R}(\mathsf{L}_{A^*})$.
(b) s is the only solution in $\mathrm{R}(\mathsf{L}_{A^*})$: if $(AA^*)u = b$, then $s = A^*u$.

Proof. Let $W = \mathrm{R}(\mathsf{L}_{A^*})$ and $W' = \mathrm{N}(\mathsf{L}_A)$. By Lemma 1, $W^\perp = W'$: $x \perp A^*y$ for all y iff $\langle Ax, y\rangle = 0$ for all y, iff $Ax = 0$.

(a) Let u be any solution and write $u = s + z$ with $s \in W$, $z \in W^\perp = W'$ (Theorem 6.6). Then $As = Au = b$. Every solution is $s + z'$ with $z' \in W' = W^\perp$ (Theorem 3.9), so $\|s + z'\|^2 = \|s\|^2 + \|z'\|^2 \ge \|s\|^2$, with equality iff $z' = 0$.

(b) If $v \in W$ is a solution, then $v - s \in W \cap W' = W \cap W^\perp = \{0\}$. If $AA^*u = b$, then $A^*u \in W$ is a solution, so it equals s. ∎

Example 3

$x + 2y + z = 4$, $x - y + 2z = -11$, $x + 5y = 19$: $AA^* = \begin{pmatrix} 6 & 1 & 11 \\ 1 & 6 & -4 \\ 11 & -4 & 26 \end{pmatrix}$, and $AA^*u = b$ has the solution $u = (1, -2, 0)$. So the minimal solution is $s = A^*u = (-1, 4, -3)$.

6.4 Normal and self-adjoint operators

Motivation. When does V have an *orthonormal* basis of eigenvectors? Over $\mathbb{C}$ exactly when T commutes with its adjoint (normal); over $\mathbb{R}$ exactly when $\mathsf{T} = \mathsf{T}^*$ (self-adjoint). Schur's triangularization is the stepping stone.

Lemma

If T (on finite-dimensional V) has an eigenvector, so does T^*.

Proof. Let $\mathsf{T}v = \lambda v$, $v \ne 0$. For all x, $0 = \langle(\mathsf{T} - \lambda\mathsf{I})v, x\rangle = \langle v, (\mathsf{T}^* - \bar\lambda\mathsf{I})x\rangle$, so $v \perp \mathrm{R}(\mathsf{T}^* - \bar\lambda\mathsf{I})$, which is therefore not all of V. Hence $\mathsf{T}^* - \bar\lambda\mathsf{I}$ is not onto, so not one-to-one: $\bar\lambda$ is an eigenvalue of T^*. ∎

Theorem 6.14 (Schur)

If the characteristic polynomial of T (on finite-dimensional inner product space V) splits, there is an orthonormal basis β with $[\mathsf{T}]_\beta$ upper triangular.

Proof. Induction on $n = \dim V$; $n = 1$ is clear. The characteristic polynomial splits, so T has an eigenvector, and by the Lemma T^* has a unit eigenvector z, $\mathsf{T}^*z = \lambda z$. $W = \{z\}^\perp$ ($\dim n - 1$) is T-invariant: for $y \in W$, $\langle \mathsf{T}y, z\rangle = \langle y, \mathsf{T}^*z\rangle = \bar\lambda\langle y, z\rangle = 0$. The characteristic polynomial of T_W divides that of T (Theorem 5.21), hence splits. By induction W has an orthonormal basis γ with $[\mathsf{T}_W]_\gamma$ upper triangular. Then $\beta = \gamma \cup \{z\}$ is orthonormal and $[\mathsf{T}]_\beta$ is upper triangular (its first $n - 1$ columns are those of $[\mathsf{T}_W]_\gamma$ padded by 0). ∎

Definition (Normal)

T is **normal** if $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T}$; A is normal if $AA^* = A^*A$. For orthonormal β, T is normal iff $[\mathsf{T}]_\beta$ is (Theorem 6.10).

Example 1

The rotation T of $\mathbb{R}^2$ by $\theta \in (0,\pi)$, $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, satisfies $AA^* = I = A^*A$, so it is normal but has no eigenvectors: normality does not give an orthonormal eigenbasis over $\mathbb{R}$.

Example 2

A real **skew-symmetric** A ($A^t = -A$) is normal: $AA^t = -A^2 = A^tA$.

Theorem 6.15

Let T be normal on V. Then
(a) $\|\mathsf{T}(x)\| = \|\mathsf{T}^*(x)\|$ for all x;
(b) $\mathsf{T} - c\mathsf{I}$ is normal for every $c \in \mathbb{F}$;
(c) if $\mathsf{T}(x) = \lambda x$, then $\mathsf{T}^*(x) = \bar{\lambda}x$;
(d) eigenvectors for distinct eigenvalues are orthogonal.

Proof.

(a) $\|\mathsf{T}x\|^2 = \langle \mathsf{T}^*\mathsf{T}x, x\rangle = \langle \mathsf{T}\mathsf{T}^*x, x\rangle = \|\mathsf{T}^*x\|^2$.

(b) $(\mathsf{T} - c\mathsf{I})^* = \mathsf{T}^* - \bar{c}\mathsf{I}$ commutes with $\mathsf{T} - c\mathsf{I}$.

(c) Apply (a) to the normal $\mathsf{U} = \mathsf{T} - \lambda\mathsf{I}$: $0 = \|\mathsf{U}x\| = \|\mathsf{U}^*x\| = \|(\mathsf{T}^* - \bar{\lambda}\mathsf{I})x\|$.

(d) If $\mathsf{T}x_1 = \lambda_1 x_1$, $\mathsf{T}x_2 = \lambda_2 x_2$, then by (c) $\lambda_1\langle x_1, x_2\rangle = \langle \mathsf{T}x_1, x_2\rangle = \langle x_1, \mathsf{T}^*x_2\rangle = \langle x_1, \bar{\lambda}_2 x_2\rangle = \lambda_2\langle x_1, x_2\rangle$, so $\langle x_1, x_2\rangle = 0$. ∎

Theorem 6.16

On a finite-dimensional complex inner product space, T is normal iff V has an orthonormal basis of eigenvectors of T.

Proof.

Step 1 (only if). Over $\mathbb{C}$ the characteristic polynomial splits, so Schur gives an orthonormal $\beta = \{v_1, \dots, v_n\}$ with $A = [\mathsf{T}]_\beta$ upper triangular. We show by induction on k that v_k is an eigenvector. v_1 is one (A is upper triangular). If $v_1, \dots, v_{k-1}$ are eigenvectors, $\mathsf{T}v_j = \lambda_j v_j$, then by Theorem 6.15(c), $A_{jk} = \langle \mathsf{T}v_k, v_j\rangle = \langle v_k, \mathsf{T}^*v_j\rangle = \langle v_k, \bar{\lambda}_j v_j\rangle = 0$ for $j < k$. So $\mathsf{T}v_k = A_{kk}v_k$.

Step 2 (if). If β is an orthonormal eigenbasis, $[\mathsf{T}]_\beta = D$ is diagonal, and $[\mathsf{T}^*]_\beta = D^*$ is diagonal too; diagonal matrices commute. ∎

Example 3

In $V = \operatorname{span}\{f_n : n \in \mathbb{Z}\} \subseteq \mathsf{H}$, $\mathsf{T}(f) = f_1 f$ and $\mathsf{U}(f) = f_{-1}f$ satisfy $\mathsf{T}(f_n) = f_{n+1}$, $\mathsf{U}(f_n) = f_{n-1}$, $\mathsf{U} = \mathsf{T}^*$ and $\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$. So T is normal, but it has no eigenvectors: if $f = \sum_{i=n}^{m} a_i f_i$ with $a_m \neq 0$ and $\mathsf{T}f = \lambda f$, then $f_{m+1} \in \operatorname{span}\{f_n, \dots, f_m\}$, contradicting independence. So Theorem 6.16 fails in infinite dimensions.

Definition (Self-adjoint)

T is **self-adjoint** (**Hermitian**) if $\mathsf{T} = \mathsf{T}^*$; A is self-adjoint if $A = A^*$. For real matrices this means symmetric.

Lemma

Let T be self-adjoint on finite-dimensional V. (a) Every eigenvalue of T is real.
(b) If V is real, the characteristic polynomial of T splits.

Proof.

(a) T is normal, so $\mathsf{T}x = \lambda x$ gives $\lambda x = \mathsf{T}x = \mathsf{T}^*x = \bar{\lambda}x$ (Theorem 6.15(c)), so $\lambda = \bar{\lambda}$.

(b) Let $A = [\mathsf{T}]_\beta$ for orthonormal β; A is real symmetric, so $A = A^*$ and L_A on $\mathbb{C}^n$ is self-adjoint. Its characteristic polynomial (that of T) splits over $\mathbb{C}$, and by (a) all its zeros are real. ■

Theorem 6.17

On a finite-dimensional real inner product space, T is self-adjoint iff V has an orthonormal basis of eigenvectors of T.

Proof. If T is self-adjoint, its characteristic polynomial splits (Lemma), Schur gives orthonormal β with $A = [\mathsf{T}]_\beta$ upper triangular, and $A^* = [\mathsf{T}^*]_\beta = A$ is lower triangular. So A is diagonal. Conversely, if $[\mathsf{T}]_\beta = D$ is real diagonal for orthonormal β, then $[\mathsf{T}^*]_\beta = D^* = D$, so $\mathsf{T}^* = \mathsf{T}$. ■

Example 4

Complex symmetric matrices need not be normal: $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$ has $(AA^*)_{12} = 1 + i \neq 1 - i = (A^*A)_{12}$.

6.5 Unitary and orthogonal operators and their matrices

Motivation. Operators preserving length are the symmetries of an inner product space: unitary over $\mathbb{C}$, orthogonal over $\mathbb{R}$. They preserve inner products, map orthonormal bases to orthonormal bases, and have matrices with $A^*A = I$. Combined with Schur and the spectral results, they give the matrix forms "unitarily/orthogonally equivalent to diagonal".

Definition (Unitary, orthogonal operator)

A linear T on finite-dimensional V with $\|\mathsf{T}(x)\| = \|x\|$ for all x is **unitary** if $\mathbb{F} = \mathbb{C}$ and **orthogonal** if $\mathbb{F} = \mathbb{R}$. (In infinite dimensions such an operator is called an isometry; if also onto, unitary/orthogonal.)

Example 1

If $h \in \mathsf{H}$ has $|h(x)| = 1$ for all x, then $\mathsf{T}(f) = hf$ satisfies $\|\mathsf{T}f\|^2 = \frac{1}{2\pi}\int |h|^2|f|^2 = \|f\|^2$.

Lemma

If U is self-adjoint on finite-dimensional V and $\langle x, \mathsf{U}(x)\rangle = 0$ for all x, then $\mathsf{U} = \mathsf{T}_0$.

Proof. Choose an orthonormal eigenbasis β (Theorems 6.16, 6.17; U is normal, and self-adjoint over $\mathbb{R}$). For $x \in \beta$ with $\mathsf{U}x = \lambda x$, $0 = \langle x, \lambda x\rangle = \bar{\lambda}$. So U vanishes on β. ■

Theorem 6.18

For linear T on finite-dimensional V the following are equivalent:
(a) $\mathsf{T}\mathsf{T}^* = \mathsf{T}^*\mathsf{T} = \mathsf{I}$;
(b) $\langle \mathsf{T}(x), \mathsf{T}(y)\rangle = \langle x, y\rangle$ for all x, y;
(c) if β is an orthonormal basis, so is $\mathsf{T}(\beta)$;
(d) there is an orthonormal basis β with $\mathsf{T}(\beta)$ orthonormal;
(e) $\|\mathsf{T}(x)\| = \|x\|$ for all x.

Proof.

Step (a)⇒(b). $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, \mathsf{T}^*\mathsf{T}y\rangle = \langle x, y\rangle$.

Step (b)⇒(c). $\langle \mathsf{T}v_i, \mathsf{T}v_j\rangle = \langle v_i, v_j\rangle = \delta_{ij}$, so $\mathsf{T}(\beta)$ is orthonormal with n vectors, hence a basis (Corollary 2 of Theorem 6.3).

Step (c)⇒(d). Orthonormal bases exist (Theorem 6.5).

Step (d)⇒(e). For $x = \sum a_i v_i$, $\|\mathsf{T}x\|^2 = \|\sum a_i \mathsf{T}v_i\|^2 = \sum |a_i|^2 = \|x\|^2$.

Step (e)⇒(a). $\langle x, (\mathsf{I} - \mathsf{T}^*\mathsf{T})x\rangle = \|x\|^2 - \|\mathsf{T}x\|^2 = 0$ for all x, and $\mathsf{I} - \mathsf{T}^*\mathsf{T}$ is self-adjoint, so $\mathsf{T}^*\mathsf{T} = \mathsf{I}$ by the Lemma. Then T is invertible with $\mathsf{T}^{-1} = \mathsf{T}^*$ (finite dimension), so $\mathsf{T}\mathsf{T}^* = \mathsf{I}$ too. ■

So unitary/orthogonal operators preserve inner products, angles and distances, and λ an eigenvalue implies $|\lambda| = 1$ ($\|\mathsf{T}x\| = |\lambda|\|x\|$).

Corollary 1

On a finite-dimensional real inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is both self-adjoint and orthogonal.

Proof. If T is self-adjoint, it has an orthonormal eigenbasis (Theorem 6.17), and if also orthogonal, eigenvalues have $|\lambda| = 1$. Conversely, such a basis makes T self-adjoint (Theorem 6.17), and $\mathsf{T}(\beta) = \{\pm v_i\}$ is orthonormal, so T is orthogonal by Theorem 6.18(d). ■

Corollary 2

On a finite-dimensional complex inner product space, V has an orthonormal basis of eigenvectors of T with eigenvalues of absolute value 1 iff T is unitary.

Proof. A unitary T is normal ($\mathsf{T}\mathsf{T}^* = \mathsf{I} = \mathsf{T}^*\mathsf{T}$), so Theorem 6.16 gives the basis, with $|\lambda| = 1$. Conversely $\mathsf{T}(\beta) = \{\lambda_i v_i\}$ is orthonormal; apply Theorem 6.18(d). ■

Example 2

The rotation T of $\mathbb{R}^2$ by θ preserves lengths, so it is orthogonal; in the standard basis $[\mathsf{T}] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $[\mathsf{T}]^*[\mathsf{T}] = I$.

Definition (Reflection of $\mathbb{R}^2$)

For a line L through 0, T is the **reflection about** L if $\mathsf{T}(x) = x$ for $x \in L$ and $\mathsf{T}(x) = -x$ for $x \in L^\perp$.

Example 3

A reflection is orthogonal: with unit $v_1 \in L$, $v_2 \in L^\perp$, $\mathsf{T}(\{v_1, v_2\}) = \{v_1, -v_2\}$ is orthonormal (Theorem 6.18(d)).

Definition (Orthogonal and unitary matrices)

A square A is **orthogonal** if $A^tA = AA^t = I$ and **unitary** if $A^*A = AA^* = I$. Equivalently, the rows (or columns) form an orthonormal basis of $\mathbb{F}^n$. T is unitary [orthogonal] iff $[\mathsf{T}]_\beta$ is, for orthonormal β.

Example 4

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal; its rows are orthonormal.

Example 5

For a reflection $\mathsf{T} = \mathsf{L}_A$ about the line L at angle α to the x-axis, $A = Q\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)Q^{-1}$ with $Q = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$, so $A = \left(\begin{smallmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{smallmatrix}\right)$, an orthogonal matrix.

Definition (Unitary/orthogonal equivalence)

A is **unitarily** [**orthogonally**] **equivalent** to D if $A = P^*DP$ for a unitary [orthogonal] P.

Theorem 6.19

A complex $n \times n$ matrix A is normal iff it is unitarily equivalent to a diagonal matrix.

Proof. If A is normal, Theorem 6.16 gives an orthonormal basis β of eigenvectors of L_A; with Q having these columns (Q unitary), $D = Q^{-1}AQ = Q^*AQ$ is diagonal. Conversely, if $A = P^*DP$, then $AA^* = P^*DD^*P = P^*D^*DP = A^*A$. ■

Theorem 6.20

A real $n \times n$ matrix A is symmetric iff it is orthogonally equivalent to a real diagonal matrix.

Proof. As in Theorem 6.19, using Theorem 6.17. Conversely $(P^tDP)^t = P^tDP$. ■

Example 6

$A = \left(\begin{smallmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{smallmatrix}\right)$ has eigenvalues $2, 2, 8$. Gram–Schmidt on the eigenvectors $(-1, 1, 0), (-1, 0, 1)$ of 2, and normalizing $(1, 1, 1)$, gives

$$P = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \qquad P^tAP = \operatorname{diag}(2, 2, 8).$$

Theorem 6.21 (Schur, matrix form)

Let the characteristic polynomial of $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ split over $\mathbb{F}$. (a) If $\mathbb{F} = \mathbb{C}$, A is unitarily equivalent to a complex upper triangular matrix.
(b) If $\mathbb{F} = \mathbb{R}$, A is orthogonally equivalent to a real upper triangular matrix.

Proof. Apply Theorem 6.14 to L_A on $\mathbb{F}^n$; the change of coordinate matrix Q from the orthonormal basis β to the standard basis has orthonormal columns, hence is unitary [orthogonal], and $Q^*AQ = [\mathsf{L}_A]_\beta$ is upper triangular. ■

6.5.1 Rigid motions

Definition (Rigid motion; translation)

On a real inner product space V, $f : V \to V$ is a **rigid motion** if $\|f(x) - f(y)\| = \|x - y\|$ for all x, y. A **translation** by v_0 is $g(x) = x + v_0$. Orthogonal operators and translations are rigid motions, and so are their composites.

Theorem 6.22

Every rigid motion f of a finite-dimensional real inner product space is $f = g \circ \mathsf{T}$ for a unique orthogonal operator T and a unique translation g.

Proof.

Step 1 (T preserves norms and inner products). Let $\mathsf{T}(x) = f(x) - f(0)$, a rigid motion with $\mathsf{T}(0) = 0$, so $\|\mathsf{T}x\| = \|x\|$. Expanding $\|\mathsf{T}x - \mathsf{T}y\|^2 = \|x - y\|^2$ gives $\langle \mathsf{T}x, \mathsf{T}y\rangle = \langle x, y\rangle$.

Step 2 (T is linear). For $x, y \in V$, $a \in \mathbb{R}$,

$$\begin{aligned}\|\mathsf{T}(x + ay) - \mathsf{T}x - a\mathsf{T}y\|^2 &= \|\mathsf{T}(x + ay) - \mathsf{T}x\|^2 + a^2\|\mathsf{T}y\|^2 - 2a\langle \mathsf{T}(x + ay) - \mathsf{T}x, \mathsf{T}y\rangle \\ &= a^2\|y\|^2 + a^2\|y\|^2 - 2a(\langle x + ay, y\rangle - \langle x, y\rangle) = 0,\end{aligned}$$

using Step 1 throughout. So T is linear and, preserving norms, orthogonal; $f = g \circ \mathsf{T}$ with g the translation by $f(0)$.

Step 3 (uniqueness). If $\mathsf{T}x + u_0 = \mathsf{U}x + v_0$ for all x, then $x = 0$ gives $u_0 = v_0$, and then $\mathsf{T} = \mathsf{U}$. ■

Theorem 6.23

For an orthogonal operator T on $\mathbb{R}^2$ with $A = [\mathsf{T}]_\beta$ (β standard), exactly one holds:
(a) T is a rotation and $\det A = 1$;
(b) T is a reflection about a line through 0 and $\det A = -1$.

Proof. $\mathsf{T}(e_1)$ is a unit vector, so $\mathsf{T}(e_1) = (\cos\theta, \sin\theta)$ for a unique $\theta \in [0, 2\pi)$, and $\mathsf{T}(e_2)$ is a unit vector orthogonal to it: $(-\sin\theta, \cos\theta)$ or $(\sin\theta, -\cos\theta)$. In the first case A is the rotation matrix, $\det A = 1$. In the second, $A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, $\det A = -1$, which is the reflection about the line at angle $\theta/2$ (Example 5). The determinants differ, so exactly one case occurs. ■

Corollary

Every rigid motion of $\mathbb{R}^2$ is a rotation followed by a translation, or a reflection about a line through 0 followed by a translation.

Proof. Theorems 6.22 and 6.23. ∎

Example 7

$A = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 1 & 2 \\ 2 & -1 \end{smallmatrix}\right)$ is orthogonal with $\det A = -\frac{1}{5} - \frac{4}{5} = -1$, so L_A is a reflection; its axis is $E_1 = \operatorname{span}\{(1+\sqrt{5}, 2)\}$, the line $y = \frac{\sqrt{5}-1}{2}x$.

6.5.2 Conic sections

A quadratic equation $ax^2 + 2bxy + cy^2 + dx + ey + f = 0$ has quadratic part X^tAX with $X = \binom{x}{y}$, $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. By Theorem 6.20 there is an orthogonal P with $P^tAP = \operatorname{diag}(\lambda_1, \lambda_2)$; replacing P by $P \cdot \operatorname{diag}(1, -1)$ if needed, $\det P = 1$, so P is a rotation. With $X = PX'$, $X^tAX = \lambda_1(x')^2 + \lambda_2(y')^2$: the xy-term disappears by rotating the axes. For example, $2x^2 - 4xy + 5y^2 - 36 = 0$ has $A = \left(\begin{smallmatrix} 2 & -2 \\ -2 & 5 \end{smallmatrix}\right)$ with eigenvalues $1, 6$ and $P = \frac{1}{\sqrt{5}}\left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$. It becomes $(x')^2 + 6(y')^2 = 36$, an ellipse.

6.6 Orthogonal projections and the spectral theorem

Motivation. A normal (complex) or self-adjoint (real) operator is a weighted sum of orthogonal projections onto its mutually orthogonal eigenspaces, $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$. This spectral decomposition gives polynomial descriptions of normality and of the projections themselves.

If $V = W_1 \oplus W_2$, the **projection on** W_1 **along** W_2 is $\mathsf{T}(x_1 + x_2) = x_1$; then $\mathrm{R}(\mathsf{T}) = W_1$, $\mathrm{N}(\mathsf{T}) = W_2$, and T is a projection iff $\mathsf{T}^2 = \mathsf{T}$.

Definition (Orthogonal projection)

A projection T on an inner product space is **orthogonal** if $\mathrm{R}(\mathsf{T})^\perp = \mathrm{N}(\mathsf{T})$ and $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}(\mathsf{T})$ (in finite dimensions one of these suffices). For finite-dimensional W, the orthogonal projection on W is $x \mapsto u$ of Theorem 6.6.

Theorem 6.24

T is an orthogonal projection iff T has an adjoint and $\mathsf{T}^2 = \mathsf{T} = \mathsf{T}^*$.

Proof.

Step 1 (only if). $\mathsf{T}^2 = \mathsf{T}$ as T is a projection. Write $x = x_1 + x_2$, $y = y_1 + y_2$ with $x_1, y_1 \in \mathrm{R}\,\mathsf{T}$, $x_2, y_2 \in \mathrm{N}\,\mathsf{T}$, orthogonal to each other. Then $\langle x, \mathsf{T}y\rangle = \langle x_1 + x_2, y_1\rangle = \langle x_1, y_1\rangle = \langle x_1, y_1 + y_2\rangle = \langle \mathsf{T}x, y\rangle$, so T^* exists and equals T.

Step 2 (if). $\mathsf{T}^2 = \mathsf{T}$ makes T the projection on $\mathrm{R}\,\mathsf{T}$ along $\mathrm{N}\,\mathsf{T}$ (for $x = \mathsf{T}x + (x - \mathsf{T}x)$). If $x \in \mathrm{R}\,\mathsf{T}$, $y \in \mathrm{N}\,\mathsf{T}$, then $\langle x, y\rangle = \langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}y\rangle = 0$, so $\mathrm{R}\,\mathsf{T} \subseteq \mathrm{N}(\mathsf{T})^\perp$. If $y \in \mathrm{N}(\mathsf{T})^\perp$, then $\|y - \mathsf{T}y\|^2 = \langle y - \mathsf{T}y, y\rangle - \langle y - \mathsf{T}y, \mathsf{T}y\rangle = 0 - \langle \mathsf{T}(y - \mathsf{T}y), y\rangle = 0$, so $y \in \mathrm{R}\,\mathsf{T}$. Hence $\mathrm{N}(\mathsf{T})^\perp = \mathrm{R}\,\mathsf{T}$; the other equality is analogous. ∎

Theorem 6.25 (Spectral theorem)

Let T on finite-dimensional V be normal ($\mathbb{F} = \mathbb{C}$) or self-adjoint ($\mathbb{F} = \mathbb{R}$), with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, eigenspaces W_i, and T_i the orthogonal projection on W_i. Then
(a) $V = W_1 \oplus \cdots \oplus W_k$;
(b) $W_i^\perp = W_i' = \bigoplus_{j \neq i} W_j$;

(c) $\mathsf{T}_i\mathsf{T}_j = \delta_{ij}\mathsf{T}_i$;
(d) $\mathsf{I} = \mathsf{T}_1 + \cdots + \mathsf{T}_k$;
(e) $\mathsf{T} = \lambda_1\mathsf{T}_1 + \cdots + \lambda_k\mathsf{T}_k$.

Proof.

(a) T is diagonalizable (Theorems 6.16, 6.17); apply Theorem 5.11.

(b) Eigenspaces are mutually orthogonal (Theorem 6.15(d)), so $W_i' \subseteq W_i^\perp$; both have dimension $\dim V - \dim W_i$ (by (a) and Theorem 6.7(c)).

(c) For $i \neq j$, $\mathrm{R}\,\mathsf{T}_j = W_j \subseteq W_i^\perp = \mathrm{N}\,\mathsf{T}_i$, so $\mathsf{T}_i\mathsf{T}_j = \mathsf{T}_0$; and $\mathsf{T}_i^2 = \mathsf{T}_i$.

(d) For $x = \sum x_j$, $x_j \in W_j$, $\mathsf{T}_i x = x_i$ by (b), so $\sum \mathsf{T}_i x = x$.

(e) $\mathsf{T}x = \sum \mathsf{T}x_i = \sum \lambda_i x_i = \sum \lambda_i \mathsf{T}_i x$. ∎

$\{\lambda_i\}$ is the **spectrum**, (d) the **resolution of the identity**, and (e) the **spectral decomposition** of T. Note $g(\mathsf{T}) = \sum g(\lambda_i)\mathsf{T}_i$ for any polynomial g, by (c).

Corollary 1

If $\mathbb{F} = \mathbb{C}$, T is normal iff $\mathsf{T}^* = g(\mathsf{T})$ for some polynomial g.

Proof. If T is normal, $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ (each T_i is self-adjoint by Theorem 6.24). Lagrange interpolation gives g with $g(\lambda_i) = \bar{\lambda}_i$, so $g(\mathsf{T}) = \sum \bar{\lambda}_i \mathsf{T}_i = \mathsf{T}^*$. Conversely $g(\mathsf{T})$ commutes with T. ∎

Corollary 2

If $\mathbb{F} = \mathbb{C}$, T is unitary iff T is normal and $|\lambda| = 1$ for every eigenvalue.

Proof. Unitary operators are normal with $|\lambda| = 1$. Conversely $\mathsf{T}\mathsf{T}^* = \sum \lambda_i\bar{\lambda}_i\mathsf{T}_i = \sum \mathsf{T}_i = \mathsf{I}$ by (c), (d). ∎

Corollary 3

If $\mathbb{F} = \mathbb{C}$ and T is normal, T is self-adjoint iff every eigenvalue is real.

Proof. $\mathsf{T}^* = \sum \bar{\lambda}_i \mathsf{T}_i$ equals $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$ iff $\bar{\lambda}_i = \lambda_i$ for all i (apply both sides to W_j). The "only if" is also the Lemma before Theorem 6.17. ∎

Corollary 4

In the spectral decomposition $\mathsf{T} = \sum \lambda_i \mathsf{T}_i$, each T_j is a polynomial in T.

Proof. Choose g_j (Lagrange) with $g_j(\lambda_i) = \delta_{ij}$; then $g_j(\mathsf{T}) = \sum g_j(\lambda_i)\mathsf{T}_i = \mathsf{T}_j$. ∎

6.7 The singular value decomposition and the pseudoinverse *

Motivation. Any linear map between inner product spaces, square or not, sends some orthonormal basis to an orthogonal set: $\mathsf{T}v_i = \sigma_i u_i$. This singular value decomposition gives the polar decomposition $A = WP$ and the pseudoinverse $A^\dagger$, which produces minimum-norm least squares solutions.

Theorem 6.26 (Singular value theorem for linear transformations)

Let $\mathsf{T}: V \to W$ (finite-dimensional inner product spaces) have rank r. There are orthonormal bases $\{v_1,\dots,v_n\}$ of V, $\{u_1,\dots,u_m\}$ of W and $\sigma_1 \geq \cdots \geq \sigma_r > 0$ with $\mathsf{T}(v_i) = \sigma_i u_i$ for $i \leq r$ and $\mathsf{T}(v_i) = 0$ for $i > r$. Conversely, given such bases and scalars, the v_i are eigenvectors of $\mathsf{T}^*\mathsf{T}$ with eigenvalues σ_i^2 ($i \leq r$) and 0 ($i > r$); hence the σ_i are uniquely determined by T.

Proof.

Step 1 (existence). The adjoint $\mathsf{T}^*: W \to V$ exists (as in Theorem 6.9), and $\mathsf{T}^*\mathsf{T}$ is self-adjoint with $\operatorname{rank}(\mathsf{T}^*\mathsf{T}) = r$ (Lemma 2 of Section 6.3 in operator form). Take an orthonormal eigenbasis $\{v_i\}$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \cdots$. They are ≥ 0: $\lambda_i = \langle \mathsf{T}^*\mathsf{T}v_i, v_i\rangle = \|\mathsf{T}v_i\|^2$. Put $\sigma_i = \sqrt{\lambda_i}$ and $u_i = \frac{1}{\sigma_i}\mathsf{T}(v_i)$ for $i \leq r$. Then $\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle \mathsf{T}^*\mathsf{T}v_i, v_j\rangle = \frac{\lambda_i}{\sigma_i\sigma_j}\delta_{ij} = \delta_{ij}$; extend to an orthonormal basis of W. For $i > r$, $\|\mathsf{T}v_i\|^2 = \lambda_i = 0$.

Step 2 (converse). $\mathsf{T}^*(u_i) = \sum_j \langle \mathsf{T}^*u_i, v_j\rangle v_j = \sum_j \langle u_i, \mathsf{T}v_j\rangle v_j = \sigma_i v_i$ for $i \leq r$ (and 0 for $i > r$), so $\mathsf{T}^*\mathsf{T}v_i = \sigma_i^2 v_i$ for $i \leq r$ and $\mathsf{T}^*\mathsf{T}v_i = 0$ for $i > r$. ∎

The σ_i are the **singular values** of T (extended by zeros up to $\min(m,n)$).

Example 1

$\mathsf{T}(f) = f'$ from $\mathsf{P}_2(\mathbb{R})$ to $\mathsf{P}_1(\mathbb{R})$, $\langle f, g\rangle = \int_{-1}^{1} fg$. In the orthonormal bases $\alpha = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{5}{8}}(3x^2-1)\}$ and $\alpha' = \{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x\}$, $A = [\mathsf{T}] = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$ and $A^*A = \operatorname{diag}(0, 3, 15)$. So $\sigma_1 = \sqrt{15}$, $\sigma_2 = \sqrt{3}$, with $v_1 = \sqrt{\frac{5}{8}}(3x^2-1)$, $v_2 = \sqrt{\frac{3}{2}}x$, $v_3 = \frac{1}{\sqrt{2}}$, $u_1 = \sqrt{\frac{3}{2}}x$, $u_2 = \frac{1}{\sqrt{2}}$.

Example 2

For invertible T on $\mathbb{R}^2$ with $\mathsf{T}v_i = \sigma_i u_i$, the unit circle $\{x_1v_1 + x_2v_2 : x_1^2 + x_2^2 = 1\}$ maps to $\{y_1u_1 + y_2u_2 : \frac{y_1^2}{\sigma_1^2} + \frac{y_2^2}{\sigma_2^2} = 1\}$, an ellipse with semi-axes σ_1, σ_2 along u_1, u_2.

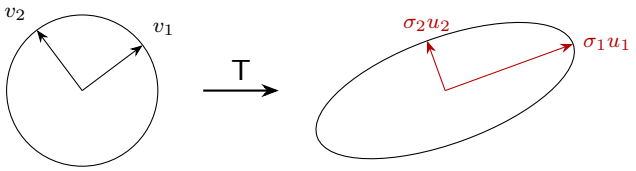


Figure. *Example 2:* T *maps the unit circle onto an ellipse whose semi-axes are* $\sigma_1 u_1$ *and* $\sigma_2 u_2$.

The singular values of a matrix A are those of L_A.

Theorem 6.27 (Singular value decomposition for matrices)

Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ have rank r and positive singular values $\sigma_1 \geq \cdots \geq \sigma_r$, and let $\Sigma \in \mathrm{M}_{m\times n}$ have $\Sigma_{ii} = \sigma_i$ ($i \leq r$) and all other entries 0. Then $A = U\Sigma V^*$ for some unitary U ($m \times m$) and V ($n \times n$).

Proof. Apply Theorem 6.26 to L_A; let V, U have columns v_j, u_i. Then AV has columns $Av_j = \sigma_j u_j$ (or 0), which are the columns of $U\Sigma$. So $AV = U\Sigma$ and $A = U\Sigma V^*$. ∎

Such a factorization is a **singular value decomposition** of A.

Example 3

$A = \left(\begin{smallmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{smallmatrix}\right)$: A^*A has eigenvalues $6, 0, 0$ with orthonormal eigenvectors $v_1 = \frac{1}{\sqrt{3}}(1,1,-1)$, $v_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, $v_3 = \frac{1}{\sqrt{6}}(1,1,2)$. So $\sigma_1 = \sqrt{6}$, $u_1 = \frac{1}{\sqrt{6}}Av_1 = \frac{1}{\sqrt{2}}(1,1)$, $u_2 = \frac{1}{\sqrt{2}}(1,-1)$, and $A = U\Sigma V^*$ with $\Sigma = \left(\begin{smallmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

Theorem 6.28 (Polar decomposition)

Every square A is $A = WP$ with W unitary and P positive semidefinite (P self-adjoint with nonnegative eigenvalues). If A is invertible, the representation is unique.

Proof. With $A = U\Sigma V^*$, put $W = UV^*$ (unitary) and $P = V\Sigma V^*$ (self-adjoint, eigenvalues $\sigma_i \geq 0$); then $WP = U\Sigma V^* = A$. If A is invertible and $A = WP = ZQ$, then $P^2 = P^*W^*WP = A^*A = Q^2$. Positive semidefinite square roots are unique (on each eigenspace of A^*A with eigenvalue μ, both P and Q must act as $\sqrt{\mu}$, since P commutes with P^2 and its eigenvalues are ≥ 0). So $P = Q$, P is invertible, and $W = Z = AP^{-1}$. ∎

Example 4

$A = \left(\begin{smallmatrix} 11 & -5 \\ -2 & 10 \end{smallmatrix}\right)$: A^*A has eigenvalues $200, 50$ with $v_1 = \frac{1}{\sqrt{2}}(1,-1)$, $v_2 = \frac{1}{\sqrt{2}}(1,1)$; $u_1 = \frac{1}{5}(4,-3)$, $u_2 = \frac{1}{5}(3,4)$. Then

$$W = UV^* = \frac{1}{5\sqrt{2}}\begin{pmatrix} 7 & -1 \\ 1 & 7 \end{pmatrix}, \qquad P = V\Sigma V^* = \frac{5}{\sqrt{2}}\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}.$$

6.7.1 The pseudoinverse

Definition (Pseudoinverse)

For linear $\mathsf{T} : V \to W$, let $\mathsf{L} : \mathrm{N}(\mathsf{T})^\perp \to \mathrm{R}(\mathsf{T})$ be the restriction of T (an isomorphism). The **pseudoinverse** (Moore–Penrose generalized inverse) $\mathsf{T}^\dagger : W \to V$ is $\mathsf{T}^\dagger(y) = \mathsf{L}^{-1}(y)$ for $y \in \mathrm{R}(\mathsf{T})$ and $\mathsf{T}^\dagger(y) = 0$ for $y \in \mathrm{R}(\mathsf{T})^\perp$. For a matrix, $A^\dagger$ is the matrix with $\mathsf{L}_{A^\dagger} = (\mathsf{L}_A)^\dagger$. If T is invertible, $\mathsf{T}^\dagger = \mathsf{T}^{-1}$.

In the setting of Theorem 6.26, $\mathsf{T}^\dagger(u_i) = \frac{1}{\sigma_i}v_i$ for $i \leq r$ and $\mathsf{T}^\dagger(u_i) = 0$ for $i > r$ ($\{v_1, \ldots, v_r\}$ spans $\mathrm{N}(\mathsf{T})^\perp$ and $\{u_1, \ldots, u_r\}$ spans $\mathrm{R}\,\mathsf{T}$).

Example 5

For $\mathsf{T} = d/dx$ of Example 1: $\mathsf{T}^\dagger(u_1) = \frac{1}{\sqrt{15}}v_1$, $\mathsf{T}^\dagger(u_2) = \frac{1}{\sqrt{3}}v_2$; e.g. $\mathsf{T}^\dagger(1) = \mathsf{T}^\dagger(\sqrt{2}u_2) = \sqrt{\frac{2}{3}}\sqrt{\frac{3}{2}}x = x$.

Theorem 6.29

If $A = U\Sigma V^*$ is a singular value decomposition and $\Sigma^\dagger \in \mathrm{M}_{n\times m}$ has $\Sigma^\dagger_{ii} = \frac{1}{\sigma_i}$ ($i \leq r$) and other entries 0, then $A^\dagger = V\Sigma^\dagger U^*$, and this is a singular value decomposition of $A^\dagger$.

Proof. $V\Sigma^\dagger U^* u_i = V\Sigma^\dagger e_i = \frac{1}{\sigma_i}v_i$ for $i \leq r$ and 0 for $i > r$, matching $(\mathsf{L}_A)^\dagger$ on the basis $\{u_i\}$. The factors are unitary and $\Sigma^\dagger$ has the required form (with values $\frac{1}{\sigma_r} \geq \cdots \geq \frac{1}{\sigma_1}$ after reordering). ∎

Example 6

For A of Example 3, $A^\dagger = V\Sigma^\dagger U^* = \frac{1}{\sqrt{6}} v_1 u_1^* = \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{pmatrix}$.

Lemma

For linear $\mathsf{T} : V \to W$ (finite-dimensional): (a) $\mathsf{T}^\dagger\mathsf{T}$ is the orthogonal projection of V on $\mathrm{N}(\mathsf{T})^\perp$;
(b) $\mathsf{T}\mathsf{T}^\dagger$ is the orthogonal projection of W on $\mathrm{R}(\mathsf{T})$.

Proof.

(a) For $x \in \mathrm{N}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger\mathsf{T}x = \mathsf{L}^{-1}\mathsf{L}x = x$; for $x \in \mathrm{N}(\mathsf{T})$, $\mathsf{T}^\dagger\mathsf{T}x = 0$.

(b) For $y \in \mathrm{R}\,\mathsf{T}$, $\mathsf{T}\mathsf{T}^\dagger y = \mathsf{L}\mathsf{L}^{-1}y = y$; for $y \in \mathrm{R}(\mathsf{T})^\perp$, $\mathsf{T}^\dagger y = 0$. ■

Theorem 6.30

For $Ax = b$ ($A \in \mathrm{M}_{m\times n}(\mathbb{F})$) let $z = A^\dagger b$.
(a) If $Ax = b$ is consistent, z is its unique solution of minimal norm.
(b) If $Ax = b$ is inconsistent, z is the unique best approximation of minimal norm: $\|Az - b\| \le \|Ay - b\|$ for all y, with equality iff $Az = Ay$, and then $\|z\| \le \|y\|$, with equality iff $z = y$.

Proof.

(a) $Az = AA^\dagger b$ is the orthogonal projection of b on $\mathrm{R}\,\mathsf{L}_A$ (Lemma), which is b itself. $z \in \mathrm{R}(\mathsf{L}_{A^\dagger}) = \mathrm{N}(\mathsf{L}_A)^\perp = \mathrm{R}(\mathsf{L}_{A^*})$, so z is the minimal solution by Theorem 6.13.

(b) Az is the projection of b on $\mathrm{R}\,\mathsf{L}_A$, hence the closest point (Corollary to Theorem 6.6): $\|Az - b\| \le \|Ay - b\|$, with equality iff $Ay = Az$. Among such y (the solutions of $Ax = Az$), $z \in \mathrm{N}(\mathsf{L}_A)^\perp$ has minimal norm by (a). ■

Example 7

For $x_1 + x_2 - x_3 = 1$, $x_1 + x_2 - x_3 = 1$, with A of Example 3: $z = A^\dagger(1,1)^t = \frac{1}{3}(1,1,-1)$ is the minimal solution. For the inconsistent right side $(1,2)^t$, $z = A^\dagger(1,2)^t = \frac{1}{2}(1,1,-1)$ is the minimal-norm best approximation.

6.8 Bilinear and quadratic forms *

Motivation. A bilinear form is represented by a matrix that changes by congruence Q^tAQ rather than by similarity. Symmetric forms can always be diagonalized, over $\mathbb{R}$ even by an orthonormal basis. This simplifies quadric surfaces, gives the second derivative test, and leads to Sylvester's law: the numbers of positive and negative squares are invariants.

Definition (Bilinear form)

$H : V \times V \to \mathbb{F}$ is a **bilinear form** if it is linear in each variable when the other is fixed: $H(ax_1 + x_2, y) = aH(x_1, y) + H(x_2, y)$ and $H(x, ay_1 + y_2) = aH(x, y_1) + H(x, y_2)$. $\mathcal{B}(V)$ is the set of bilinear forms on V.

Example 1

$H(a,b) = 2a_1b_1 + 3a_1b_2 + 4a_2b_1 - a_2b_2$ on $\mathbb{R}^2$ equals x^tAy with $A = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$, hence is bilinear.

Example 2

For any $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $H(x,y) = x^tAy$ is a bilinear form on $\mathbb{F}^n$. An inner product on a real space is bilinear; on a complex space it is not (conjugate linear in y).

Properties: $H(0,x) = H(x,0) = 0$; for fixed x, $L_x(y) = H(x,y)$ and $R_x(y) = H(y,x)$ are linear.

Theorem 6.31

Sums and scalar multiples of bilinear forms are bilinear forms, and $\mathcal{B}(V)$ is a vector space under $(H_1 + H_2)(x,y) = H_1(x,y) + H_2(x,y)$, $(aH)(x,y) = aH(x,y)$.

Proof. Bilinearity is preserved by these pointwise operations, and $\mathcal{B}(V)$ is a subspace of $\mathcal{F}(V \times V, \mathbb{F})$ containing the zero form. ■

Definition (Matrix of a bilinear form)

For an ordered basis $\beta = \{v_1, \ldots, v_n\}$, $\psi_\beta(H) = A$ with $A_{ij} = H(v_i, v_j)$ is the **matrix representation** of H.

Example 3

For H of Example 1 and $\beta = \{(1,1),(1,-1)\}$: $\psi_\beta(H) = \left(\begin{smallmatrix} 8 & 4 \\ 2 & -6 \end{smallmatrix}\right)$ (e.g. $B_{11} = 2+3+4-1$); in the standard basis $\psi(H) = \left(\begin{smallmatrix} 2 & 3 \\ 4 & -1 \end{smallmatrix}\right)$.

Theorem 6.32

$\psi_\beta : \mathcal{B}(V) \to \mathrm{M}_{n\times n}(\mathbb{F})$ is an isomorphism ($\dim V = n$).

Proof. ψ_β is linear. If $\psi_\beta(H) = O$, then $H(v_i, \cdot) = 0$ on β, so $H(v_i, y) = 0$ for all y; thus $H(\cdot, y) = 0$ on β, hence $H = 0$. Onto: given A, $H(x,y) = [\phi_\beta(x)]^t A [\phi_\beta(y)]$ is bilinear with $H(v_i, v_j) = e_i^t A e_j = A_{ij}$. ■

Corollary 1

$\dim \mathcal{B}(V) = n^2$.

Proof. Theorems 6.32 and 2.19. ■

Corollary 2

$\psi_\beta(H) = A$ iff $H(x,y) = [\phi_\beta(x)]^t A [\phi_\beta(y)]$ for all x, y.

Proof. The form on the right has matrix A (proof of Theorem 6.32); apply injectivity of ψ_β. ■

Corollary 3

For every $H \in \mathcal{B}(\mathbb{F}^n)$ there is a unique A with $H(x,y) = x^t A y$, namely $A = \psi_\beta(H)$ for the standard β.

Proof. Corollary 2 with $\phi_\beta = \mathsf{I}$. ■

Example 4

$H(a,b) = \det\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = a_1 b_2 - a_2 b_1$ has $A_{ij} = H(e_i, e_j)$, i.e. $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Definition (Congruence)

B is **congruent** to A if $B = Q^t A Q$ for some invertible Q; congruence is an equivalence relation.

Theorem 6.33

If Q changes γ-coordinates into β-coordinates, then $\psi_\gamma(H) = Q^t \psi_\beta(H) Q$ for every $H \in \mathcal{B}(V)$.

Proof. $w_j = \sum_k Q_{kj} v_k$, so $H(w_i, w_j) = \sum_{k,l} Q_{ki} H(v_k, v_l) Q_{lj} = (Q^t \psi_\beta(H) Q)_{ij}$. ■

Corollary

If $B = Q^t \psi_\beta(H) Q$ with Q invertible, then $B = \psi_\gamma(H)$ for the ordered basis γ with $w_j = \sum_i Q_{ij} v_i$ (so Q changes γ- into β-coordinates).

Proof. γ is a basis since Q is invertible; apply Theorem 6.33. ■

Definition (Symmetric, diagonalizable forms)

H is **symmetric** if $H(x,y) = H(y,x)$; it is **diagonalizable** if $\psi_\beta(H)$ is diagonal for some β.

Theorem 6.34

H is symmetric iff $\psi_\beta(H)$ is symmetric.

Proof. If H is symmetric, $A_{ij} = H(v_i, v_j) = H(v_j, v_i) = A_{ji}$. Conversely, if A is symmetric, $\bar{H}(x,y) = H(y,x)$ is bilinear with $\psi_\beta(\bar{H}) = A^t = A$, so $\bar{H} = H$. ■

Corollary

A diagonalizable bilinear form is symmetric.

Proof. Diagonal matrices are symmetric; apply Theorem 6.34. ■

Example 5

Over $\mathbb{F} = \mathbb{Z}_2$, $H(a,b) = a_1 b_2 + a_2 b_1$ on $\mathbb{F}^2$ is symmetric with $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ of rank 2, but it is not diagonalizable: a congruent diagonal matrix would be I (the only nonzero scalar is 1). But

$H(x,x) = 2a_1a_2 = 0$ for all x, whereas a form with diagonal matrix I has $H(e_1,e_1) = 1$.

Lemma

If $H \neq 0$ is a symmetric bilinear form on V over a field of characteristic $\neq 2$, then $H(x,x) \neq 0$ for some x.

Proof. Take u, v with $H(u,v) \neq 0$. If $H(u,u) = H(v,v) = 0$, then $H(u+v, u+v) = 2H(u,v) \neq 0$. ■

Theorem 6.35

Over a field of characteristic $\neq 2$, every symmetric bilinear form on finite-dimensional V is diagonalizable.

Proof. Induction on $n = \dim V$; $n = 1$ is trivial, and $H = 0$ is diagonal. Otherwise choose x with $H(x,x) \neq 0$ (Lemma). $L_x(y) = H(x,y)$ is a nonzero functional, so $\dim \mathrm{N}(L_x) = n - 1$, and $x \notin \mathrm{N}(L_x)$. The restriction of H to $\mathrm{N}(L_x)$ is symmetric, so by induction $\mathrm{N}(L_x)$ has a basis $\{v_1, \ldots, v_{n-1}\}$ with $H(v_i, v_j) = 0$ $(i \neq j)$. With $v_n = x$, $H(v_i, v_n) = H(x, v_i) = 0$ for $i < n$, and $\{v_1, \ldots, v_n\}$ is a basis diagonalizing H. ■

Corollary

Over a field of characteristic $\neq 2$, every symmetric matrix is congruent to a diagonal matrix.

Proof. Apply Theorem 6.35 to $H(x,y) = x^tAy$ on $\mathbb{F}^n$ and use Theorem 6.33. ■

Computing. Write $Q = E_1 \cdots E_k$ (elementary). Then $D = Q^tAQ = E_k^t \cdots E_1^t A E_1 \cdots E_k$: perform each elementary column operation followed by the same row operation.

Example 6

$A = \begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix}$: add column 1 to column 2 (and row 1 to row 2); subtract 3×column 1 from column 3 (and the rows likewise); subtract 4×column 2 from column 3 (and the rows likewise). With

$$Q = \begin{pmatrix} 1 & 1 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Q^tAQ = \operatorname{diag}(1, 1, -24).$$

Definition (Quadratic form)

$K : V \to \mathbb{F}$ is a **quadratic form** if $K(x) = H(x,x)$ for a symmetric $H \in \mathcal{B}(V)$. In characteristic $\neq 2$, H is recovered: $H(x,y) = \frac{1}{2}(K(x+y) - K(x) - K(y))$.

Example 7

A homogeneous polynomial of degree 2, $f(t_1, \ldots, t_n) = \sum_{i \leq j} a_{ij} t_i t_j$, is a quadratic form on $\mathbb{R}^n$: $f(x) = x^tAx$ with $A_{ii} = a_{ii}$ and $A_{ij} = A_{ji} = \frac{1}{2}a_{ij}$ for $i < j$.

Theorem 6.36

For a symmetric bilinear form H on a finite-dimensional real inner product space there is an orthonormal basis β with $\psi_\beta(H)$ diagonal.

Proof. Let γ be orthonormal and $A = \psi_\gamma(H)$, symmetric. By Theorem 6.20, $Q^tAQ = D$ for an orthogonal Q. Let β be defined by Q as in the Corollary to Theorem 6.33; its vectors $w_j = \sum_i Q_{ij}v_i$ are orthonormal (the columns of Q are orthonormal), and $\psi_\beta(H) = D$. ∎

Corollary

For a quadratic form K on a finite-dimensional real inner product space there are an orthonormal basis $\{v_1, \dots, v_n\}$ and scalars λ_i with $K(\sum s_i v_i) = \sum \lambda_i s_i^2$. (The λ_i are the eigenvalues of the matrix of K in any orthonormal basis.)

Proof. Theorem 6.36: $K(x) = H(x,x) = \sum_{i,j} s_i s_j H(v_i, v_j) = \sum \lambda_i s_i^2$. ∎

Example 8

$f(t_1, t_2) = 5t_1^2 + 2t_2^2 + 4t_1t_2$ has $A = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right)$ with eigenvalues $6, 1$ and unit eigenvectors $x_1 = \frac{1}{\sqrt{5}}(2,1)$, $x_2 = \frac{1}{\sqrt{5}}(-1,2)$. In coordinates $(t_1,t_2) = s_1x_1 + s_2x_2$, $f = 6s_1^2 + s_2^2$.

Example 9

The surface $2t_1^2 + 6t_1t_2 + 5t_2^2 - 2t_2t_3 + 2t_3^2 + 3t_1 - 2t_2 - t_3 + 14 = 0$: the quadratic part has matrix $\left(\begin{smallmatrix} 2 & 3 & 0 \\ 3 & 5 & -1 \\ 0 & -1 & 2 \end{smallmatrix}\right)$ with eigenvalues $2, 7, 0$ and unit eigenvectors $v_1 = \frac{1}{\sqrt{10}}(1,0,3)$, $v_2 = \frac{1}{\sqrt{35}}(-3,-5,1)$, $v_3 = \frac{1}{\sqrt{14}}(-3,2,1)$. In the coordinates s_i relative to $\{v_1, v_2, v_3\}$ the linear part is $-\sqrt{14}\, s_3$, so the equation becomes $s_3 = \frac{2s_1^2 + 7s_2^2}{\sqrt{14}} + \sqrt{14}$: an elliptic paraboloid.

6.8.1 The second derivative test

For $f : \mathbb{R}^n \to \mathbb{R}$ with continuous third partial derivatives, p is a **critical point** if all $\frac{\partial f}{\partial t_i}(p) = 0$; the **Hessian** $A(p)$ has entries $\frac{\partial^2 f}{\partial t_i \partial t_j}(p)$ and is symmetric.

Theorem 6.37 (Second derivative test)

Let p be a critical point of f with Hessian $A(p)$.
(a) If all eigenvalues of $A(p)$ are positive, f has a local minimum at p.
(b) If all are negative, f has a local maximum at p.
(c) If $A(p)$ has a positive and a negative eigenvalue, p is a saddle point (no local extremum).
(d) If rank $A(p) < n$ and $A(p)$ does not have eigenvalues of both signs, the test is inconclusive.

Proof. Translating, assume $p = 0$ and $f(0) = 0$. By Taylor's theorem, $f(x) = \frac{1}{2}x^tA(0)x + S(x)$ with $S(x)/\|x\|^2 \to 0$ as $x \to 0$. Choose an orthonormal eigenbasis $\{v_i\}$ of $A(0)$ (Theorem 6.20) with eigenvalues λ_i; for $x = \sum s_i v_i$, $\frac{1}{2}x^tA(0)x = \frac{1}{2}\sum \lambda_i s_i^2$.

(a) Let $\lambda = \min \lambda_i > 0$. Choose $\varepsilon > 0$ with $|S(x)| < \frac{\lambda}{4}\|x\|^2$ for $0 < \|x\| < \varepsilon$. Then $f(x) \geq \frac{\lambda}{2}\|x\|^2 - \frac{\lambda}{4}\|x\|^2 > 0 = f(0)$.

(b) Apply (a) to $-f$.

(c) If $\lambda_1 > 0 > \lambda_2$, then for small $s \neq 0$, $f(sv_1) = \frac{1}{2}\lambda_1 s^2 + S(sv_1) > 0$ and $f(sv_2) = \frac{1}{2}\lambda_2 s^2 + S(sv_2) < 0$, so 0 is not a local extremum.

(d) $f = t_1^2 + t_2^4$, $f = t_1^2 - t_2^4$ and $f = t_1^2 + t_2^3$ all have Hessian $\operatorname{diag}(2, 0)$ at 0, and they have, respectively, a minimum, no extremum, and no extremum there. ■

6.8.2 Sylvester's law of inertia

Theorem 6.38 (Sylvester's law of inertia)

For a symmetric bilinear form H on a finite-dimensional real vector space, the numbers of positive and of negative diagonal entries are the same in every diagonal matrix representation of H.

Proof. The number of nonzero diagonal entries is rank of the matrix, which is invariant under congruence (Theorem 3.4); so it suffices to treat the positive entries. Let $\beta = \{v_i\}$ and $\gamma = \{w_i\}$ diagonalize H, with $H(v_i, v_i) > 0$ exactly for $i \leq p$ and $H(w_i, w_i) > 0$ exactly for $i \leq q$, and suppose $p < q$. Let $L(x) = (H(x, v_1), \ldots, H(x, v_p), H(x, w_{q+1}), \ldots, H(x, w_r))$, $r =$ rank, a linear map $V \to \mathbb{R}^{p+r-q}$. Since $p + r - q < r$, nullity $L > n - r$, so some $x_0 \notin \operatorname{span}\{v_{r+1}, \ldots, v_n\}$ has $L(x_0) = 0$. Write $x_0 = \sum a_j v_j = \sum b_j w_j$. Then $H(x_0, v_i) = a_i H(v_i, v_i) = 0$ gives $a_i = 0$ for $i \leq p$, and $b_i = 0$ for $q < i \leq r$. Hence

$$H(x_0, x_0) = \sum_{j=p+1}^{r} a_j^2 H(v_j, v_j) \leq 0 \quad \text{and} \quad H(x_0, x_0) = \sum_{j=1}^{q} b_j^2 H(w_j, w_j) \geq 0,$$

so both vanish. The first gives $a_j = 0$ for $p < j \leq r$, so $x_0 \in \operatorname{span}\{v_{r+1}, \ldots, v_n\}$, a contradiction. By symmetry $p = q$. ■

Definition (Index, signature)

The number of positive diagonal entries in a diagonal representation is the **index**; (positive) − (negative) is the **signature**. Rank, index and signature are the **invariants**; any two determine the third. The same terms apply to real symmetric matrices via congruence.

Example 10

The form of Example 9 has a diagonal representation $\operatorname{diag}(2, 7, 0)$: rank 2, index 2, signature 2.

Example 11

$K(x, y) = x^2 - y^2$ has matrix $\operatorname{diag}(1, -1)$, and in the basis $\{(1, 1), (1, -1)\}$ the matrix $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ (not diagonal); rank 2, index 1, signature 0.

Corollary 1 (Sylvester's law for matrices)

For a real symmetric A, the numbers of positive and negative diagonal entries of a diagonal matrix congruent to A do not depend on the choice of the diagonal matrix.

Proof. Apply Theorem 6.38 to $H(x, y) = x^t A y$ and use the Corollary to Theorem 6.33. ■

Corollary 2

Two real symmetric $n \times n$ matrices are congruent iff they have the same invariants.

Proof. Congruent matrices are congruent to the same diagonal matrices, so they have the same invariants (Corollary 1). Conversely, a real symmetric matrix with index p and rank r is congruent to some $D = \operatorname{diag}(d_1, \ldots, d_n)$ with $d_i > 0$ $(i \le p)$, $d_i < 0$ $(p < i \le r)$, $d_i = 0$ $(i > r)$ (reorder by a permutation Q). With $Q = \operatorname{diag}(|d_i|^{-1/2} \text{ or } 1)$, $Q^t D Q = J_{pr} = \operatorname{diag}(I_p, -I_{r-p}, O)$. Matrices with the same invariants are both congruent to J_{pr}. ■

Corollary 3

A real symmetric $n \times n$ matrix has index p and rank r iff it is congruent to J_{pr}.

Proof. The proof of Corollary 2, and the fact that J_{pr} has index p and rank r. ■

Example 12

A of Example 6 is congruent to $\operatorname{diag}(1, 1, -24)$: rank 3, index 2. $B = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ has eigenvalues $2, -1, -1$: rank 3, index 1. $C = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & -1 \end{smallmatrix}\right)$ has eigenvalues $1, \frac{1 \pm \sqrt{13}}{2}$: rank 3, index 2. So A and C are congruent, and B is congruent to neither (eigenvalues give a diagonal congruent matrix by Theorem 6.20).

6.9 Einstein's special theory of relativity *

Motivation. If the speed of light is the same in every inertial frame, the coordinate change between two frames in relative motion must preserve the light cone $x^2 + y^2 + z^2 - t^2 = 0$. A few axioms then force the Lorentz transformation, and time dilation follows.

Frames S, S' have parallel axes, S' moves along the x-axis of S with velocity $0 < v < 1$ (units: light seconds and seconds, so the speed of light is 1), and the clocks C, C' read 0 when the origins coincide. An event has **space–time coordinates** $(x, y, z, t)^t$ relative to S, C and $(x', y', z', t')^t$ relative to S', C'; $\mathsf{T}_v : \mathbb{R}^4 \to \mathbb{R}^4$ maps the former to the latter.

Axioms. (R1) Light has speed 1 in either frame. (R2) T_v is an isomorphism. (R3) $y' = y$ and $z' = z$. (R4) x', t' do not depend on y, z. (R5) The origin of S moves along the x'-axis with velocity $-v$ as measured in S'. T_v is the **Lorentz transformation** in direction x.

Theorem 6.39

On $\mathbb{R}^4$ (standard inner product):
(a) $\mathsf{T}_v(e_i) = e_i$ for $i = 2, 3$;
(b) $\operatorname{span}\{e_2, e_3\}$ is T_v-invariant;
(c) $\operatorname{span}\{e_1, e_4\}$ is T_v-invariant;
(d) both are T_v^*-invariant;
(e) $\mathsf{T}_v^*(e_i) = e_i$ for $i = 2, 3$.

Proof.

(a) $\mathsf{T}_v(0) = 0$, so by (R4) the first and fourth coordinates of $\mathsf{T}_v(0, a, b, 0)^t$ are 0 for all a, b, and by (R3) $\mathsf{T}_v(e_2) = e_2$, $\mathsf{T}_v(e_3) = e_3$.

(b) Follows from (a).

(c) [Supplement] By (R3) the second and third coordinates of $\mathsf{T}_v(x,0,0,t)^t$ are 0.

(d) [Supplement] $W = \operatorname{span}\{e_2, e_3\}$ and $W^\perp = \operatorname{span}\{e_1, e_4\}$ are both T_v-invariant. If $\mathsf{T}_v(W) \subseteq W$, then for $x \in W^\perp$, $w \in W$: $\langle \mathsf{T}_v^* x, w\rangle = \langle x, \mathsf{T}_v w\rangle = 0$, so $\mathsf{T}_v^*(W^\perp) \subseteq W^\perp$; symmetrically for W.

(e) For $j \neq 2$, $\langle \mathsf{T}_v^* e_2, e_j\rangle = \langle e_2, \mathsf{T}_v e_j\rangle = 0$ by (a), (c); so $\mathsf{T}_v^* e_2 = k e_2$ with $k = \langle \mathsf{T}_v^* e_2, e_2\rangle = \langle e_2, \mathsf{T}_v e_2\rangle = 1$. Similarly for e_3. ■

A light flash at the common origin at time 0 is seen at (x,y,z) at time $t \geq 0$ iff $x^2+y^2+z^2-t^2 = 0$, and by (R1) the same holds in primed coordinates. Let $A = \operatorname{diag}(1,1,1,-1)$, so $\langle \mathsf{L}_A(w), w\rangle = x^2+y^2+z^2-t^2$.

Theorem 6.40

If $\langle \mathsf{L}_A(w), w\rangle = 0$, then $\langle \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w), w\rangle = 0$.

Proof. If $t \geq 0$, w is an event of the flash, so its primed coordinates $\mathsf{T}_v w$ satisfy $\langle \mathsf{L}_A \mathsf{T}_v w, \mathsf{T}_v w\rangle = 0$, which is $\langle \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v w, w\rangle$. If $t < 0$, apply this to $-w$. ■

Let $w_1 = (1,0,0,1)^t$, $w_2 = (1,0,0,-1)^t$, an orthogonal basis of $\operatorname{span}\{e_1, e_4\}$, which is invariant under $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ by Theorem 6.39.

Theorem 6.41

There are nonzero scalars a, b with (a) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_1) = a w_2$;
(b) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_2) = b w_1$.

Proof.

(a) $\langle \mathsf{L}_A w_1, w_1\rangle = 0$, so by Theorem 6.40 $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v(w_1) \perp w_1$, and it lies in $\operatorname{span}\{w_1, w_2\}$; hence it is $a w_2$. Since $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ is invertible, $a \neq 0$.

(b) Same, with $\langle \mathsf{L}_A w_2, w_2\rangle = 0$. ■

Corollary

With $B_v = [\mathsf{T}_v]_\beta$ (β standard): (a) $B_v^* A B_v = A$;
(b) $\mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v = \mathsf{L}_A$.

Proof [Supplement] (left as an exercise in the book). $M = \mathsf{T}_v^* \mathsf{L}_A \mathsf{T}_v$ is self-adjoint, so $2a = \langle M w_1, w_2\rangle = \langle w_1, M w_2\rangle = 2b$, i.e. $a = b$. Since $A w_1 = w_2$ and $A w_2 = w_1$, $M = a \mathsf{L}_A$ on $\operatorname{span}\{e_1, e_4\}$, and $M = \mathsf{L}_A$ on $\operatorname{span}\{e_2, e_3\}$ by Theorem 6.39. So $\langle A \mathsf{T}_v w, \mathsf{T}_v w\rangle = a\langle Aw, w\rangle$ for $w \in \operatorname{span}\{e_1, e_4\}$. The axioms as stated do not force $a = 1$: rescaling T_v on $\operatorname{span}\{e_1, e_4\}$ by $\sqrt{a}$ preserves (R1)–(R5). We therefore add the physically natural *reciprocity* assumption: $\mathsf{T}_{-v} = \mathsf{T}_v^{-1}$, and $\mathsf{T}_{-v} = R \mathsf{T}_v R$ for the reflection $R = \operatorname{diag}(-1,1,1,1)$ (reversing the x-axes). Then $a(-v) = a(v)$ (since R preserves $\langle A\cdot,\cdot\rangle$), and $a(v)a(-v) = 1$ (since $\mathsf{T}_{-v}\mathsf{T}_v = \mathsf{I}$). So $a^2 = 1$. Moreover $a > 0$: by (R5), $\mathsf{T}_v(e_4) = (-vt'', 0, 0, t'')^t$ with $t'' > 0$, and then $a \cdot (-1) = a\langle Ae_4, e_4\rangle = (v^2-1)t''^2 < 0$. Hence $a = 1$, which is (b), and (a) is its matrix form. ■

One second (on C) after the origins coincide, the origin of S' has S-coordinates $(v,0,0,1)^t$ and S'-coordinates $(0,0,0,t')^t$ with $t' > 0$. By the Corollary, $v^2 - 1 = \langle \mathsf{L}_A(v,0,0,1)^t, (v,0,0,1)^t\rangle = \langle \mathsf{L}_A(0,0,0,t')^t, (0,0,0,t')^t\rangle = -t'^2$, so $t' = \sqrt{1-v^2}$ and $\mathsf{T}_v(v,0,0,1)^t = (0,0,0,\sqrt{1-v^2})^t$. Likewise (R5) gives $\mathsf{T}_v(e_4) = (-vt'', 0, 0, t'')^t$ with $t'' = 1/\sqrt{1-v^2}$.

Theorem 6.42

In the standard basis,

$$[\mathsf{T}_v]_\beta = B_v = \begin{pmatrix} \frac{1}{\sqrt{1-v^2}} & 0 & 0 & \frac{-v}{\sqrt{1-v^2}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-v}{\sqrt{1-v^2}} & 0 & 0 & \frac{1}{\sqrt{1-v^2}} \end{pmatrix}.$$

Proof. Columns $2, 3$ are e_2, e_3 (Theorem 6.39(a)); column 4 is $\mathsf{T}_v(e_4)$ above. Column 1 is $\mathsf{T}_v(e_1) = \frac{1}{v}(\mathsf{T}_v(v,0,0,1)^t - \mathsf{T}_v(e_4)) = \frac{1}{v}(v/\sqrt{1-v^2}, 0, 0, \sqrt{1-v^2} - 1/\sqrt{1-v^2})^t = (\frac{1}{\sqrt{1-v^2}}, 0, 0, \frac{-v}{\sqrt{1-v^2}})^t$. ∎

Time contraction. A ship moving with velocity v from the solar system has S-coordinates $(vt, 0, 0, t)^t$ and S'-coordinates $(0, 0, 0, t')^t$. From $B_v(vt, 0, 0, t)^t = (0, 0, 0, t')^t$, $t' = \frac{-v^2 t + t}{\sqrt{1-v^2}} = t\sqrt{1-v^2}$: time t on Earth corresponds to $t\sqrt{1-v^2}$ on the ship. In general units, $t' = t\sqrt{1 - v^2/c^2}$.

6.10 Conditioning and the Rayleigh quotient *

Motivation. Small errors in the data b of $Ax = b$ may produce large relative errors in x. The amplification is controlled by the condition number $\|A\|\|A^{-1}\|$, which the Rayleigh quotient computes from the extreme eigenvalues of A^*A.

Example 1

$x_1 + x_2 = 5$, $x_1 - x_2 = 1 + h$ has solution $(3 + \frac{h}{2}, 2 - \frac{h}{2})$. The relative changes are $\frac{\|\delta b\|}{\|b\|} = \frac{|h|}{\sqrt{26}} = \frac{\|\delta x\|}{\|x\|}$: the system is **well-conditioned**.

Example 2

$x_1 + x_2 = 3$, $x_1 + 1.00001x_2 = 3.00001 + h$ has solution $(2 - 10^5 h,\ 1 + 10^5 h)$, so $\frac{\|\delta x\|}{\|x\|} = 10^5\sqrt{\frac{2}{5}}\,|h| \geq 10^4|h|$ while $\frac{\|\delta b\|}{\|b\|} \approx \frac{|h|}{3\sqrt{2}}$: **ill-conditioned** (the two lines are nearly parallel).

Definition (Matrix norm; Rayleigh quotient)

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\|A\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|}$. For self-adjoint B, the **Rayleigh quotient** is $R(x) = \frac{\langle Bx, x\rangle}{\|x\|^2}$ $(x \neq 0)$.

Theorem 6.43

For self-adjoint B, $\max_{x\neq 0} R(x)$ is the largest and $\min_{x\neq 0} R(x)$ the smallest eigenvalue of B.

Proof. Take an orthonormal eigenbasis $\{v_i\}$, $Bv_i = \lambda_i v_i$, $\lambda_1 \geq \cdots \geq \lambda_n$ (real). For $x = \sum a_i v_i$, $R(x) = \frac{\sum \lambda_i |a_i|^2}{\sum |a_i|^2} \leq \lambda_1$, with $R(v_1) = \lambda_1$. The minimum is analogous. ∎

Corollary 1

For any square A, $\|A\| = \sqrt{\lambda}$, where λ is the largest eigenvalue of A^*A.

Proof. $\frac{\|Ax\|^2}{\|x\|^2} = \frac{\langle A^*Ax, x\rangle}{\|x\|^2}$ is the Rayleigh quotient of the self-adjoint A^*A; apply Theorem 6.43. ■

Lemma

For square A, λ is an eigenvalue of A^*A iff it is an eigenvalue of AA^*.

Proof. If $\lambda \neq 0$ and $A^*Ax = \lambda x$, $x \neq 0$, then $Ax \neq 0$ and $AA^*(Ax) = \lambda Ax$; symmetrically. For $\lambda = 0$: A^*A is singular iff A is (Lemma 2 of Section 6.3), iff A^* is, iff AA^* is. ■

Corollary 2

If A is invertible, $\|A^{-1}\| = \frac{1}{\sqrt{\lambda}}$, where λ is the smallest eigenvalue of A^*A.

Proof. By Corollary 1, $\|A^{-1}\|^2$ is the largest eigenvalue of $(A^{-1})^*A^{-1} = (AA^*)^{-1}$, i.e. the reciprocal of the smallest eigenvalue of AA^*, which equals that of A^*A by the Lemma. ■

Example 3

For $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $B = A^*A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ has eigenvalues $3, 3, 0$, so $\|A\| = \sqrt{3}$; indeed $R(x) = \frac{2(a^2+b^2+c^2-ab+ac+bc)}{a^2+b^2+c^2} \leq 3$.

The **condition number** of invertible A is $\text{cond}(A) = \|A\| \cdot \|A^{-1}\|$. From $\|Ax\| \leq \|A\|\|x\|$: $\|b\| \leq \|A\|\|x\|$, $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, $\|\delta b\| \leq \|A\|\|\delta x\|$, $\|x\| \leq \|A^{-1}\|\|b\|$.

Theorem 6.44

For $Ax = b$ with A invertible and $b \neq 0$:
(a) for any norm, $\frac{1}{\text{cond}(A)} \frac{\|\delta b\|}{\|b\|} \leq \frac{\|\delta x\|}{\|x\|} \leq \text{cond}(A) \frac{\|\delta b\|}{\|b\|}$;
(b) for the Euclidean norm, $\text{cond}(A) = \sqrt{\lambda_1/\lambda_n}$, the largest and smallest eigenvalues of A^*A.

Proof.

(a) Multiply the inequalities above: $\frac{\|\delta x\|}{\|x\|} \leq \frac{\|A^{-1}\|\|\delta b\|}{\|b\|/\|A\|}$ and $\frac{\|\delta x\|}{\|x\|} \geq \frac{\|\delta b\|/\|A\|}{\|A^{-1}\|\|b\|}$.

(b) Corollaries 1 and 2. ■

Thus $\text{cond}(A) \geq 1$, with equality iff A is a scalar multiple of a unitary (orthogonal) matrix; a large condition number signals potential loss of accuracy.

6.11 The geometry of orthogonal operators *

Motivation. Every orthogonal operator on a real inner product space splits into rotations of mutually orthogonal planes and at most one reflection. In $\mathbb{R}^3$, an orthogonal operator is a rotation about an axis, possibly followed by a reflection.

Definition (Rotation; reflection)

An operator T on finite-dimensional real V is a **rotation** if $\mathsf{T} = \mathsf{I}$ or there are a 2-dimensional subspace W, an orthonormal basis $\{x_1, x_2\}$ of W and θ with $\mathsf{T}(x_1) = (\cos\theta)x_1 + (\sin\theta)x_2$, $\mathsf{T}(x_2) = (-\sin\theta)x_1 + (\cos\theta)x_2$, and $\mathsf{T}(y) = y$ for $y \in W^\perp$ (a rotation of W about $W^\perp$). T is a **reflection** if for some 1-dimensional W, $\mathsf{T}(x) = -x$ on W and $\mathsf{T}(y) = y$ on $W^\perp$ (a reflection

about $W^\perp$). Both are orthogonal.

Example 1

On a 1-dimensional real space, an orthogonal operator is $\mathsf{T} = \pm\mathsf{I}$: $\mathsf{T}(x) = \lambda x$ with $|\lambda| = 1$. So it is the identity (a rotation) or a reflection.

Example 2

(a) $\mathsf{T}(a, b) = (-a, b)$ is the reflection of $\mathbb{R}^2$ about the y-axis ($W = \operatorname{span}\{e_1\}$).
(b) $\mathsf{T}(a, b, c) = (a, b, -c)$ is the reflection of $\mathbb{R}^3$ about the xy-plane.

Theorem 6.45

An orthogonal operator T on a 2-dimensional real inner product space is a rotation or a reflection; it is a rotation iff $\det \mathsf{T} = 1$ and a reflection iff $\det \mathsf{T} = -1$.

Proof. Choose an orthonormal basis and argue as in Theorem 6.23: the matrix is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ (a rotation, $\det = 1$) or $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ (a symmetric orthogonal matrix with eigenvalues $1, -1$ and orthogonal eigenvectors: a reflection, $\det = -1$). ■

Corollary

On a 2-dimensional real inner product space, a composite of a reflection and a rotation (in either order) is a reflection.

Proof. It is orthogonal with determinant $(-1) \cdot 1 = -1$; apply Theorem 6.45. ■

Lemma

Every linear operator T on a nonzero finite-dimensional real vector space has an invariant subspace W with $1 \le \dim W \le 2$.

Proof [Supplement]. Factor the characteristic polynomial f over $\mathbb{R}$ into linear and irreducible quadratic factors $g_1 \cdots g_k$. By Cayley–Hamilton $f(\mathsf{T}) = g_1(\mathsf{T}) \cdots g_k(\mathsf{T}) = \mathsf{T}_0$, so some $g_j(\mathsf{T})$ is not one-to-one; take $x \neq 0$ with $g_j(\mathsf{T})x = 0$. If $g_j = t - c$, $W = \operatorname{span}\{x\}$; if $g_j = t^2 + bt + c$, then $\mathsf{T}^2 x = -b\mathsf{T}x - cx$, so $W = \operatorname{span}\{x, \mathsf{T}x\}$ is invariant. ■

Theorem 6.46

For an orthogonal operator T on a nonzero finite-dimensional real inner product space V, there are pairwise orthogonal T-invariant subspaces $W_1, \dots, W_m$ with (a) $1 \le \dim W_i \le 2$ and (b) $V = W_1 \oplus \cdots \oplus W_m$.

Proof. Induction on $\dim V$. By the Lemma there is an invariant W_1 of dimension 1 or 2. T is injective, so $\mathsf{T}(W_1) = W_1$, and then $W_1^\perp$ is invariant: for $y \in W_1^\perp$, $x \in W_1$, $x = \mathsf{T}x'$ with $x' \in W_1$ and $\langle \mathsf{T}y, x\rangle = \langle \mathsf{T}y, \mathsf{T}x'\rangle = \langle y, x'\rangle = 0$. $\mathsf{T}_{W_1^\perp}$ is orthogonal, so induction decomposes $W_1^\perp$, and $V = W_1 \oplus W_1^\perp$. ■

Theorem 6.47

In Theorem 6.46:
(a) the number of W_i on which T_{W_i} is a reflection is even or odd according as $\det \mathsf{T} = 1$ or -1;
(b) the decomposition can be chosen so that this number is 0 or 1 according as $\det \mathsf{T} = 1$ or -1, and so that every 1-dimensional W_i on which T is a rotation (i.e. the identity) is orthogonal to all the others.

Proof.

(a) $\det \mathsf{T} = \prod \det \mathsf{T}_{W_i}$ (Theorem 5.25: block-diagonal matrix), and $\det \mathsf{T}_{W_i} = -1$ exactly for reflections (Example 1, Theorem 6.45).

(b) Split each 2-dimensional W_i on which T is a reflection into the two orthogonal lines E_1, E_{-1} of T_{W_i}. Now all reflection pieces are lines with $\mathsf{T} = -\mathsf{I}$; pair them up two at a time, each pair spanning a plane on which $\mathsf{T} = -\mathsf{I}$, a rotation by π. At most one line with $\mathsf{T} = -\mathsf{I}$ remains, and its presence is decided by (a). ■

Corollary

Every orthogonal operator T on a finite-dimensional real inner product space is a product $\mathsf{T} = \mathsf{T}_1 \cdots \mathsf{T}_m$ of commuting orthogonal operators T_i, each a rotation or a reflection, at most one of them a reflection.

Proof. With $V = W_1 \oplus \cdots \oplus W_m$ as in Theorem 6.47(b), let $\mathsf{T}_i = \mathsf{T}$ on W_i and $\mathsf{T}_i = \mathsf{I}$ on $W_i^{\perp} = \bigoplus_{j \neq i} W_j$. Each T_i is a rotation or a reflection, they commute (they act on different summands), and $\mathsf{T}_1 \cdots \mathsf{T}_m$ agrees with T on every W_i. ■

Example 3

For orthogonal T on a 3-dimensional space, Theorem 6.47(b) gives $m = 2$ (a line W_1 and a plane W_2, with T_{W_2} a rotation) or $m = 3$ (three lines). In the first case T is a rotation about the axis W_1, followed by a reflection in $W_1^{\perp}$ if $\mathsf{T}_{W_1} = -\mathsf{I}$. In the second, at most one line carries $\mathsf{T} = -\mathsf{I}$, so T is the identity or a reflection. Hence: an orthogonal operator on $\mathbb{R}^3$ is a rotation about an axis or such a rotation composed with a reflection, according as $\det \mathsf{T} = 1$ or -1.

Appendices. Sets, Functions, Fields, Complex Numbers, Polynomials

A Sets

Motivation. The language of sets, unions, intersections and equivalence relations used throughout the notes.

A **set** is a collection of objects (**elements**); $x \in A$ means x is an element of A. $A = B$ if they have the same elements; $A \subseteq B$ (A is a **subset** of B) if every element of A is in B; $A \subsetneq B$ is a **proper** subset. The **empty set** $\emptyset$ has no elements. Sets are described by listing, $\{1, 2, 3\}$, or by a property, $\{x \in \mathbb{R} : 1 < x < 2\}$.

Example 1

$A = \{x \in \mathbb{R} : 1 < x < 2\}$, the reals between 1 and 2; $\{1, 2\} = \{2, 1\}$ (order and repetition do not matter).

The **union** $A \cup B = \{x : x \in A \text{ or } x \in B\}$, the **intersection** $A \cap B = \{x : x \in A \text{ and } x \in B\}$; A, B are **disjoint** if $A \cap B = \emptyset$. For a family $\{A_\alpha : \alpha \in \Lambda\}$, $\bigcup_\alpha A_\alpha$ and $\bigcap_\alpha A_\alpha$ are defined likewise.

Example 2

$A = \{1, 3, 5\}$, $B = \{1, 5, 7, 8\}$: $A \cup B = \{1, 3, 5, 7, 8\}$, $A \cap B = \{1, 5\}$.

Example 3

For $\alpha > 1$ let $A_\alpha = \{x \in \mathbb{R} : -\frac{1}{\alpha} \leq x \leq 1 + \alpha\}$. Then $\bigcup_{\alpha>1} A_\alpha = \{x : -1 < x\}$ and $\bigcap_{\alpha>1} A_\alpha = \{x : 0 \leq x \leq 2\}$.

A **relation** on A is a set S of ordered pairs of elements of A; write $x \sim y$ if $(x, y) \in S$. It is an **equivalence relation** if it is **reflexive** ($x \sim x$), **symmetric** ($x \sim y \Rightarrow y \sim x$) and **transitive** ($x \sim y$, $y \sim z \Rightarrow x \sim z$). Examples: equality; congruence mod m on $\mathbb{Z}$; similarity and congruence of matrices; isomorphism of vector spaces.

B Functions

Motivation. One-to-one, onto and invertible functions, composition and restriction.

A **function** $f : A \to B$ assigns to each $x \in A$ a unique $f(x) \in B$; A is the **domain**, B the **codomain**, and $f(A) = \{f(x) : x \in A\}$ the **range**. For $S \subseteq A$, $f(S)$ is the **image**; for $T \subseteq B$, $f^{-1}(T) = \{x : f(x) \in T\}$ is the **preimage**. $f = g$ if $f(x) = g(x)$ for all x.

Example 1

$f : [-10, 10] \to \mathbb{R}$, $f(x) = x^2 + 1$: the range is $[1, 101]$, $f([2, 3]) = [5, 10]$, and $f^{-1}([5, 10]) = [-3, -2] \cup [2, 3]$.

f is **one-to-one** if $f(x) = f(y)$ implies $x = y$, and **onto** if $f(A) = B$. The **restriction** f_S of f to $S \subseteq A$ is f on S.

Example 2

$f : [-1, 1] \to [0, 1]$, $f(x) = x^2$ is onto but not one-to-one ($f(-1) = f(1)$); $f_{[0,1]}$ is one-to-one and onto.

The **composite** of $f : A \to B$ and $g : B \to C$ is $g \circ f : A \to C$, $(g \circ f)(x) = g(f(x))$; composition is associative but not commutative. f is **invertible** if some $g : B \to A$ has $g \circ f = \mathsf{I}_A$ and $f \circ g = \mathsf{I}_B$; then $g = f^{-1}$ is unique. Facts: f is invertible iff it is one-to-one and onto; $(f^{-1})^{-1} = f$; if f, g are invertible, so is $g \circ f$, with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Example 3

$f(x) = 3x + 1$ on $\mathbb{R}$ is one-to-one and onto, with $f^{-1}(x) = \frac{x-1}{3}$.

C Fields

Motivation. Scalars may come from any field: a set where one can add, subtract, multiply and divide by nonzero elements. The characteristic of a field (for example 2 in $\mathbb{Z}_2$) occasionally matters.

Definition (Field)

A **field** $\mathbb{F}$ is a set with operations $+$ and $\cdot$ such that for all $a, b, c \in \mathbb{F}$: (F1) $a + b = b + a$, $ab = ba$; (F2) $(a + b) + c = a + (b + c)$, $(ab)c = a(bc)$; (F3) there are distinct $0, 1$ with $0 + a = a$, $1 \cdot a = a$; (F4) each a has c with $a + c = 0$, and each $b \neq 0$ has d with $bd = 1$; (F5) $a(b + c) = ab + ac$.

Example 1

$\mathbb{R}$ is a field.

Example 2

$\mathbb{Q}$ is a field.

Example 3

$\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a field: $(a + b\sqrt{2})^{-1} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}$ ($a^2 \neq 2b^2$ for rationals not both zero).

Example 4

$\mathbb{Z}_2 = \{0, 1\}$ with $0 + 0 = 1 + 1 = 0$, $0 + 1 = 1$, $0 \cdot 0 = 0 \cdot 1 = 0$, $1 \cdot 1 = 1$ is a field.

Example 5

$\mathbb{N}$ and $\mathbb{Z}$ are not fields: (F4) fails (2 has no multiplicative inverse in $\mathbb{Z}$).

Theorem C.1 (Cancellation laws)

(a) If $a+b=c+b$, then $a=c$.
(b) If $ab=cb$ and $b\neq 0$, then $a=c$.

Proof.

(a) Add d with $b+d=0$: $a=a+(b+d)=(a+b)+d=(c+b)+d=c$.

(b) Multiply by b^{-1} similarly. ■

Corollary

The elements 0, 1 of (F3) and the c, d of (F4) are unique.

Proof. If $0'$ also works, $0'+b=b=0+b$, so $0'=0$ by (a); if $a+c=0=a+c'$, then $c=c'$ by (a). Similarly for 1 and d using (b). ■

Write $-a$ and b^{-1}; $a-b=a+(-b)$, $a/b=ab^{-1}$.

Theorem C.2

(a) $a\cdot 0=0$.
(b) $(-a)b=a(-b)=-(ab)$.
(c) $(-a)(-b)=ab$.

Proof.

(a) $0+a\cdot 0=a\cdot 0=a(0+0)=a\cdot 0+a\cdot 0$; cancel.

(b) $ab+(-a)b=(a+(-a))b=0\cdot b=0$, so $(-a)b=-(ab)$; similarly $a(-b)$.

(c) By (b) twice, $(-a)(-b)=-(a(-b))=-(-(ab))=ab$. ■

Corollary

0 has no multiplicative inverse.

Proof. $0\cdot d=0\neq 1$ by (a) and (F3). ■

The **characteristic** of $\mathbb{F}$ is the least $p\geq 1$ with $1+\cdots+1$ (p summands) $=0$, or 0 if no such p exists. $\mathbb{Z}_2$ has characteristic 2; $\mathbb{Q},\mathbb{R},\mathbb{C}$ have characteristic 0. In characteristic 2, $x+x=0$ for all x, so 2 cannot be inverted; some results (e.g. Theorem 6.35) exclude this case. A nonzero characteristic is prime.

D Complex numbers

Motivation. $\mathbb{C}$ is the field in which every nonconstant polynomial has a zero. This is why complex scalars make eigenvalue theory complete.

Definition (Complex numbers)

A **complex number** is $z=a+bi$ with $a,b\in\mathbb{R}$ (real and imaginary parts); $(a+bi)+(c+di)=(a+c)+(b+d)i$ and $(a+bi)(c+di)=(ac-bd)+(bc+ad)i$. Then $i^2=-1$, and $\mathbb{R}\subseteq\mathbb{C}$ via $a=a+0i$.

Example 1

$z = 3 - 5i$, $w = 9 + 7i$: $z + w = 12 + 2i$, $zw = (27 + 35) + (-45 + 21)i = 62 - 24i$.

Example 2

$(-5 + 2i)(1 - 3i) = -5 + 15i + 2i - 6i^2 = 1 + 17i$.

Theorem D.1

$\mathbb{C}$ is a field.

Proof [Supplement]. (F1), (F2), (F5) are direct computations; $0 = 0 + 0i$, $1 = 1 + 0i$; $-(a + bi) = -a - bi$; and for $a + bi \neq 0$, $(a + bi)\frac{a-bi}{a^2+b^2} = 1$. ■

Definition (Conjugate)

$\overline{a + bi} = a - bi$.

Example 3

$\overline{-3 + 2i} = -3 - 2i$, $\overline{4 - 7i} = 4 + 7i$, $\bar{6} = 6$.

Theorem D.2

(a) $\bar{\bar{z}} = z$;
(b) $\overline{z + w} = \bar{z} + \bar{w}$;
(c) $\overline{zw} = \bar{z}\bar{w}$;
(d) $\overline{z/w} = \bar{z}/\bar{w}$ $(w \neq 0)$;
(e) $z \in \mathbb{R}$ iff $\bar{z} = z$.

Proof. Direct computation with $z = a + bi$, $w = c + di$; e.g. (c): $\overline{zw} = (ac - bd) - (bc + ad)i = (a - bi)(c - di)$. (d) follows from (c) applied to $(z/w)w = z$. ■

Definition (Absolute value)

$|a + bi| = \sqrt{a^2 + b^2}$; then $z\bar{z} = |z|^2$, and $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$.

Example 4

$\frac{1+4i}{3-2i} = \frac{(1+4i)(3+2i)}{9+4} = \frac{-5+14i}{13}$.

Theorem D.3

(a) $|zw| = |z||w|$;
(b) $|z/w| = |z|/|w|$ $(w \neq 0)$;
(c) $|z + w| \leq |z| + |w|$;
(d) $|z| - |w| \leq |z + w|$.

Proof.

(a) $|zw|^2 = zw\overline{zw} = z\bar{z}w\bar{w}$.

(b) From (a).

(c) $|z+w|^2 = |z|^2 + 2\,\mathrm{Re}(z\bar{w}) + |w|^2 \le |z|^2 + 2|z||w| + |w|^2$, since $\mathrm{Re}\,u \le |u|$.

(d) $|z| = |(z+w)+(-w)| \le |z+w| + |w|$. ■

Polar form: $z = |z|(\cos\theta + i\sin\theta) = |z|e^{i\theta}$, and zw multiplies absolute values and adds angles; $z^n = |z|^n e^{in\theta}$, so every complex number has n-th roots.

Theorem D.4 (Fundamental theorem of algebra)

Every $p(z) = a_n z^n + \cdots + a_0 \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ has a zero.

Proof.

Step 1 (a minimum exists). Let $\mu = \inf_z |p(z)|$. For $|z| = s$ large, $|p(z)| \ge |a_n|s^n - \sum_{k<n}|a_k|s^k > \mu + 1$, say for $s > R$. On the closed disk $|z| \le R$, the continuous function $|p|$ attains its minimum at some z_0 (compactness), and $|p(z_0)| = \mu$.

Step 2 (d'Alembert). Suppose $p(z_0) \ne 0$. Let $q(z) = p(z+z_0)/p(z_0)$, a polynomial of degree n with $q(0) = 1$ and $|q| \ge 1$ everywhere. Write $q(z) = 1 + b_k z^k + \cdots + b_n z^n$ with $b_k \ne 0$. Choose θ with $e^{ik\theta}b_k = -|b_k|$ (k-th roots exist). For $r > 0$ small with $r^k|b_k| < 1$,

$$|q(re^{i\theta})| \le 1 - r^k|b_k| + r^{k+1}\sum_{j>k}|b_j|r^{j-k-1} = 1 - r^k\Big(|b_k| - r\sum_{j>k}|b_j|r^{j-k-1}\Big) < 1$$

for r small enough, contradicting $|q| \ge 1$. So $p(z_0) = 0$. ■

Corollary

Every $p \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ factors as $p(z) = a_n(z-c_1)\cdots(z-c_n)$.

Proof. Induction on n: by Theorem D.4 and Corollary 1 of Theorem E.1, $p(z) = (z-c_1)q(z)$ with $\deg q = n-1$ and leading coefficient a_n. ■

A field in which every nonconstant polynomial splits is **algebraically closed**; $\mathbb{C}$ is, $\mathbb{R}$ is not.

E Polynomials

Motivation. Division with remainder, the Euclidean property "relatively prime $\Rightarrow q_1f_1 + q_2f_2 = 1$" and unique factorization into irreducibles are the polynomial facts used for eigenvalues, Cayley–Hamilton and decompositions of operators.

$f \mid g$ (f **divides** g) if $g = fq$ for some polynomial q.

Theorem E.1 (Division algorithm)

If $\deg f = n$ and $\deg g = m \ge 0$, there are unique q, r with $f = qg + r$ and $\deg r < m$.

Proof.

Step 1 (existence). Induction on n. If $n < m$, take $q = 0$, $r = f$. Otherwise, with leading coefficients a_n of f and b_m of g, $h = f - a_n b_m^{-1}x^{n-m}g$ has degree $< n$, so $h = q_1 g + r$ by induction, and $f = (q_1 + a_n b_m^{-1}x^{n-m})g + r$.

Step 2 (uniqueness). If $qg + r = q'g + r'$, then $(q - q')g = r' - r$ has degree $< m$, which forces $q = q'$ (otherwise the left side has degree $\geq m$) and then $r = r'$. ■

Corollary 1

For $\deg f > 0$ and $a \in \mathbb{F}$: $f(a) = 0$ iff $(x - a) \mid f$.

Proof. $f = q \cdot (x - a) + r$ with r constant; evaluating at a, $r = f(a)$. ■

Corollary 2

A polynomial of degree $n \geq 1$ has at most n distinct zeros.

Proof. Induction: if $f(a) = 0$, $f = (x - a)q$ with $\deg q = n - 1$, and every other zero of f is a zero of q (a field has no zero divisors). ■

Definition (Relatively prime)

Nonzero f, g are **relatively prime** if no polynomial of positive degree divides both. E.g. $x^2(x - 1)$ and $(x - 1)(x - 2)$ are not; $x^2(x - 1)$ and $(x - 2)(x - 3)$ are.

Theorem E.2

If f_1, f_2 are relatively prime, there are q_1, q_2 with $q_1 f_1 + q_2 f_2 = 1$.

Proof. Let $I = \{q_1 f_1 + q_2 f_2\}$, a set closed under addition and under multiplication by polynomials, containing nonzero elements. Let $d \in I$ be nonzero of least degree. By the division algorithm, $f_1 = qd + r$ with $\deg r < \deg d$ and $r = f_1 - qd \in I$; minimality forces $r = 0$, so $d \mid f_1$, and likewise $d \mid f_2$. As f_1, f_2 are relatively prime, $\deg d = 0$: $d = c \neq 0$, and $c^{-1}d = 1 \in I$. ■

Example 1

$f_1 = x^3 - x^2 + 1$ and $f_2 = (x - 1)^2$ are relatively prime; indeed $(-x + 2)f_1 + (x^2 - x - 1)f_2 = 1$.

Definition (Polynomials of operators)

For $f(x) = a_0 + a_1 x + \cdots + a_n x^n$, $f(\mathsf{T}) = a_0 \mathsf{I} + a_1 \mathsf{T} + \cdots + a_n \mathsf{T}^n$ and $f(A) = a_0 I + a_1 A + \cdots + a_n A^n$.

Example 2

$\mathsf{T}(a, b) = (2a + b, a - b)$ and $f(x) = x^2 + 2x - 3$: $\mathsf{T}^2(a, b) = (5a + b, a + 2b)$, so $f(\mathsf{T})(a, b) = (5a + b, a + 2b) + (4a + 2b, 2a - 2b) - (3a, 3b) = (6a + 3b, 3a - 3b)$.

Theorem E.3

(a) $f(\mathsf{T})$ is a linear operator.
(b) If β is a finite ordered basis and $A = [\mathsf{T}]_\beta$, then $[f(\mathsf{T})]_\beta = f(A)$.

Proof.

(a) Linear combinations of composites of linear maps are linear (Theorems 2.7, 2.9).

(b) $[\mathsf{T}^k]_\beta = A^k$ (Corollary to Theorem 2.11), and Theorem 2.8. ■

Theorem E.4

$f_1(\mathsf{T})f_2(\mathsf{T}) = f_2(\mathsf{T})f_1(\mathsf{T})$ and $f_1(A)f_2(A) = f_2(A)f_1(A)$.

Proof. Both equal $(f_1 f_2)(\mathsf{T})$, since powers of T commute and composition distributes over sums (Theorem 2.10). ■

Theorem E.5

If f_1, f_2 are relatively prime, there are q_1, q_2 with (a) $q_1(\mathsf{T})f_1(\mathsf{T}) + q_2(\mathsf{T})f_2(\mathsf{T}) = \mathsf{I}$; (b) $q_1(A)f_1(A) + q_2(A)f_2(A) = I$.

Proof. Apply $g \mapsto g(\mathsf{T})$, which respects sums and products, to the identity of Theorem E.2. ■

Definition (Monic, irreducible)

f is **monic** if its leading coefficient is 1; it is **irreducible** if $\deg f > 0$ and f is not a product of two polynomials of positive degree. (Irreducibility depends on $\mathbb{F}$: x^2+1 is irreducible over $\mathbb{R}$, not over $\mathbb{C}$.) Degree-one polynomials are irreducible.

Theorem E.6

If ϕ is irreducible and $\phi \nmid f$, then ϕ and f are relatively prime.

Proof. A common divisor g of positive degree divides ϕ, so $\phi = cg$ with c constant (irreducibility); then $\phi \mid f$, a contradiction. ■

Theorem E.7

Two distinct irreducible monic polynomials are relatively prime.

Proof. If $\phi_1 \mid \phi_2$, then $\phi_2 = c\phi_1$ (irreducible), and $c = 1$ (monic), so $\phi_1 = \phi_2$. Hence $\phi_1 \nmid \phi_2$; apply Theorem E.6. ■

Theorem E.8

If ϕ is irreducible and $\phi \mid fg$, then $\phi \mid f$ or $\phi \mid g$.

Proof. If $\phi \nmid f$, then $q_1\phi + q_2 f = 1$ (Theorems E.6, E.2), so $g = q_1\phi g + q_2 fg$ is divisible by ϕ. ■

Corollary

If irreducible monic ϕ divides $\phi_1 \cdots \phi_n$ (irreducible monic), then $\phi = \phi_i$ for some i.

Proof. Induction on n: $\phi \mid \phi_1 \cdots \phi_{n-1}$ (then induction) or $\phi \mid \phi_n$ (then $\phi = \phi_n$ by Theorem E.7). ■

Theorem E.9 (Unique factorization)

Every f of positive degree is uniquely $f = c\,\phi_1^{n_1}\cdots\phi_k^{n_k}$ with c a constant, ϕ_i distinct irreducible monic polynomials and $n_i \geq 1$.

Proof.

Step 1 (existence). Induction on $\deg f$. If f is irreducible, $f = c\cdot(c^{-1}f)$. Otherwise $f = gh$ with $0 < \deg g, \deg h < \deg f$; factor g and h by induction and collect equal factors.

Step 2 (uniqueness). c is the leading coefficient. If $\prod \phi_i^{n_i} = \prod \psi_j^{m_j}$, then ϕ_1 divides the right side, so $\phi_1 = \psi_j$ for some j (Corollary to Theorem E.8). Cancel one factor (no zero divisors) and induct on the total degree. ■

Theorem E.10

If $\mathbb{F}$ is infinite and $f(a) = g(a)$ for all $a \in \mathbb{F}$, then $f = g$.

Proof. $h = f - g$ vanishes at infinitely many points. If $\deg h \geq 1$, this contradicts Corollary 2 of Theorem E.1. So h is constant, and $h(a) = 0$ gives $h = 0$. ■

Over a finite field this fails: x^2 and x define the same function on $\mathbb{Z}_2$.